

THE EYE

VOLUME 5

AI Architecture

The governed model, context, multi-agent, safety, evaluation, learning,
and AI operations architecture for The Eye

OBSERVE / UNDERSTAND / PREDICT / SIMULATE / DECIDE / LEARN

CONTROLLED AI ARCHITECTURE BASELINE / V1.0

Inherited from Volume 0 — Product Constitution, Volume 3 — Technical Architecture, and Volume 4 — Engineering
Specification

1 August 2026

Document Control

Volume 5 is the controlled AI architecture of The Eye. It governs every model, method, agent, prompt, context, retrieval, tool, evaluator, learned artifact, AI product, safety control, and AI operating process inside the frozen Strategic Intelligence Operating System.

CONTROL	VALUE
Product	The Eye
Tagline	The Operating System for Strategic Intelligence
Controlled document	Volume 5 — AI Architecture
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Status	Controlled AI Architecture Baseline
Binding sources	Volume 0 — Product Constitution v1.0; Volume 3 — Technical Architecture v1.0; Volume 4 — Engineering Specification v1.0
Primary audience	CTO, Chief AI Officer, principal architects and engineers, AI and data engineering, model risk, security, privacy, SRE, governance, domain leadership, product and programme leadership
Normative scope	AI system planes; model and inference platform; context and grounding; multi-agent runtime; named agent architecture; safety; explainability; evaluation; learning; release; deployment; incidents
Downstream inheritance	Volume 6 — Infrastructure Architecture; Volume 7 — Data Platform; model and agent specifications; evaluation plans; safety cases; implementation repositories and operating procedures

Normative language

SHALL, MUST, SHALL NOT, and MUST NOT are binding. MAY permits an implementation choice inside a frozen semantic and authority boundary. Requirements carry stable AI-CC-NNN identities. Conformance is scoped to the exact task, artifact, configuration, data, population, consequence class, environment, deployment profile, and period under review.

CONTROLLING RULE

Volume 0 controls constitutional meaning. Volume 3 controls product architecture. Volume 4 controls engineering realization. Volume 5 controls AI architecture beneath those authorities. No model, agent, evaluator, or provider can create a parallel truth system or final decision authority.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-008 · C-028 · C-051 · C-052

AI Architecture Charter

The Eye uses artificial intelligence to extend human strategic judgment while keeping institutional authority, truth, memory, evidence, policy, and accountability outside the control of any model or agent.

AI COMMITMENT	BINDING ARCHITECTURE CONSEQUENCE
Humans decide	AI recommends, forecasts, simulates, reasons, explains, and prepares; named humans retain final authority for consequential decisions and irreversible effects.
Ground before inference	Context resolves through governed evidence, Enterprise Memory, the Knowledge Graph, temporal state, and Digital Twins before decision-active products are admitted.
Explain by construction	Every material result carries evidence, method, assumptions, uncertainty, alternatives, dissent, limitations, policy, and execution lineage.
Models are plural dependencies	Task and product contracts remain stable while approved providers, model families, rules, statistics, graph methods, optimizers, and simulators evolve.
Agents are bounded runtime packages	Identity, role, task, context, models, tools, authority, budgets, supervision, gates, failure, evidence, and lifecycle are explicit.
No silent learning	Feedback and outcomes produce governed change proposals; production behavior changes only through evaluated, approved, reversible releases.
Trust is enforced outside AI	Policy, capabilities, side-effect boundaries, human gates, audit, safety monitoring, and emergency stop do not depend on model compliance.
One product across deployments	SaaS, Private Cloud, and On-Premise preserve the same task, product, authority, provenance, uncertainty, safety, replay, and lifecycle semantics.

Definition of AI architecture complete

- Every AI capability maps to one canonical layer, owning component, task contract, product schema, consequence class, accountable owner, and evidence profile.
- Every execution can reconstruct its instructions, context, retrieval, models, agents, tools, policies, evaluations, human actions, outputs, effects, and terminal state.
- Uncertainty, abstention, partial results, dissent, correction, degraded service, and revocation remain explicit and testable product states.
- Agent and model changes pass scope-specific evaluation, red teaming, safety, privacy, security, deployment-parity, release, monitoring, rollback, and retirement controls.
- Decision Replay preserves the original information and authority boundary while later outcomes and lessons remain distinct comparison state.

CONSTITUTIONAL INHERITANCE / C-001 · C-004 · C-005 · C-024 · C-028 · C-035 · C-044 · C-052

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Native Word heading styles provide navigation. Chapter, requirement, component, agent, model, risk, and decision identifiers are stable references for architecture, engineering, assurance, operations, and downstream volumes.

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Executive AI Architecture Statement

Volume 5 defines the governed intelligence machinery that allows The Eye to observe, understand, predict, simulate, recommend, explain, and learn continuously while preserving human authority and institutional accountability.

The AI architecture is not a chatbot framework and it is not a collection of autonomous digital employees. It is a controlled computational system embedded in the ten canonical layers. Models, agents, graph methods, temporal methods, optimizers, simulators, retrieval systems, and rules operate as plural, replaceable dependencies behind stable strategic product contracts. The Knowledge Graph, Enterprise Memory, Digital Twins, evidence, policy, and temporal truth provide grounding; owning layers admit canonical state; named humans decide.

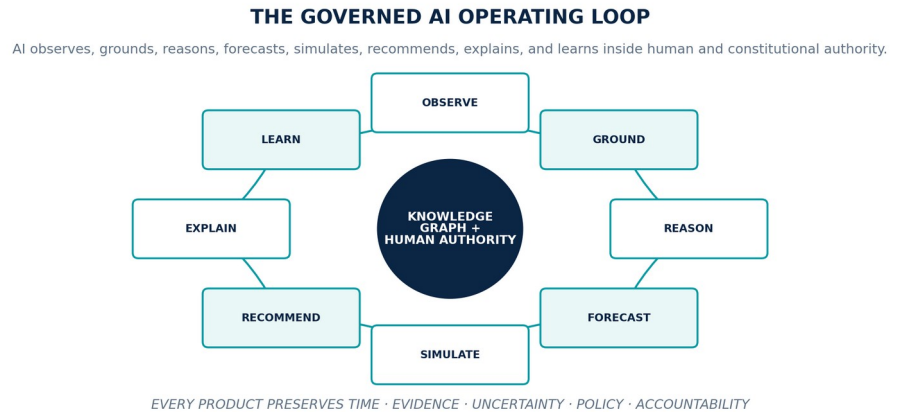


Figure 1. The governed AI operating loop. AI capabilities remain grounded in the Knowledge Graph and subordinate to human final authority.

AI PLANE	ARCHITECTURE OBLIGATION
Model control plane	Own task contracts, artifact identity, resolution, adapters, instructions, inference security, structured output, capacity, and economics.
Agent control plane	Own agent packages, planning, supervision, durable workflow, messages, capabilities, tools, human gates, checkpoints, and replay.
Context control plane	Assemble minimum authorized temporal context from retrieval, graph, memory, twins, evidence, and policy with a reproducible manifest.
AI product plane	Produce typed evidence-bound assessments, forecasts, scenarios, simulations, warnings, explanations, and decision packages for owning-layer admission.
Trust and governance plane	Classify risk, enforce privacy and security, constrain effects, preserve audit, enable contest, and keep safety controls external to AI.
Evaluation and learning plane	Establish scope-specific fitness, monitor drift and outcomes, govern change, release immutable configurations, and retire or revoke safely.

AI ARCHITECTURE THESIS

The Eye gains strategic power by connecting methods, evidence, memory, graph, twins, futures, and human decisions into one controlled operating loop. It remains trustworthy only when uncertainty, disagreement, authority, effects, correction, and learning are explicit state—not implied behavior.

PART I

AI Authority

Normative scope, doctrine, human sovereignty, explanation, pluralism, and traceability.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

1 AI Architecture Authority and Scope

Volume 5 defines the complete artificial-intelligence architecture of The Eye within the frozen product constitution, ten-layer technical architecture, and controlled engineering baseline. It governs models, agents, context, tools, evaluation, safety, learning, and AI operations without creating an autonomous decision system or an alternate source of strategic truth.

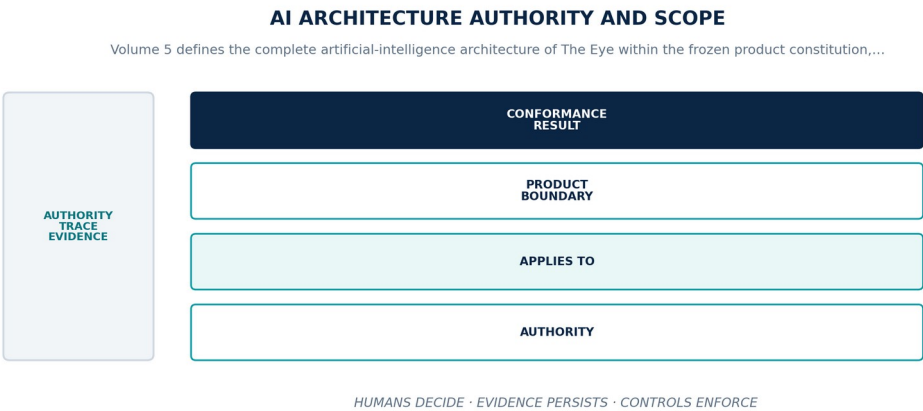


Figure 2. AI Architecture Authority and Scope. The diagram organizes authority, applies to, product boundary, conformance result as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Authority	Volume 0 controls constitutional meaning; Volume 3 controls product architecture; Volume 4 controls engineering realization; Volume 5 controls AI architecture.
Applies to	All model families, agent packages, prompts, instructions, retrieval, context assembly, tools, evaluators, learned artifacts, AI products, and AI operating controls.
Product boundary	AI capabilities operate inside the ten canonical layers and shared control services; they are not an eleventh layer, a separate copilot product, or a decision authority.
Conformance result	Conformant, conditionally conformant under a recorded exception, or non-conformant for an exact artifact, task, consequence class, customer domain, and deployment profile.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-01-001	Every AI capability SHALL identify its owning canonical layer, V3 logical component, V4 deployable owner, task contract, consequence class, product schema, control profile, evaluation profile, and accountable human owner.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-01-002	No model, agent, evaluator, retrieval service, or learned artifact SHALL establish independent strategic truth, final decision authority, policy authority, or canonical state ownership.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-01-003	AI output SHALL enter shared product state only through the owning component's admission contract with explicit truth state, uncertainty, provenance, policy, quality, and lifecycle.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-01-004	An implementation SHALL NOT represent itself as The Eye when it bypasses human final authority, temporal truth, end-to-end provenance, tenant isolation, deployment semantic parity, or governed learning.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-01-005	Production release SHALL be blocked when the applicable AI architecture, engineering, evaluation, safety, data, deployment, or operational evidence is absent, stale, or bound to another artifact version.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unmapped AI capability	Deny production registration until layer, component, task, owner, controls, and evidence are recorded.
Conflict with higher authority	Stop promotion and resolve through constitutional or architecture change control; local AI governance cannot waive the conflict.
Undeclared AI behavior	Disable or quarantine the behavior, identify affected products and decisions, and initiate conformance review.

Assurance evidence

- AI inventory resolves every production artifact to canonical layer, component, object, interface, owner, and release manifest.
- End-to-end fixtures prove model and agent outputs cannot bypass canonical admission or human authority.
- Architecture review finds no AI-owned parallel truth, policy, decision, or cross-deployment semantic fork.
- Production registry rejects unsigned, unmapped, unfit, expired, or revoked AI artifacts.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Architecture Authority and Scope is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 1, Ch. 2, Ch. 4, Ch. 6, Ch. 14, Ch. 43, Ch. 45, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-008 · C-028 · C-030 · C-052

2 AI Specification Hierarchy and Conformance

AI behavior is controlled through a versioned authority chain. Task specifications, evaluation profiles, prompts, model policies, agent manifests, and learned artifacts add precision beneath Volume 5 but may not silently weaken its semantics or the authorities above it.

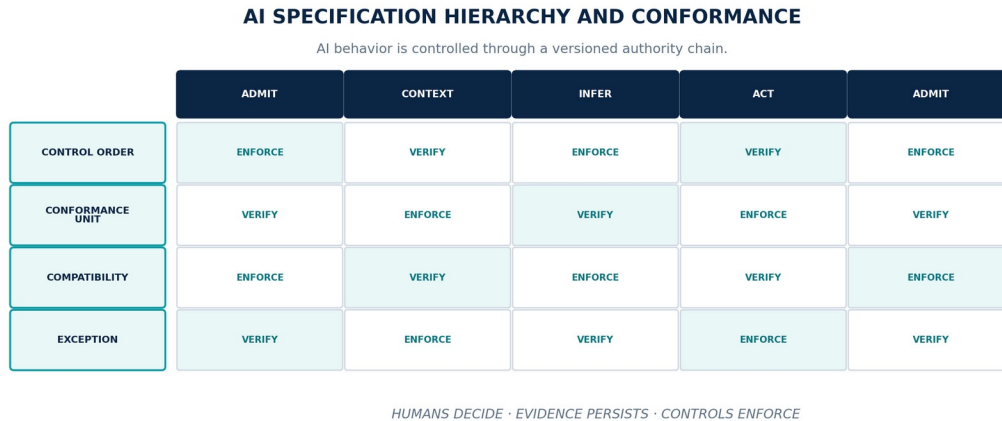


Figure 3. AI Specification Hierarchy and Conformance. The diagram organizes control order, conformance unit, compatibility, exception as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Control order	Constitution -> Technical Architecture -> Engineering Specification -> AI Architecture -> AI domain and task specifications -> artifacts -> evidence.
Conformance unit	One exact combination of artifact, task, data class, consequence, policy, customer domain, environment, and deployment profile.
Compatibility	Wire compatibility is insufficient; semantic, policy, temporal, uncertainty, explanation, and side-effect compatibility are required.
Exception	Narrow, owned, risk-accepted, time-bounded, monitored, reversible, and prohibited from changing human or constitutional authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-02-001	Every AI artifact SHALL declare controlling specifications, owner, status, version, effective time, task scope, consequence scope, evaluation expiry, dependencies, and supersession.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-02-002	Conformance SHALL be re-evaluated when model, agent, prompt, context source, retrieval method, tool, policy, schema, evaluation population, deployment, or external dependency changes materially.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-02-003	Derived AI specifications SHALL reference parent requirements and SHALL NOT replace explicit uncertainty, provenance, abstention, human gate, or deployment-parity obligations with aggregate quality claims.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-02-004	Conditional conformance SHALL identify the exact affected tenants, tasks, products, decisions, controls, monitoring, expiry, and closure criteria.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-02-005	Expired, revoked, or indeterminate conformance SHALL stop new resolution of the affected artifact and initiate impact analysis for active runs and decision-active products.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Conflicting AI artifacts	The higher controlling version prevails; quarantine the lower artifact and its dependent release.
Evidence cannot be reproduced	Return indeterminate conformance and prohibit promotion or scope expansion.
Exception scope expands	Mark non-conformant, stop affected use, and require explicit amendment or closure.

Assurance evidence

- Machine-readable manifests prove closed references across specifications, artifacts, tasks, policies, evaluations, and releases.
- Compatibility suites cover meaning, uncertainty, policy, provenance, side effects, human gates, and degraded behavior.
- Exception registry proves independent approval, compensating controls, telemetry, expiry, and non-expansion.
- Historical resolution reconstructs the exact conformance state effective for any AI product or decision replay.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Specification Hierarchy and Conformance is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 3, Ch. 6, Ch. 23, Ch. 45, Ch. 46, Ch. 56, Ch. 66, Ch. 68, Ch. 72.

3 Human Decision Sovereignty

AI may recommend, forecast, simulate, reason, explain, prioritize, draft, and prepare. A named human retains final authority for consequential institutional decisions, material commitments, irreversible effects, and changes to the authority system itself.

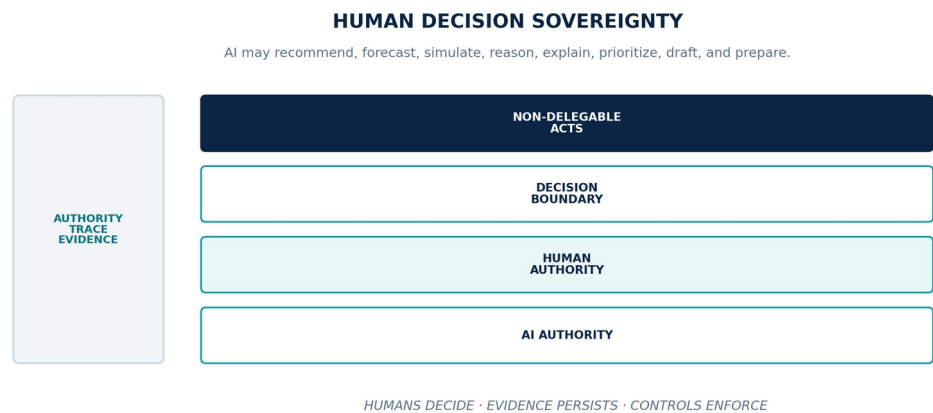


Figure 4. Human Decision Sovereignty. The diagram organizes ai authority, human authority, decision boundary, non-delegable acts as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
AI authority	Bounded analytical and operational preparation under purpose, task, data, tool, model, budget, policy, supervision, and stop conditions.
Human authority	Authenticated role with current mandate to decide, approve, reject, abstain, delegate, challenge, reopen, or withdraw.
Decision boundary	The exact point at which a recommendation becomes an institutional decision or an external side effect becomes authorized.
Non-delegable acts	Final high-impact decision, approval of irreversible effect, change to policy authority, and acceptance of unresolved material risk where policy requires a human.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-03-001	Every consequential AI-assisted decision SHALL terminate at a durable human gate that presents the exact proposed act, alternatives, evidence, uncertainty, dissent, policy, expected effects, reversibility, and missing information.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-03-002	Model confidence, agent consensus, elapsed time, workflow timeout, system administration, service identity, or prior approval SHALL NOT be interpreted as current human consent.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-03-003	AI MAY propose an action or decision but SHALL NOT broaden its scope, alter the approved effect, select the approving human, or execute an approval-required effect after the gate context changes.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-03-004	The human decision record SHALL bind the approver, authority, information boundary, exact choice, rationale, reservations, overrides, obligations, time, and downstream commitment state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-03-005	Emergency operation SHALL preserve named human command, bounded authority, independent audit, later reconciliation, and explicit prohibition against silent conversion into normal autonomous behavior.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Authorized human unavailable	Escalate or reassign under policy; elapsed time produces expiry or suspension, never approval.
Decision package materially changes	Invalidate the pending act and require refreshed human review of the changed boundary.
AI attempts prohibited effect	Deny at the tool and workflow enforcement points, terminate or suspend the execution, and open safety review.

Assurance evidence

- Human-gate suites cover role, delegation, quorum, separation, stale context, revocation, expiry, emergency, and replay.
- Adversarial tests prove models, agents, administrators, and service accounts cannot synthesize or reuse human approval.
- Decision Replay reconstructs exactly what the human saw, what AI produced, and which authority applied.
- Side-effect receipts prove the dispatched action matches the approved request digest and permitted execution window.

CHAPTER ACCEPTANCE AND INHERITANCE

Human Decision Sovereignty is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 39, Ch. 42, Ch. 43, Ch. 44, Ch. 48, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-026 · C-028 · C-029

4 Explainability by Construction

Explanation is produced from the same evidence, models, assumptions, tools, policies, and transformations that produced the recommendation. It is not a persuasive narrative generated after the decision boundary and it never conceals uncertainty or disagreement.

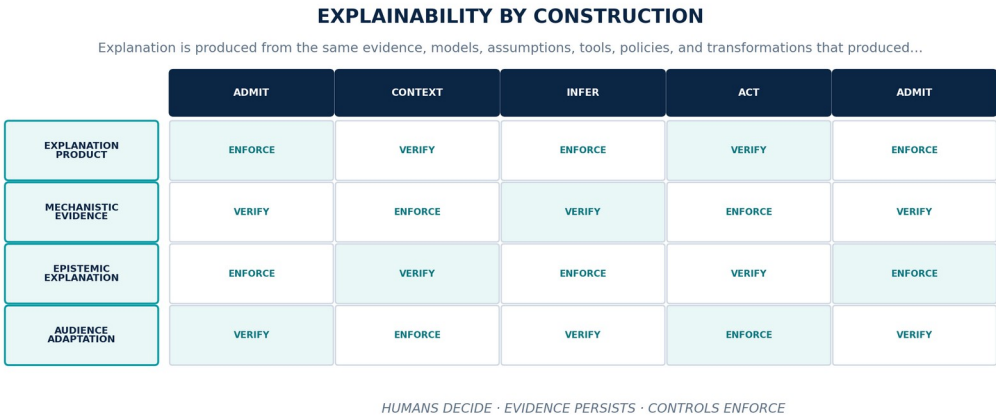


Figure 5. Explainability by Construction. The diagram organizes explanation product, mechanistic evidence, epistemic explanation, audience adaptation as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Explanation product	Versioned, policy-filtered object linking conclusion, evidence, method, assumptions, alternatives, uncertainty, limitations, dissent, and responsible owners.
Mechanistic evidence	Execution trace, retrieval manifest, model and agent versions, tool results, transformation lineage, and governing policy decisions.
Epistemic explanation	What is known, claimed, inferred, assessed, simulated, uncertain, contradicted, missing, or out of distribution.
Audience adaptation	Role-appropriate language and detail that preserves the same meaning, caveats, evidence, and decision-use constraints.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-04-001	Every recommendation, forecast, scenario, simulation interpretation, warning, score, and decision package SHALL carry an explanation product derived from preserved execution and provenance state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-04-002	Explanations SHALL distinguish source evidence, deterministic transformation, model inference, agent judgment, human assessment, assumptions, synthetic state, and later correction.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-04-003	An explanation SHALL expose material counter-evidence, competing hypotheses, model or agent disagreement, sensitivity, failure conditions, abstention, and unknowns.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-04-004	Natural-language rendering MAY summarize an explanation but SHALL NOT introduce unsupported reasons, remove material limitations, or imply access to hidden model cognition.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-04-005	Explanation completeness and faithfulness SHALL be evaluated for each task and consequence class and SHALL constrain decision-active use when below threshold.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Required lineage missing	Return explanation incomplete, mark the product unfit for consequential use, and repair or withdraw it.
Explanation conflicts with trace	Treat as a faithfulness failure, preserve both artifacts, and initiate product impact review.
Audience lacks access to evidence	Provide a policy-safe explanation with explicit withheld scope; do not invent substitute justification.

Assurance evidence

- Golden traces compare explanation claims with exact inputs, retrieval, tools, models, policies, and output transformations.
- Counterfactual tests change material evidence and verify the explanation and recommendation respond coherently.
- Human review measures comprehensibility, contestability, uncertainty communication, and decision usefulness by role.
- Security tests prove explanations do not disclose unauthorized content, hidden prompts, secrets, or cross-domain information.

CHAPTER ACCEPTANCE AND INHERITANCE

Explainability by Construction is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 21, Ch. 24, Ch. 27, Ch. 28, Ch. 39, Ch. 46, Ch. 55, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-012 · C-021 · C-024 · C-035 · C-047

5 Model, Method, and Agent Pluralism

The Eye is model- and vendor-plural while preserving one strategic product contract. Models, rules, statistical methods, graph algorithms, optimizers, simulators, retrieval systems, and agents are replaceable governed dependencies selected by task fitness and deployment policy.

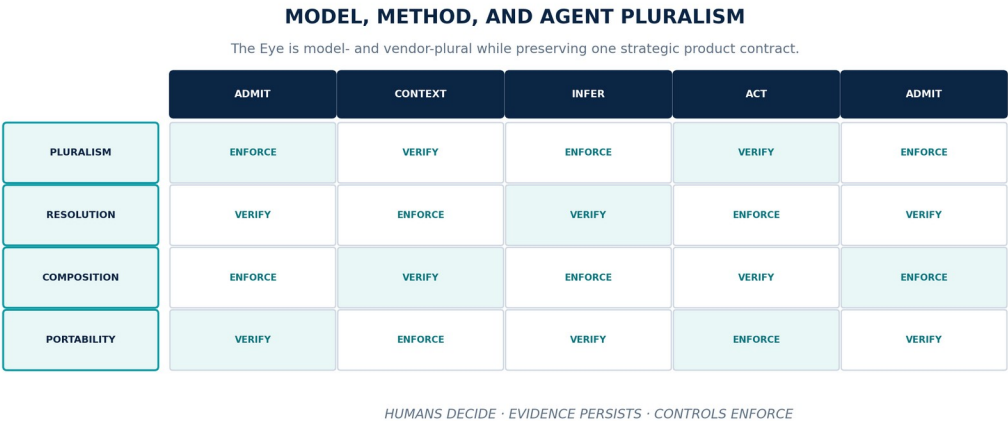


Figure 6. Model, Method, and Agent Pluralism. The diagram organizes pluralism, resolution, composition, portability as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Pluralism	More than one approved computational method may satisfy a task contract without changing canonical product meaning.
Resolution	Policy-bound selection using task, consequence, data, deployment, sovereignty, quality, latency, cost, capacity, and availability constraints.
Composition	Declared ensemble, cascade, debate, retrieval, tool, or human-review pattern with preserved component results and aggregation method.
Portability	Canonical contracts remain independent of provider-specific prompts, tokens, embeddings, tool formats, safety labels, and operational APIs.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-05-001	AI architecture SHALL separate task contracts and canonical products from provider-specific model, agent, retrieval, embedding, and tool interfaces.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-05-002	Resolution SHALL select only artifacts with current task-specific fitness, data rights, sovereignty, safety, compatibility, and operational approval for the requested scope.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-05-003	Ensembles and agent groups SHALL preserve individual outputs, confidence, dissent, failure, and aggregation lineage rather than exposing consensus alone.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-05-004	Fallback SHALL satisfy the same policy and declared minimum product quality or return explicit abstention, degradation, or unavailability; silent semantic downgrade is prohibited.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-05-005	Provider or artifact replacement SHALL pass canonical conformance, historical replay, safety, performance, cost, correction, and deployment-parity tests before promotion.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
No conforming artifact	Return unavailable or abstained with the unsatisfied constraints; do not use an unapproved substitute.
Methods materially disagree	Preserve alternatives and uncertainty and route the product through declared challenge or human review.
Provider semantics leak	Quarantine the adapter path and restore the canonical contract before downstream admission.

Assurance evidence

- Cross-provider fixtures produce semantically equivalent canonical envelopes and product states.
- Resolution tests cover quality, sovereignty, rights, capacity, cost, latency, outage, withdrawal, and fallback constraints.
- Ensemble tests preserve component evidence, disagreement, aggregation, and calibration by population slice.
- Replacement rehearsals reproduce historical tasks and decision packages against frozen information boundaries.

CHAPTER ACCEPTANCE AND INHERITANCE

Model, Method, and Agent Pluralism is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 20, Ch. 23, Ch. 27, Ch. 42, Ch. 45, Ch. 46, Ch. 56, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-005 · C-021 · C-028 · C-030 · C-035 · C-041 · C-042

6 AI Traceability and Documentation Inheritance

Every AI product must be traversable from constitutional authority to runtime execution and back. Traceability binds requirements, canonical layers, components, tasks, schemas, models, agents, prompts, context, tools, policies, evaluations, releases, decisions, outcomes, and later learning.

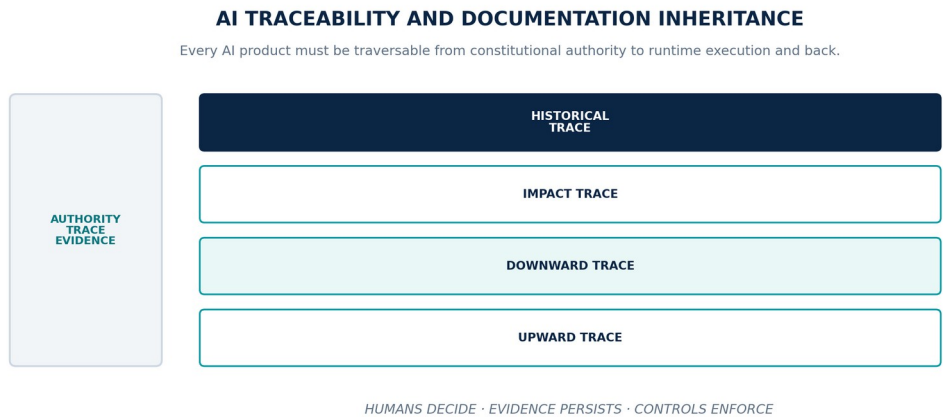


Figure 7. AI Traceability and Documentation Inheritance. The diagram organizes upward trace, downward trace, impact trace, historical trace as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Upward trace	AI artifact and behavior -> V5 requirement -> V4 engineering contract -> V3 architecture -> V0 invariant.
Downward trace	Authority -> task and product contract -> artifacts and configuration -> tests and release -> runtime evidence and outcomes.
Impact trace	Changed source, model, agent, prompt, policy, tool, context, schema, evaluation, or package -> affected products, decisions, and deployments.
Historical trace	Exact versions and effective authority resolved for any execution, product, human decision, incident, correction, or replay time.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-06-001	Every normative AI requirement SHALL have a stable identity, owning authority, verification method, required evidence, implementation mapping, and lifecycle status.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-06-002	Every production AI execution SHALL preserve references to task, product schema, model, agent, prompt, context, retrieval, tool, policy, evaluation, configuration, release, and deployment versions.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-06-003	Traceability SHALL support backward explanation and forward impact analysis across synchronous, event, workflow, agent, model, tool, and human boundaries.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-06-004	A change to any decision-relevant AI dependency SHALL identify and classify affected products as current, stale, challenged, withdrawn, recomputation-required, or human-review-required.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-06-005	Documentation, manifests, registries, schemas, and test evidence SHALL be released as controlled artifacts and SHALL remain historically resolvable after supersession.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Trace link unresolved	Mark dependent conformance and explanation incomplete and prohibit unsupported closure.
Impact analysis partial	Keep the change workflow open and expose unresolved consumers, deployments, and decisions.
Historical artifact unavailable	Return replay incomplete with the exact missing dependency and accountable recovery state.

Assurance evidence

- Bidirectional graph queries traverse invariant-to-evidence and product-to-authority paths without unresolved mandatory edges.
- Seeded dependency changes identify all affected models, agents, products, warnings, decisions, deployments, and documents.
- Release manifests reproduce exact AI execution environments and conformance populations.
- Decision Replay resolves the complete AI and documentation context without hindsight contamination.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Traceability and Documentation Inheritance is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 3, Ch. 6, Ch. 20, Ch. 23, Ch. 28, Ch. 30, Ch. 46, Ch. 48, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-005 · C-011 · C-012 · C-025 · C-045 · C-051 · C-052

PART II

AI System Architecture

AI system context, planes, products, control boundaries, and layer inheritance.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

7 AI System Context

The AI subsystem is an embedded reasoning and execution fabric serving the ten canonical layers. It receives governed tasks and context, invokes approved models and agents, returns typed intelligence products, and remains bounded by canonical state owners, human authority, and cross-cutting controls.

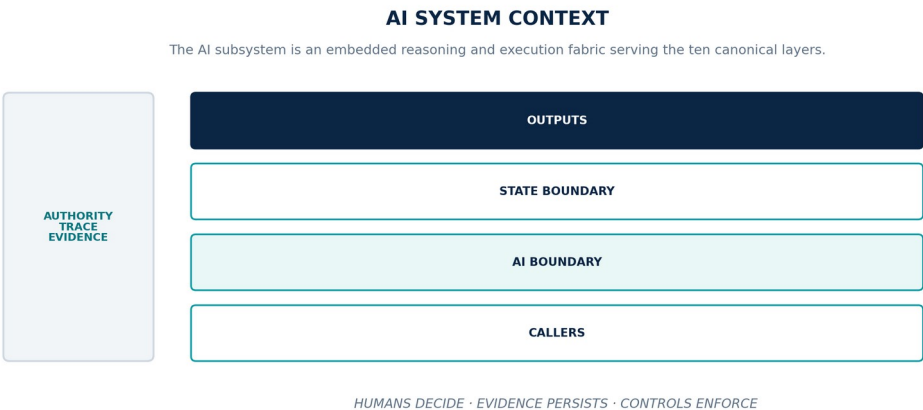


Figure 8. AI System Context. The diagram organizes callers, ai boundary, state boundary, outputs as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Callers	Canonical layer components, durable workflows, governed human workspaces, evaluation systems, and approved external integrations.
AI boundary	Model gateway, context services, agent runtime, tool broker, evaluation plane, safety controls, and artifact registries.
State boundary	AI registries and execution evidence are authoritative for AI operation; strategic truth remains with canonical layer owners.
Outputs	Typed proposals, transformations, assessments, forecasts, scenarios, simulations, recommendations, explanations, warnings, and evaluation evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-07-001	AI services SHALL accept work only through authenticated, authorized, versioned task contracts carrying purpose, tenant, domain, consequence, deadline, data, side-effect, and human-gate context.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-07-002	AI services SHALL consume canonical objects by immutable reference or policy-filtered projection and SHALL NOT write directly into another component's authoritative store.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-07-003	AI results SHALL be proposals or typed products whose owning canonical component validates and commits decision-active state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-07-004	External model, tool, search, and data providers SHALL remain outside the trusted customer domain until gateway, policy, provenance, minimization, integrity, and sovereignty controls complete.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-07-005	AI system health SHALL distinguish model, agent, context, tool, evaluation, control, product quality, and decision-use state rather than exposing a single availability indicator.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Canonical owner unavailable	Queue or return explicit uncommitted result; AI runtime cannot assume state authority.
Control plane unavailable	Stop admission or apply only declared restrictive continuity for bounded read-only tasks.
External provider returns untrusted content	Quarantine as observation and prohibit product admission until validation completes.

Assurance evidence

- System-context tests enumerate every caller, trust boundary, state owner, external dependency, output, and human authority transition.
- Data-flow tests prove tenant, purpose, classification, rights, and provenance survive every AI boundary.
- Negative tests detect direct store writes, bypassed gateways, ambient credentials, and unregistered provider calls.
- Health fixtures expose independent dependency and product-quality states under partial failure.

CHAPTER ACCEPTANCE AND INHERITANCE

AI System Context is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 4, Ch. 7, Ch. 8, Ch. 10, Ch. 11, Ch. 13, Ch. 15, Ch. 43, Ch. 45.

CONSTITUTIONAL INHERITANCE / C-001 · C-003 · C-004 · C-008 · C-012 · C-035 · C-039

8 AI Plane Decomposition

AI architecture is decomposed into interoperating operational planes beneath the canonical product layers. The decomposition creates clear ownership and control without introducing alternate product layers or shared mutable authority.

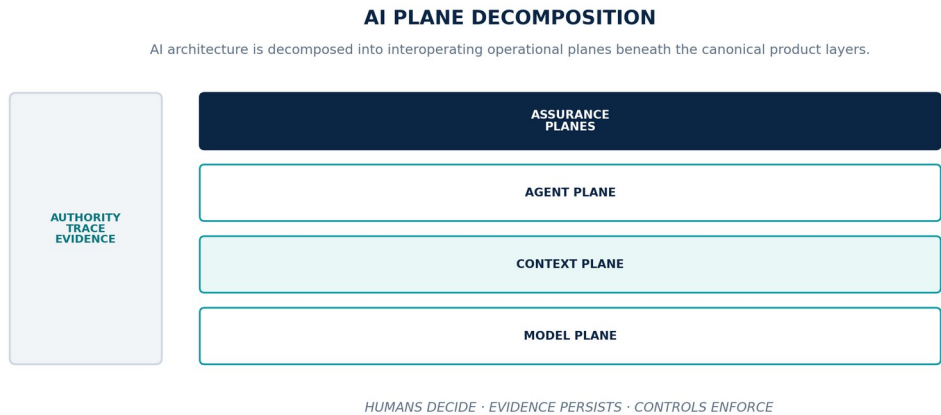


Figure 9. AI Plane Decomposition. The diagram organizes model plane, context plane, agent plane, assurance planes as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Model plane	Model registry, adapters, routing, inference, embeddings, optimization, and model lifecycle.
Context plane	Retrieval, memory access, graph and temporal grounding, twin context, policy filtering, and citation binding.
Agent plane	Identity, planning, supervision, workflow, communication, tools, sandboxes, checkpoints, and human gates.
Assurance planes	Evaluation, safety, policy, provenance, audit, observability, learning, release, deployment, and sovereignty.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-08-001	Each AI plane SHALL have one accountable platform owner, explicit state classes, versioned interfaces, deployment boundary, SLOs, failure domains, and conformance suites.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-08-002	Plane boundaries SHALL use canonical envelopes and SHALL prohibit shared-write databases, ambient context, implicit tenant state, and provider-specific semantics outside adapters.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-08-003	Task execution SHALL compose planes through a durable execution manifest that identifies every resolved dependency and control decision.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-08-004	Control and assurance planes SHALL remain independently enforceable and observable from model, agent, context, and tool implementations.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-08-005	Plane consolidation into a deployable unit MAY occur only when state authority, telemetry, policy, resource, failure, and audit boundaries remain independently testable.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Ownership overlap	Assign a single state owner and replace shared writes with declared contracts or rebuildable projections.

FAILURE CONDITION	REQUIRED BEHAVIOR
Plane dependency cycle	Reject the design until authoritative direction and failure recovery are deterministic.
Assurance plane bypass	Stop affected processing, identify admitted products, and execute safety and conformance review.

Assurance evidence

- Architecture dependency graph contains no unauthorized write cycle or hidden control dependency.
- Service manifests map every deployable unit and datastore to one or more explicit plane responsibilities and owners.
- Failure injection isolates each plane and verifies truthful downstream degradation.
- Deployment tests reproduce the same plane contracts and authority across SaaS, Private Cloud, and On-Premise.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Plane Decomposition is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 4, Ch. 10, Ch. 11, Ch. 12, Ch. 13, Ch. 14, Ch. 43, Ch. 45, Ch. 46.

CONSTITUTIONAL INHERITANCE / C-003 · C-008 · C-028 · C-030 · C-035 · C-042 · C-043

9 AI Product and Value Flow

AI work begins with a governed task and ends with a canonical product decision, not with raw model output. The flow preserves intent, evidence, context, methods, uncertainty, explanation, review, admission, use, correction, and outcome through one causally connected execution record.

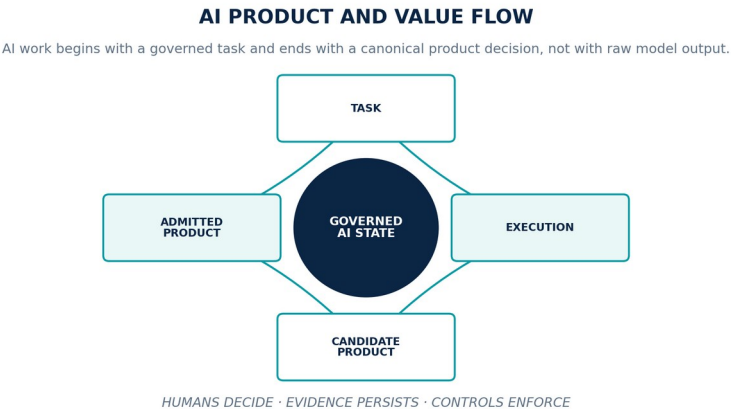


Figure 10. AI Product and Value Flow. The diagram organizes task, execution, candidate product, admitted product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Task	Declared analytical objective, caller, purpose, consequence, inputs, context policy, output schema, quality, deadline, and side-effect class.
Execution	Resolved plan, artifacts, context, tools, controls, intermediate products, evaluations, challenges, and terminal state.
Candidate product	Schema-valid proposed intelligence object with truth state, uncertainty, provenance, policy, quality, and explanation.
Admitted product	Owning component's committed version with lifecycle, consumer constraints, monitoring, correction, and replay references.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-09-001	AI task admission SHALL bind the exact requested product schema, consequence class, quality threshold, explanation profile, human review, and permitted side effects before execution.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-09-002	Intermediate model and agent outputs SHALL remain execution evidence and SHALL NOT become shared memory or graph truth without owning-component admission.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-09-003	Candidate products SHALL distinguish observed, claimed, inferred, assessed, synthetic, simulated, recommended, decided, withdrawn, and superseded state as applicable.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-09-004	Product admission SHALL evaluate schema, evidence, temporal scope, provenance, uncertainty, policy, quality, duplication, contradiction, correction, and accountable ownership.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-09-005	Use, challenge, override, correction, decision, commitment, and outcome events SHALL link back to the admitted product and initiate evaluation or governed learning where applicable.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Task cannot satisfy output contract	Return abstained, partial, or failed state with unsatisfied conditions; do not fabricate a complete product.
Candidate validation fails	Quarantine execution output and preserve raw evidence for correction or investigation.
Admitted product later invalidated	Propagate stale, challenged, withdrawn, or recomputation-required state to dependent consumers and decisions.

Assurance evidence

- End-to-end golden cases trace tasks through execution, candidate validation, admission, use, correction, and outcome.
- Negative tests prove raw model and agent output cannot enter canonical stores through caches, logs, vectors, or user surfaces.
- Correction fixtures identify every dependent context, product, warning, decision, briefing, and learning proposal.
- Replay reconstructs the exact product versions, information boundary, AI execution, and human review effective at use time.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Product and Value Flow is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 21, Ch. 24, Ch. 27, Ch. 28, Ch. 30, Ch. 43, Ch. 48.

CONSTITUTIONAL INHERITANCE / C-002 · C-005 · C-010 · C-011 · C-012 · C-025 · C-044

10 Model Control Plane

The model control plane governs model identity, provenance, task fitness, policy, compatibility, resolution, release, monitoring, withdrawal, and historical use across first-party, customer-provided, open, embedded, and external models.

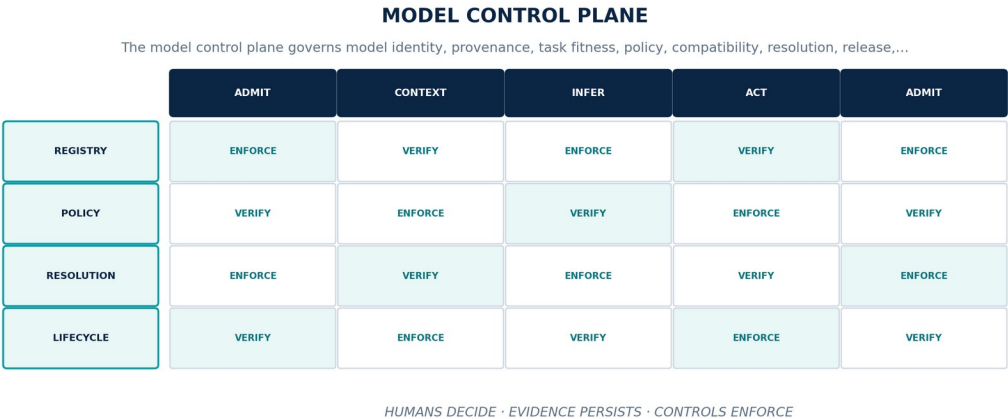


Figure 11. Model Control Plane. The diagram organizes registry, policy, resolution, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Registry	Authoritative model profiles, versions, composition, provenance, rights, evaluations, task approvals, constraints, and lifecycle.
Policy	Permitted tasks, data classes, domains, deployments, regions, providers, context, retention, logging, fallback, and consequence.
Resolution	Deterministic selection of an approved model or ensemble for an exact invocation contract.
Lifecycle	Proposed, quarantined, evaluated, approved, shadow, constrained, active, suspended, withdrawn, deprecated, or retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-10-001	Every model artifact SHALL have stable identity, content digest, version, provenance, composition, owner, licenses, data terms, architecture, task profiles, deployment support, evaluations, and lifecycle state.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-10-002	The control plane SHALL resolve models only from signed current registry state and SHALL record the policy inputs, candidate set, selected artifact, fallback chain, and decision reason.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-10-003	Task approval SHALL be consequence- and population-specific and SHALL expire or be revoked independently of the model's general registry presence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-10-004	Model withdrawal SHALL stop new resolution, identify active executions and decision-active products, and initiate recomputation, review, or disclosure according to consequence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-10-005	Historical registry state SHALL remain resolvable for audit, incident analysis, correction impact, reproducibility, and Decision Replay.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Registry unavailable	Use only short-lived verified cache within declared continuity or return model resolution unavailable.
Model signature or provenance invalid	Quarantine the artifact and all unverified derived packages; prohibit inference.
Task fitness expires	Stop new task resolution and mark affected active runs for bounded completion, suspension, or review.

Assurance evidence

- Registry tests validate identity, digest, provenance, composition, ownership, task approval, expiry, revocation, and historical resolution.
- Resolution fixtures cover data, deployment, region, task, consequence, capacity, quality, cost, safety, and fallback constraints.
- Withdrawal drills trace active runs and products through recomputation, review, notification, and closure.
- Cross-deployment tests prove registry meaning and lifecycle remain identical while custody and physical placement vary.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Control Plane is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 10, Ch. 14, Ch. 23, Ch. 45, Ch. 46, Ch. 52, Ch. 56, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-012 · C-028 · C-030 · C-031 · C-035 · C-041 · C-045

11 Agent Control Plane

The agent control plane governs agent identity, package provenance, role, task scope, capability grants, supervision, evaluation, release, monitoring, suspension, and revocation. It never grants an agent authority to redefine its own purpose or controls.



Figure 12. Agent Control Plane. The diagram organizes agent registry, capability policy, supervision policy, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Agent registry	Signed package, role, owner, version, instructions, models, tools, data scopes, budgets, evaluations, deployment support, and lifecycle.
Capability policy	Task-specific permitted data, tools, models, network, compute, storage, side effects, duration, and human gates.
Supervision policy	Planner, supervisor, reviewer, escalation, challenge, checkpoint, stop, and terminal-state obligations.
Lifecycle	Draft, quarantined, evaluated, approved, shadow, restricted, active, suspended, revoked, deprecated, or retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-11-001	Every agent SHALL execute under a signed registered package and a run-specific capability grant that is narrower than or equal to its approved role profile.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-11-002	Agent packages SHALL declare instructions, models, tools, context sources, data classes, network destinations, side-effect classes, budgets, dependencies, evaluations, owners, and failure behavior.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-11-003	No agent SHALL create, approve, extend, or conceal its own capability grant, task purpose, human authority, evaluation status, or lifecycle state.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-11-004	The control plane SHALL support immediate suspension and revocation and SHALL define treatment of active executions, credentials, checkpoints, products, and derived decisions.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-11-005	Agent registry and execution history SHALL remain historically resolvable for explanation, incident analysis, impact assessment, audit, and Decision Replay.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Agent package unverified	Quarantine and prohibit admission, execution, and marketplace publication.
Capability service unavailable	Stop new executions and side effects; continue only bounded safe work under verified cached grants where policy allows.
Agent revoked during execution	Suspend new work and effects, checkpoint evidence, revoke credentials, and route active products for review.

Assurance evidence

- Registry tests cover package identity, dependencies, instructions, capabilities, task approval, evaluation, expiry, and revocation.
- Capability tests attempt data, tool, model, network, tenant, budget, and side-effect escalation.
- Revocation drills measure closure across runtimes, credentials, tools, queues, checkpoints, and active human gates.
- Historical replay resolves exact agent role, package, grant, supervisor, execution, and control versions.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Control Plane is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 10, Ch. 23, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 46, Ch. 55, Ch. 56.

CONSTITUTIONAL INHERITANCE / C-004 · C-028 · C-029 · C-031 · C-035 · C-045 · C-046

12 Context Control Plane

The context control plane assembles the exact policy-filtered information boundary presented to a model or agent. It controls source authority, temporal scope, truth state, retrieval method, ranking, coverage, token or feature budgets, citations, and context retention.

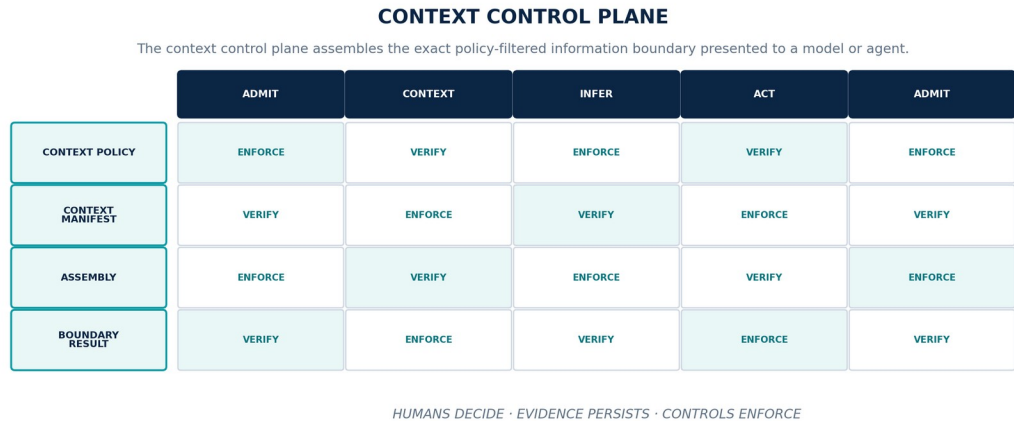


Figure 13. Context Control Plane. The diagram organizes context policy, context manifest, assembly, boundary result as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Context policy	Allowed sources, object types, truth states, time, graph scope, twin state, data class, purpose, model, agent, retention, and export.
Context manifest	Immutable ordered inventory of every object, excerpt, graph path, feature, tool result, transformation, omission, and version supplied.
Assembly	Policy filter, retrieval, ranking, diversification, contradiction preservation, temporal resolution, minimization, formatting, and budget allocation.
Boundary result	Complete, partial, stale, coverage-limited, policy-limited, unavailable, or challenged with explicit quality and use constraints.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-12-001	Every model and agent invocation SHALL bind an immutable context manifest sufficient to reconstruct the exact information boundary.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-12-002	Context assembly SHALL enforce tenant, purpose, classification, rights, residency, retention, truth state, temporal, and model or agent use policy before content enters the invocation.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-12-003	Ranking and compression SHALL preserve material contradictions, dissent, uncertainty, source diversity, recency, strategic relevance, and known coverage gaps.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-12-004	Context SHALL distinguish canonical objects from untrusted observations, retrieved text, synthetic examples, tool output, prior model output, and user-provided instructions.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-12-005	Context caches and embeddings SHALL remain policy-aware, tenant-isolated, revocation-aware, versioned, and reconstructable from declared source state.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Required context unavailable	Return partial or abstained state with missing scope; do not imply complete evidence coverage.
Policy changes during execution	Invalidate or constrain uncommitted context and require re-evaluation before consequential continuation.
Context integrity fails	Quarantine the invocation, preserve evidence, and prohibit product admission.

Assurance evidence

- Context replay rebuilds exact manifests from object, graph, twin, source, policy, ranking, and formatter versions.
- Policy tests attempt prohibited retrieval through search, graph paths, embeddings, caches, summaries, and prior run memory.
- Coverage tests measure recall, source diversity, contradiction preservation, temporal validity, and known omissions.
- Adversarial tests distinguish data from instructions and prevent retrieved or tool-supplied instruction escalation.

CHAPTER ACCEPTANCE AND INHERITANCE

Context Control Plane is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 33, Ch. 34, Ch. 45.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-010 · C-011 · C-012 · C-015 · C-040

13 Knowledge, Memory, and Twin Grounding Fabric

AI reasoning is grounded in governed enterprise memory, temporal Knowledge Graph identity, Strategy Graph context, and versioned Digital Twins. Grounding supplies connected state while preserving the distinct authority, truth, and lifecycle of each substrate.

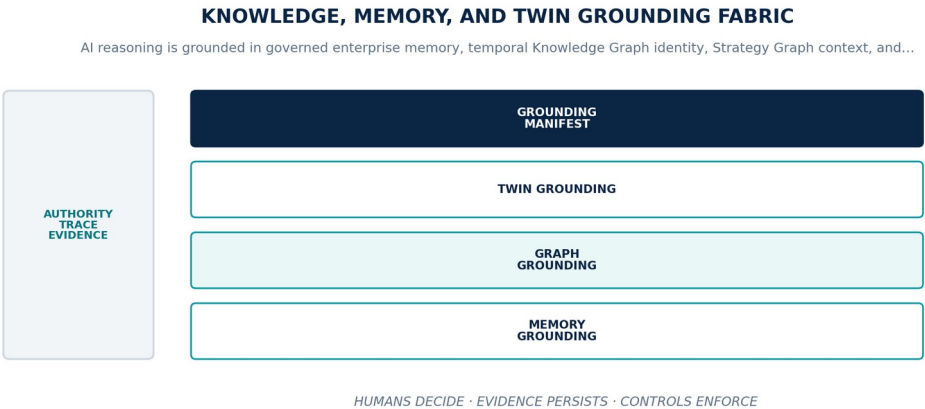


Figure 14. Knowledge, Memory, and Twin Grounding Fabric. The diagram organizes memory grounding, graph grounding, twin grounding, grounding manifest as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Memory grounding	Policy-filtered historical objects, assessments, decisions, commitments, outcomes, and lessons with retention and correction state.
Graph grounding	Canonical identities, typed temporal relationships, ontology, provenance paths, contradictions, and strategy connections.
Twin grounding	Observed, estimated, forecast, scenario, and simulated states with explicit branch, model, evidence, validity, and uncertainty.
Grounding manifest	Exact revisions, snapshots, query semantics, projections, policies, watermarks, and quality supplied to an execution.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-13-001	AI tasks requiring shared identity or relationships SHALL resolve them through governed Knowledge Graph contracts rather than private entity or relationship stores.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-13-002	AI retrieval from Enterprise Memory SHALL preserve object version, truth state, temporal boundary, retention, correction, policy, and source lineage.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-13-003	AI use of Digital Twins SHALL preserve observed, estimated, forecast, scenario, and simulated states as separate branches and SHALL NOT write synthetic results into observed state.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-13-004	Strategy Graph grounding SHALL connect AI products to objectives, assumptions, risks, opportunities, capabilities, commitments, measures, and outcomes without creating opaque strategic scores.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-13-005	Grounding artifacts SHALL expose revision watermarks, freshness, coverage, uncertainty, projection lag, and invalidation state to downstream products and explanations.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity unresolved	Preserve candidates and uncertainty; do not fabricate a canonical entity or merge.
Grounding revisions incompatible	Return conflict or bounded historical result and route reconciliation to the owning substrate.
Twin branch contamination	Quarantine affected products, restore branch separation, and execute forward impact analysis.

Assurance evidence

- Grounding fixtures reconstruct exact memory versions, graph revisions, twin snapshots, policies, and projection watermarks.
- Identity and relationship tests preserve candidates, evidence, confidence, stewardship, merges, splits, and corrections.
- Twin tests prove synthetic branches cannot alter observed state and retain reproducible model and assumption lineage.
- Strategy tests decompose every AI relevance or health claim to objectives, measures, evidence, uncertainty, and owners.

CHAPTER ACCEPTANCE AND INHERITANCE

Knowledge, Memory, and Twin Grounding Fabric is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 33, Ch. 34, Ch. 35.

CONSTITUTIONAL INHERITANCE / C-009 · C-011 · C-015 · C-017 · C-018 · C-019 · C-034

14 Evaluation, Governance, and Learning Planes

Evaluation, governance, and learning are independent control planes around AI production. Evaluation determines fitness, governance authorizes use and change, and learning proposes evidence-backed improvement; none of them silently modifies production behavior.

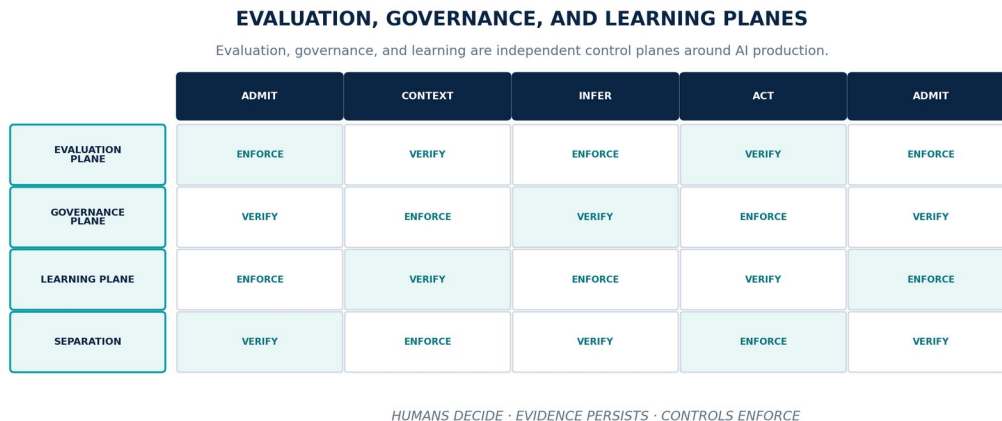


Figure 15. Evaluation, Governance, and Learning Planes. The diagram organizes evaluation plane, governance plane, learning plane, separation as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Evaluation plane	Task profiles, evidence sets, metrics, challenge methods, runners, judges, fitness decisions, runtime monitors, and expiry.
Governance plane	Authorities, approvals, exceptions, risk acceptance, policy, accountability, audit, marketplace, deprecation, and change control.
Learning plane	Outcomes, corrections, feedback, overrides, incidents, lesson proposals, curation, training, evaluation, release, and rollback.
Separation	Artifact creators, evaluators, risk acceptors, release approvers, and production operators remain independently attributable where consequence requires.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-14-001	Every production AI task SHALL have a current evaluation profile and fitness decision for the exact artifact composition, population, consequence, and deployment scope.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-14-002	Governance SHALL approve task use, risk, exception, marketplace publication, learned changes, and retirement through named authorities whose scope is independent of artifact execution.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-14-003	Runtime feedback SHALL enter governed evidence stores and MAY create lesson proposals but SHALL NOT directly alter models, agents, prompts, policies, tools, thresholds, or retrieval behavior.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-14-004	Learning releases SHALL pass the same architecture, evaluation, safety, privacy, deployment, release, monitoring, rollback, and acceptance controls as other production artifacts.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-14-005	Evaluation and governance evidence SHALL remain immutable, versioned, historically resolvable, tenant-appropriate, and protected against manipulation by the artifacts being evaluated.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Evaluation evidence insufficient	Return indeterminate fitness and restrict or deny production use according to consequence.
Governance authority unavailable	Hold the decision; no model, agent, evaluator, or operator may self-authorize.
Learning release degrades behavior	Stop rollout, rollback or constrain, preserve impact evidence, and initiate incident and re-evaluation.

Assurance evidence

- Control tests prove artifact developers cannot rewrite evidence, thresholds, fitness decisions, approval, or audit history.
- Outcome fixtures trace feedback through lesson proposal, evaluation, approval, release, monitoring, and rollback.
- Separation-of-duties matrices cover artifact, evaluation, risk, policy, release, operations, and exception authorities.
- Historical queries resolve exact evaluation, governance, and learning state for every AI execution and decision.

CHAPTER ACCEPTANCE AND INHERITANCE

Evaluation, Governance, and Learning Planes is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 13, Ch. 46, Ch. 48, Ch. 55, Ch. 56, Ch. 63, Ch. 68, Ch. 71, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-028 · C-031 · C-035 · C-044 · C-045 · C-051

PART III

Model and Inference Platform

Model identity, routing, adapters, instructions, structured output, security, and capacity.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

15 Model Identity, Registry, and Lifecycle

A model is a governed architecture artifact with stable identity and a task-specific lifecycle. Registry state connects model bytes or external versions to provenance, architecture, data terms, evaluations, policies, deployments, owners, incidents, derived products, and retirement obligations.

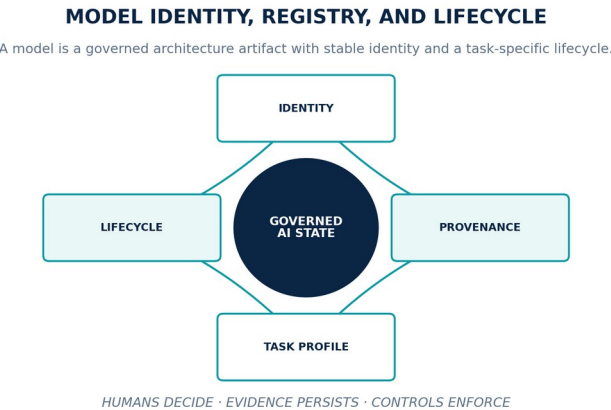


Figure 16. Model Identity, Registry, and Lifecycle. The diagram organizes identity, provenance, task profile, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Identity	Stable model ID plus immutable version digest, provider identifier, architecture class, tokenizer or feature contract, and packaging format.
Provenance	Publisher, source, training and adaptation lineage, build environment, composition, dependencies, licenses, attestations, and custody.
Task profile	Approved tasks, populations, consequence classes, inputs, outputs, limitations, evaluation, explanation, and human review.
Lifecycle	Submitted -> quarantined -> evaluated -> approved/shadow/constrained -> active -> suspended/withdrawn -> deprecated -> retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-15-001	Every model version SHALL have a unique immutable identity and SHALL NOT reuse a prior version identifier for changed weights, provider behavior, tokenizer, safety configuration, context handling, or serving implementation.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-15-002	Registry metadata SHALL record architecture, provenance, data and license terms, intended tasks, prohibited uses, evaluations, limitations, security, privacy, sovereignty, resource, compatibility, owner, and support lifecycle.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-15-003	External hosted models SHALL be pinned to a provider version or verified behavior profile and SHALL be treated as changed when the provider cannot guarantee stable semantics.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-15-004	Fine-tunes, adapters, quantizations, distillations, ensembles, cascades, and retrieval-augmented configurations SHALL have independent version identities linked to every parent artifact.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-15-005	Retirement SHALL preserve historical resolution, impact analysis, export or replacement obligations, Decision Replay, audit verification, and customer exit requirements.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Model bytes or endpoint do not match registry	Quarantine the serving unit, stop resolution, and investigate all affected executions.
Provenance incomplete	Restrict to a declared non-production evaluation state; do not infer safety or rights from publisher reputation.
Provider changes unversioned behavior	Treat as an emergency new version, suspend consequential use, and run drift and conformance evaluation.

Assurance evidence

- Artifact verification recomputes digests, signatures, attestations, composition, and provider-behavior fingerprints.
- Registry completeness tests reject missing owner, task, evaluation, rights, deployment, support, or retirement fields.
- Lineage queries traverse every adapted and composed model to parent artifacts, data authorities, evaluations, releases, and products.
- Retirement exercises prove resolution closure and historical replay across connected and disconnected deployments.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Identity, Registry, and Lifecycle is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 28, Ch. 45, Ch. 46, Ch. 52, Ch. 54, Ch. 56, Ch. 64, Ch. 68, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-012 · C-030 · C-035 · C-038 · C-041 · C-045 · C-048

16 Model Gateway, Routing, and Resolution

The Model Gateway is the exclusive inference boundary. It normalizes task intent, evaluates policy and fitness, resolves an approved execution path, enforces budgets and data controls, records complete invocation lineage, and returns typed status without exposing provider semantics to callers.

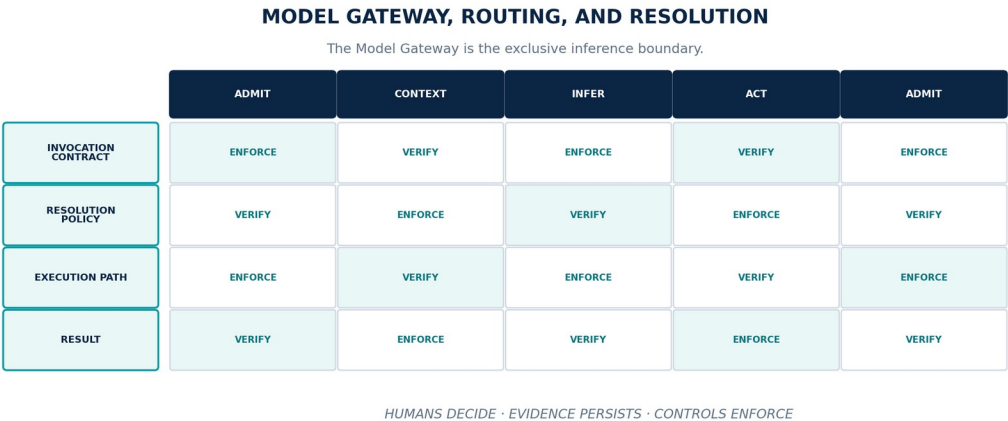


Figure 17. Model Gateway, Routing, and Resolution. The diagram organizes invocation contract, resolution policy, execution path, result as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Invocation contract	Task, product schema, caller, purpose, consequence, data class, context manifest, quality, latency, cost, deployment, and fallback.
Resolution policy	Candidate eligibility, priority, ensemble or cascade, capacity, sovereignty, safety, failure, and abstention rules.
Execution path	Selected model or composition, adapter, endpoint, region, parameters, safety profile, credentials, and resource reservation.
Result	Complete, partial, abstained, refused, filtered, timed out, unavailable, failed, or challenged with normalized usage and quality evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-16-001	All production inference SHALL pass through the Model Gateway, including local, embedded, external, batch, streaming, multimodal, evaluator, judge, and agent-internal calls.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-16-002	Resolution SHALL use signed current registry, task fitness, policy, data, consequence, deployment, sovereignty, capacity, latency, cost, safety, and fallback state.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-16-003	The gateway SHALL record candidate models, rejection reasons, selected path, exact parameters, context and instruction versions, provider receipt, usage, safety events, and terminal status.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-16-004	Retries and fallbacks SHALL preserve the invocation idempotency and information boundary and SHALL NOT cross policy, quality, sovereignty, or human-review constraints.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-16-005	Callers SHALL receive canonical result and error envelopes and SHALL NOT depend on provider-specific tokens, refusal codes, tool formats, hidden system prompts, or safety labels.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
No eligible model	Return no_conforming_model with unsatisfied constraints and safe retry guidance.
Provider outcome unknown	Return reconciliation-required state; do not duplicate a metered or side-effecting invocation blindly.
Gateway policy stale	Stop new consequential resolution and use only restrictive bounded continuity where expressly approved.

Assurance evidence

- Routing fixtures deterministically cover task, data, consequence, region, deployment, capacity, quality, cost, and outage conditions.
- Provider conformance suites normalize outputs, errors, refusals, streaming, usage, safety events, and cancellation.
- Fallback fault tests prove no crossing of classification, sovereignty, quality, or task-approval boundaries.
- Invocation replay resolves every dependency and receipt without relying on provider-retained customer content.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Gateway, Routing, and Resolution is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 21, Ch. 22, Ch. 23, Ch. 45, Ch. 49, Ch. 50, Ch. 53, Ch. 58, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-005 · C-028 · C-030 · C-035 · C-041 · C-042 · C-049

17 Model Adapters and Provider Abstraction

Adapters isolate provider protocols and model-specific behavior behind canonical task, context, tool, output, error, usage, and safety contracts. They translate representation but may not reinterpret product meaning or conceal provider limitations.

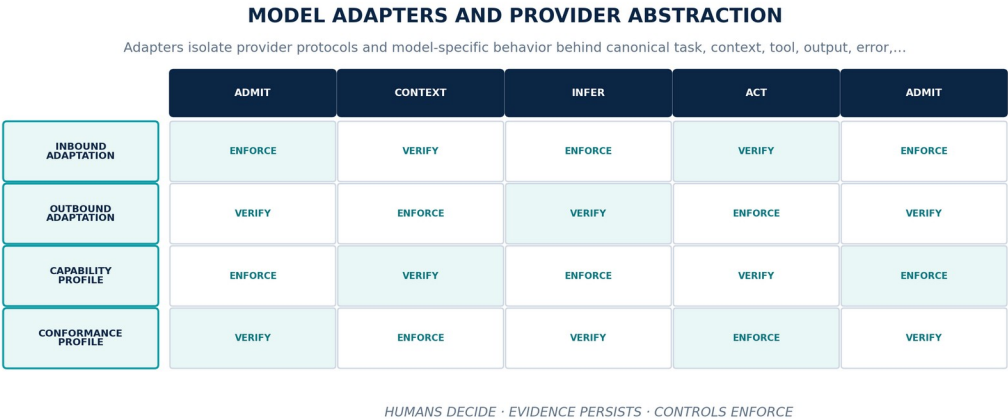


Figure 18. Model Adapters and Provider Abstraction. The diagram organizes inbound adaptation, outbound adaptation, capability profile, conformance profile as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Inbound adaptation	Canonical instruction, context, schema, tools, parameters, streaming, deadlines, and controls to provider request.
Outbound adaptation	Provider result, status, safety, usage, citations, tool calls, and errors to canonical envelope.
Capability profile	Supported modalities, context, tools, structured output, streaming, determinism, privacy, regions, and limits.
Conformance profile	Golden fixtures, semantic tolerances, error mapping, safety behavior, observability, and provider-version support.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-17-001	Each model or provider family SHALL use a versioned adapter registered independently from the model artifact and serving endpoint.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-17-002	Adapters SHALL preserve canonical roles, instruction precedence, context identity, tool schemas, deadlines, cancellation, output schema, status, usage, policy, trace, and provenance.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-17-003	Adapters SHALL NOT invent citations, confidence, explanation, deterministic guarantees, safety approval, or tool completion not provided by verified upstream evidence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-17-004	Provider-specific content retention, logging, training use, region, subprocessors, safety intervention, and support access SHALL be enforced or reflected as resolution constraints.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-17-005	Adapter upgrades SHALL run cross-version golden, adversarial, performance, failure, cancellation, and replay suites before model resolution can select them.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Provider response unparseable	Preserve raw protected evidence, return structured failure, and prohibit partial field inference.

FAILURE CONDITION	REQUIRED BEHAVIOR
Capability misreported	Suspend affected adapter-model combinations and re-evaluate impacted executions.
Semantic mismatch	Quarantine the mapping and restore canonical meaning before provider use resumes.

Assurance evidence

- Golden fixtures compare canonical requests and results across adapters and provider versions.
- Schema mutation tests exercise missing, extra, reordered, malformed, streamed, and truncated content.
- Policy tests verify retention, training use, region, safety intervention, and support-access constraints.
- Contract replay proves identical canonical interpretation independent of provider wire representation.

CHAPTER ACCEPTANCE AND INHERITANCE

Model Adapters and Provider Abstraction is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 21, Ch. 23, Ch. 45, Ch. 53, Ch. 54, Ch. 67.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-030 · C-035 · C-038 · C-041 · C-042

18 Prompt, Instruction, and Policy Artifacts

Prompts and instructions are executable architecture artifacts. They define role, task, precedence, output, tool, safety, explanation, and interaction behavior and therefore receive the same identity, review, testing, release, monitoring, and rollback discipline as code and policy.

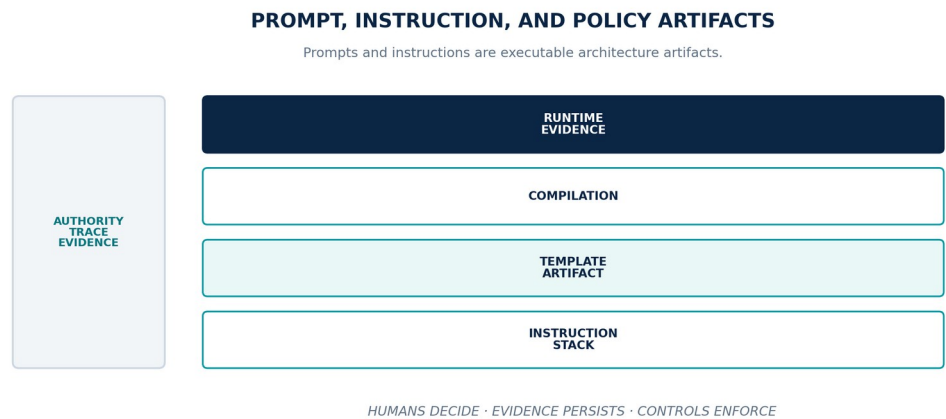


Figure 19. Prompt, Instruction, and Policy Artifacts. The diagram organizes instruction stack, template artifact, compilation, runtime evidence as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Instruction stack	Constitutional controls, system policy, task contract, agent role, workflow step, approved user intent, context, and tool results in explicit precedence.
Template artifact	Stable ID, version, owner, task, model compatibility, parameters, variables, output schema, safety, evaluation, and lifecycle.
Compilation	Deterministic assembly from approved fragments with escaping, context/data separation, token budget, and final digest.
Runtime evidence	Resolved fragments, variables, redactions, context references, model, policy, compiler version, digest, and result linkage.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-18-001	Every production instruction fragment and template SHALL be versioned, owned, reviewed, signed, evaluated, deployment-compatible, and historically resolvable.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-18-002	Instruction precedence SHALL be explicit and enforced outside the model; retrieved content, source documents, tool output, prior model output, and user data SHALL NOT acquire higher authority by textual form.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-18-003	Templates SHALL bind an output schema, allowed tools, side-effect class, explanation profile, uncertainty behavior, prohibited content, and escalation or abstention conditions.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-18-004	Dynamic variables SHALL be typed, length-bounded, classified, escaped or structured, policy-filtered, and recorded by digest or protected reference.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-18-005	Prompt and instruction changes SHALL pass task, safety, injection, regression, cost, latency, model-compatibility, deployment-parity, and rollback gates before promotion.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Instruction conflict	Apply declared precedence, record the rejected instruction, and escalate material ambiguity.
Template version unavailable	Stop execution or use an explicitly compatible approved version; do not reconstruct from memory.
Injection suspected	Treat content as untrusted data, constrain tools and outputs, and route the execution through safety review.

Assurance evidence

- Compiler fixtures reproduce byte- or token-equivalent instruction stacks from immutable dependencies.
- Injection suites cover sources, documents, retrieval, images, code, tool results, user content, and inter-agent messages.
- Regression evaluations compare product meaning, uncertainty, explanation, safety, cost, and latency across template versions.
- Historical replay resolves the exact instruction stack without disclosing protected content to unauthorized reviewers.

CHAPTER ACCEPTANCE AND INHERITANCE

Prompt, Instruction, and Policy Artifacts is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 23, Ch. 28, Ch. 43, Ch. 44, Ch. 46, Ch. 50, Ch. 55, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-012 · C-028 · C-031 · C-035 · C-045

19 Context Assembly and Budgeting

Context assembly converts governed information into a bounded model- or agent-ready representation. It maximizes decision value while preserving policy, evidence diversity, contradiction, temporal validity, provenance, uncertainty, and known omissions.

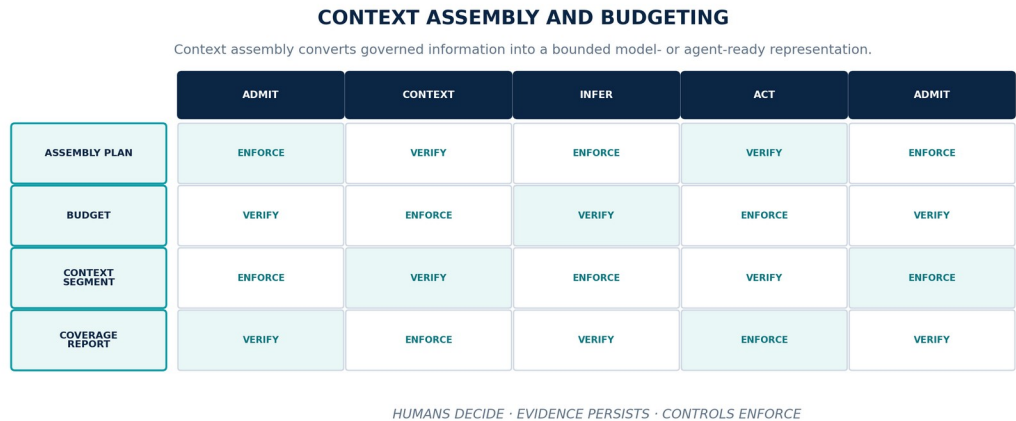


Figure 20. Context Assembly and Budgeting. The diagram organizes assembly plan, budget, context segment, coverage report as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Assembly plan	Task-derived source families, retrieval methods, filters, ranking, diversification, compression, ordering, and token or feature allocation.
Budget	Maximum content, history, tools, images, features, latency, cost, and protected-data exposure by context family.
Context segment	Typed content plus source object, version, temporal scope, truth state, classification, quality, rank reason, and citation key.
Coverage report	Requested and retrieved scope, omissions, truncation, contradictions, diversity, freshness, and confidence in context sufficiency.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-19-001	Context assembly SHALL begin from a versioned task-specific plan and SHALL record every retrieval, filter, ranking, compression, omission, and ordering decision.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-19-002	Budget allocation SHALL prioritize material evidence, counter-evidence, source diversity, uncertainty, objectives, and decision relevance rather than lexical similarity alone.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-19-003	Compression and summarization SHALL retain citation mapping, material qualifications, temporal scope, contradictions, and uncertainty and SHALL expose information loss.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-19-004	Conversation and agent history SHALL be admitted as governed context with actor, authority, version, truth state, retention, and relevance rather than ambient session memory.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-19-005	The final manifest SHALL bind each supplied segment to immutable evidence and identify content omitted by policy, unavailability, ranking, truncation, or budget.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Budget exceeded	Apply declared prioritization and expose truncated scope; never silently drop mandatory evidence classes.
Coverage below threshold	Return coverage-limited or abstained state and request additional retrieval or human analysis.
Compression loses citation alignment	Reject the compressed segment and preserve the uncompressed evidence reference.

Assurance evidence

- Assembly replay reproduces ordered segments, digests, ranks, omissions, token or feature counts, and coverage state.
- Ablation tests measure product sensitivity to removed evidence classes and contradictory sources.
- Policy tests verify minimization and exclusion across history, retrieval, tools, images, and cached representations.
- Long-context tests cover truncation, instruction/data separation, lost-in-context effects, and citation stability.

CHAPTER ACCEPTANCE AND INHERITANCE

Context Assembly and Budgeting is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 45, Ch. 46, Ch. 57, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-014 · C-024 · C-040

20 Structured Output and Product Admission

AI outputs are untrusted until transformed into a canonical candidate and admitted by the owning component. Structured output separates model generation from schema validity, semantic validity, evidence sufficiency, policy compliance, quality, and product authority.

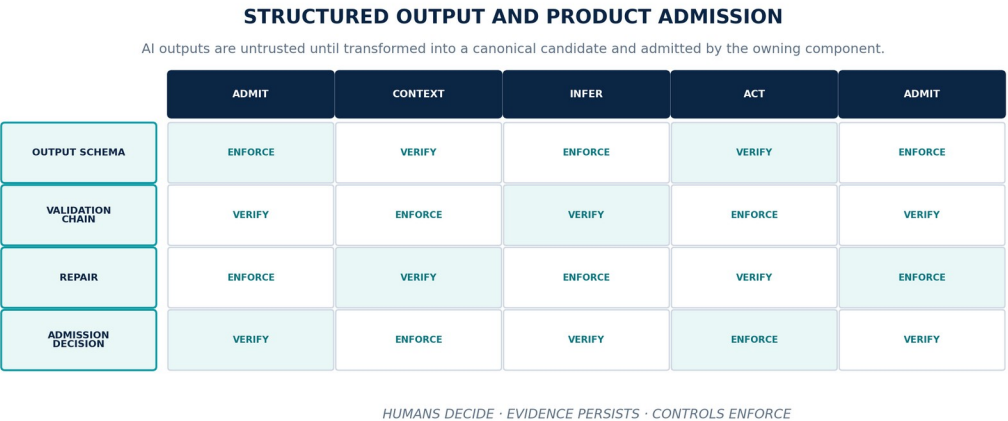


Figure 21. Structured Output and Product Admission. The diagram organizes output schema, validation chain, repair, admission decision as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Output schema	Versioned fields, types, enums, units, cardinality, temporal rules, truth state, uncertainty, citations, and prohibited combinations.
Validation chain	Parse, schema, semantic, policy, evidence, temporal, ontology, quality, duplication, contradiction, and ownership checks.
Repair	Bounded deterministic or model-assisted correction that preserves original output, repair instruction, changes, and evaluation.
Admission decision	Admit, admit-with-constraint, return for review, quarantine, abstain, reject, or withdraw for an exact candidate version.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-20-001	Every AI task producing a material product SHALL bind a canonical output schema and SHALL reject free-form output as authoritative state.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-20-002	Validation SHALL occur outside the producing model or agent and SHALL verify schema, semantics, policy, provenance, evidence, time, truth state, uncertainty, ontology, quality, and owner.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-20-003	Repair attempts SHALL be bounded, versioned, observable, and prohibited from creating missing evidence, citations, confidence, authorization, or human approval.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-20-004	Partial and abstained outputs SHALL preserve which fields or claims are supported, unsupported, omitted, refused, or unavailable and SHALL remain distinct from complete success.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-20-005	Canonical admission SHALL be performed by the owning layer component and SHALL emit immutable product, audit, lineage, quality, and correction references.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Output cannot be parsed	Preserve raw protected output, attempt bounded repair if permitted, then return explicit failure.
Semantic validation fails	Quarantine the candidate and route to regeneration, correction, or human review under task policy.
Evidence does not support claims	Reject decision-active admission and create an explainability or safety finding.

Assurance evidence

- Schema mutation suites cover missing, extra, invalid, conflicting, truncated, streamed, and malicious fields.
- Evidence-grounding tests verify every material claim and citation against admitted context and source objects.
- Repair tests prove original output and every change remain visible and no unsupported authority is introduced.
- Owning-component tests prove no alternate gateway, cache, UI, or agent path can commit canonical product state.

CHAPTER ACCEPTANCE AND INHERITANCE

Structured Output and Product Admission is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 21, Ch. 24, Ch. 27, Ch. 28, Ch. 32, Ch. 43, Ch. 46.

21 Inference Security and Privacy Boundary

Inference is a high-value trust boundary connecting protected context, executable instructions, model artifacts, external providers, tool schemas, output, telemetry, and billing. Security and privacy controls constrain data before invocation and treat all returned content as untrusted.

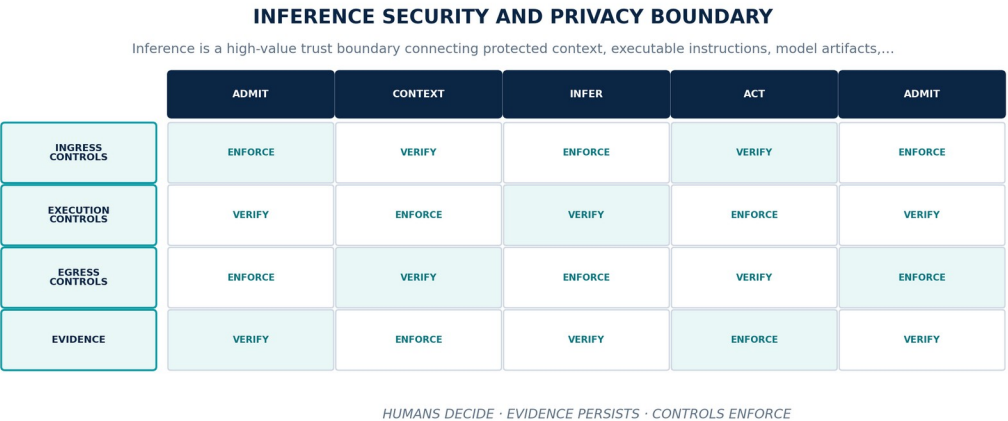


Figure 22. Inference Security and Privacy Boundary. The diagram organizes ingress controls, execution controls, egress controls, evidence as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Ingress controls	Identity, purpose, authorization, classification, rights, minimization, residency, instruction integrity, schema, and malware/content checks.
Execution controls	Workload identity, isolated endpoint, encrypted channel, key custody, provider policy, context retention, logging, and resource limits.
Egress controls	Schema, policy, content safety, provenance, secrets, data loss, citation, classification, and destination checks.
Evidence	Invocation digest, policy decisions, data manifest, endpoint identity, key references, provider receipt, output digest, and audit state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-21-001	Protected context SHALL be minimized and authorized for the exact model, provider, region, purpose, task, and retention policy before crossing the inference boundary.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-21-002	Customer secrets, credentials, hidden policy, internal keys, unrelated tenant data, and uncontrolled support content SHALL be prohibited from model context and telemetry.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-21-003	External inference SHALL use controlled egress, authenticated destinations, contractually and technically enforced data handling, and receipts sufficient for audit and incident response.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-21-004	Model output, citations, tool calls, code, links, and embedded instructions SHALL be treated as untrusted until validation and policy enforcement complete.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-21-005	Privacy-preserving local, customer-managed, redacted, tokenized, or aggregated execution paths SHALL be resolved when classification or sovereignty policy requires them.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Required protection cannot be enforced	Deny invocation and identify the unsatisfied policy obligation.
Sensitive output detected	Quarantine result, prevent downstream admission, and initiate exposure assessment.
Provider custody uncertain	Stop new transfer and treat affected invocations under incident, retention, and customer-notification policy.

Assurance evidence

- Data-flow inventories reconcile every context field, provider, region, retention, log, support, cache, and deletion path.
- Exfiltration tests cover prompt, context, output, citation, tool, network, telemetry, error, and model-behavior channels.
- Privacy tests validate minimization, tokenization, redaction, subject actions, deletion, and provider retention controls.
- Tenant-isolation suites attempt cross-domain leakage through batching, caches, embeddings, routing, logs, and evaluator context.

CHAPTER ACCEPTANCE AND INHERITANCE

Inference Security and Privacy Boundary is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 55, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-035 · C-037 · C-039 · C-040 · C-041 · C-047 · C-050

22 Inference Capacity, Latency, and Economics

Inference resources are allocated by strategic consequence, deadline, quality, and customer policy. Capacity and economics are observable constraints inside the task contract; cost or congestion may cause explicit degradation or abstention but never hidden quality loss or control bypass.

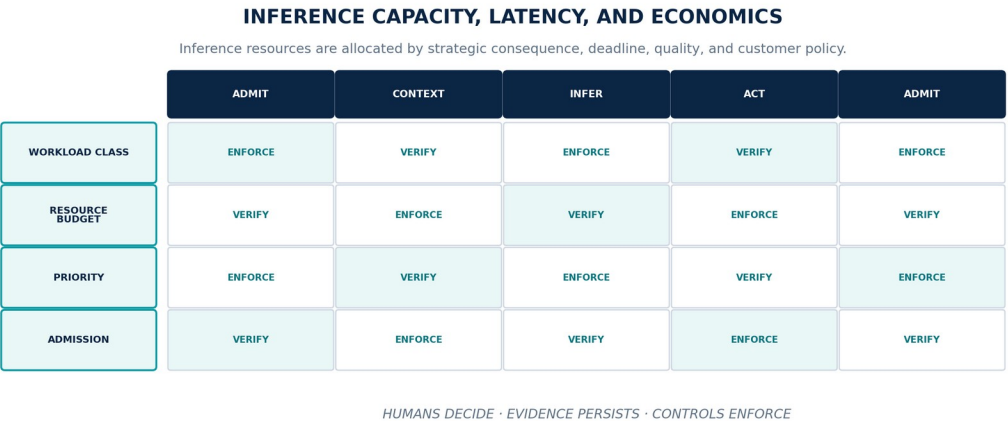


Figure 23. Inference Capacity, Latency, and Economics. The diagram organizes workload class, resource budget, priority, admission as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Workload class	Interactive, warning, decision, workflow, agent, batch, embedding, evaluation, simulation, replay, or recovery inference.
Resource budget	Tokens or features, calls, concurrency, accelerator, memory, storage, network, latency, cost, energy, and provider quota.
Priority	Consequence, urgency, response window, customer policy, fairness, deadline, and recovery reserve.
Admission	Accept, reserve, queue, batch, reduce optional work, choose conforming alternate, defer, or reject with visible reason.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-22-001	Every inference task SHALL carry explicit latency, throughput, concurrency, quality, cost, resource, and deadline budgets appropriate to consequence and product use.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-22-002	Admission control SHALL protect high-consequence interactive, warning, human-gate, and recovery workloads from bulk, evaluation, training, simulation, and low-priority agent demand.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-22-003	Batching, caching, speculative execution, quantization, and smaller-model routing SHALL preserve tenant isolation, policy, product semantics, evaluation approval, and explanation.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-22-004	Cost optimization SHALL NOT silently reduce evidence coverage, context, model fitness, uncertainty quality, safety controls, provenance, or human review.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-22-005	Capacity models SHALL cover normal, burst, warning storm, provider outage, regional failover, evaluation campaign, replay, and customer-domain isolation conditions.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Capacity exhausted	Queue, defer, or return constrained/unavailable under declared priority; do not accept work without a credible deadline.
Cost budget exceeded	Stop optional continuation or seek authorized budget change; preserve partial work and product state.
Provider quota collapses	Apply conforming fallback or isolate affected workload and expose service and product degradation.

Assurance evidence

- Load tests measure percentile latency, throughput, queue age, fairness, quality, cost, energy, and resource saturation by workload class.
- Burst and outage tests validate priority, reservation, backpressure, fallback, and deadline behavior.
- Optimization evaluations compare product quality, calibration, safety, explanation, and policy before and after efficiency changes.
- Capacity forecasts reconcile demand by tenant, task, model, context size, agent graph, horizon, and recovery reserve.

CHAPTER ACCEPTANCE AND INHERITANCE

Inference Capacity, Latency, and Economics is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 45, Ch. 46, Ch. 57, Ch. 58, Ch. 59, Ch. 60, Ch. 62.

CONSTITUTIONAL INHERITANCE / C-014 · C-021 · C-032 · C-042 · C-043 · C-049 · C-050

PART IV

Context, Retrieval, and Grounding

Hybrid retrieval, graph and temporal context, memory, twins, provenance, and citations.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

23 Hybrid Retrieval Architecture

Hybrid retrieval combines lexical, semantic, graph, temporal, geospatial, structured, and source-native methods under one task and policy contract. It preserves method-specific evidence and uses fusion to improve coverage without erasing contradiction or retrieval uncertainty.

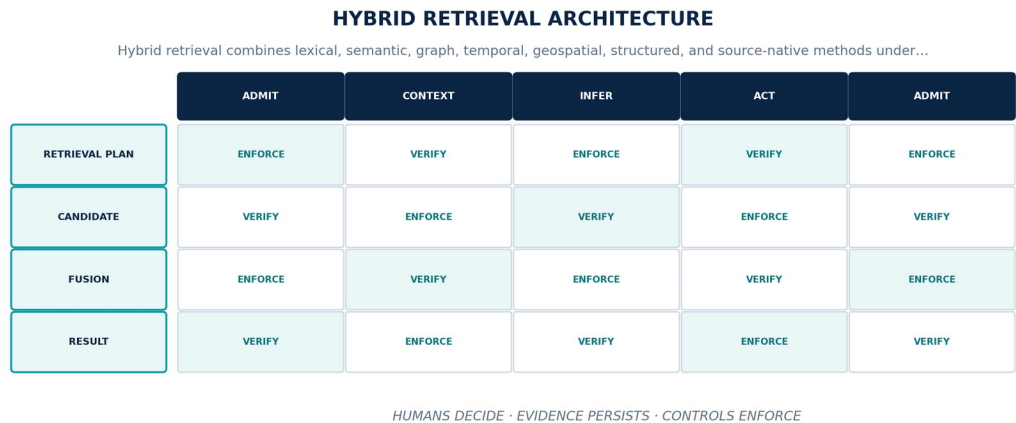


Figure 24. Hybrid Retrieval Architecture. The diagram organizes retrieval plan, candidate, fusion, result as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Retrieval plan	Question decomposition, source classes, methods, filters, time, graph scope, geography, languages, depth, budgets, and stopping conditions.
Candidate	Object or segment reference with method, rank, score, reason, source authority, version, truth state, time, quality, and policy.
Fusion	Versioned normalization, deduplication, diversification, contradiction preservation, reranking, coverage assessment, and cutoff.
Result	Ordered context candidates plus method contributions, coverage, omissions, freshness, projection watermarks, and retrieval uncertainty.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-23-001	Retrieval SHALL operate over governed object and projection references and SHALL NOT treat embeddings, indexes, cached snippets, or provider results as independent canonical truth.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-23-002	The retrieval plan SHALL select methods based on task, object type, language, temporal need, graph structure, source authority, consequence, and known failure modes.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-23-003	Fusion SHALL preserve source and method diversity, material contradiction, minority evidence, temporal scope, and policy rather than optimizing a single relevance score.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-23-004	Semantic indexes and embeddings SHALL be versioned, tenant-isolated, policy-aware, correction-aware, rebuildable, and linked to exact source objects and models.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-23-005	Retrieval results SHALL expose coverage, freshness, truncation, ranking method, uncertainty, and known unavailable or policy-excluded scope to context assembly.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Index stale	Expose watermark and use canonical or alternate retrieval only within declared freshness policy.
Methods disagree	Preserve method-specific candidates and uncertainty; do not collapse to unsupported consensus.
Coverage indeterminate	Return coverage-limited state and constrain downstream claims or request additional search.

Assurance evidence

- Retrieval benchmarks measure recall, precision, diversity, freshness, temporal validity, contradiction, and citation resolution by task slice.
- Correction tests remove or supersede seeded objects and verify indexes, caches, and candidates update within contract.
- Policy tests attempt inference of protected existence through counts, ranks, timing, embeddings, graph paths, and errors.
- Method-ablation tests show fusion behavior and identify populations dependent on one retrieval channel.

CHAPTER ACCEPTANCE AND INHERITANCE

Hybrid Retrieval Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 21, Ch. 26, Ch. 28, Ch. 29, Ch. 30, Ch. 33, Ch. 34, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-009 · C-011 · C-012 · C-014 · C-015 · C-035 · C-038

24 Graph Retrieval and Graph Reasoning

Graph retrieval grounds AI in canonical identity, typed temporal relationships, provenance, ontology, and strategy context. Graph reasoning produces explicit paths, subgraphs, rules, hypotheses, and uncertainty that can be inspected and challenged.

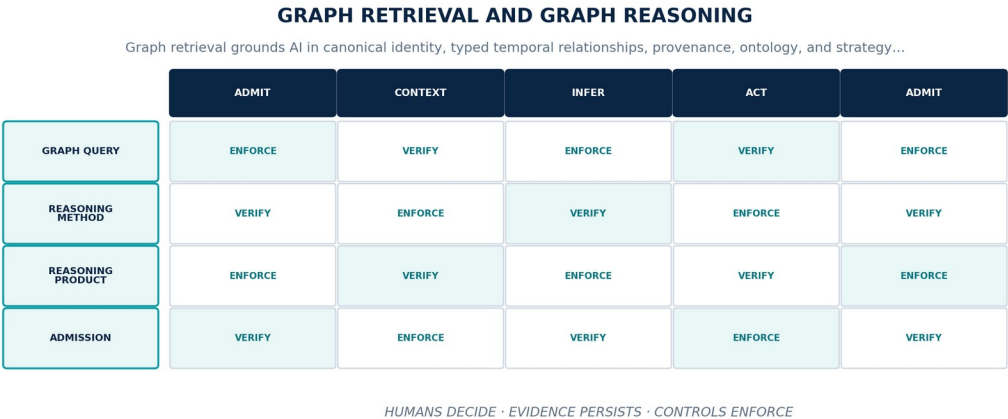


Figure 25. Graph Retrieval and Graph Reasoning. The diagram organizes graph query, reasoning method, reasoning product, admission as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Graph query	Typed starting identities, relationship and node classes, direction, depth, time, truth states, policy, provenance, and result limits.
Reasoning method	Traversal, pattern, rule, constraint, community, path, causal hypothesis, graph embedding, or hybrid algorithm with version.
Reasoning product	Subgraph, paths, inferred relations, supporting and contradicting evidence, confidence, assumptions, limits, and revision watermark.
Admission	Proposed inference remains distinct from observed or stewarded graph state until graph-owner validation and commit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-24-001	AI graph access SHALL use policy-filtered graph contracts that preserve identity, edge direction, temporal scope, truth state, provenance, confidence, and ontology version.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-24-002	Graph reasoning SHALL record query, subgraph boundary, algorithm, parameters, rules, embeddings, graph revision, pruning, and result lineage.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-24-003	Inferred nodes and relationships SHALL remain proposals with explicit method and confidence until Knowledge Graph stewardship validates canonical admission.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-24-004	Graph explanations SHALL expose material paths, relationships, evidence, temporal validity, policy-limited gaps, and alternative or contradictory structures.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-24-005	Graph embeddings and approximations SHALL NOT replace canonical relationships for identity, authorization, provenance, temporal truth, or explanation.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Graph revision changes mid-run	Complete against the pinned revision or restart; do not combine unrecorded revisions.

FAILURE CONDITION	REQUIRED BEHAVIOR
Traversal exceeds policy or budget	Stop at the authorized boundary and return truncation, coverage, and affected reasoning state.
Inferred relation contradicts evidence	Preserve both claims and route for adjudication; never overwrite stewarded state.

Assurance evidence

- Golden graph fixtures reproduce paths, subgraphs, rules, pruning, confidence, and revision watermarks.
- Temporal tests validate relationships before, during, and after validity and correction boundaries.
- Authorization tests detect protected-node and edge inference through path length, counts, timing, and embeddings.
- Stewardship tests prove inferred graph products cannot commit without owning-component validation.

CHAPTER ACCEPTANCE AND INHERITANCE

Graph Retrieval and Graph Reasoning is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 34, Ch. 46, Ch. 50.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-011 · C-012 · C-017 · C-035 · C-040

25 Enterprise, Strategic, and Agent Memory Boundaries

Enterprise Memory preserves governed institutional records; Strategic Memory connects evidence, assumptions, decisions, commitments, outcomes, and lessons; agent execution memory supports bounded work. These memory classes remain separate in authority, retention, reuse, and correction.

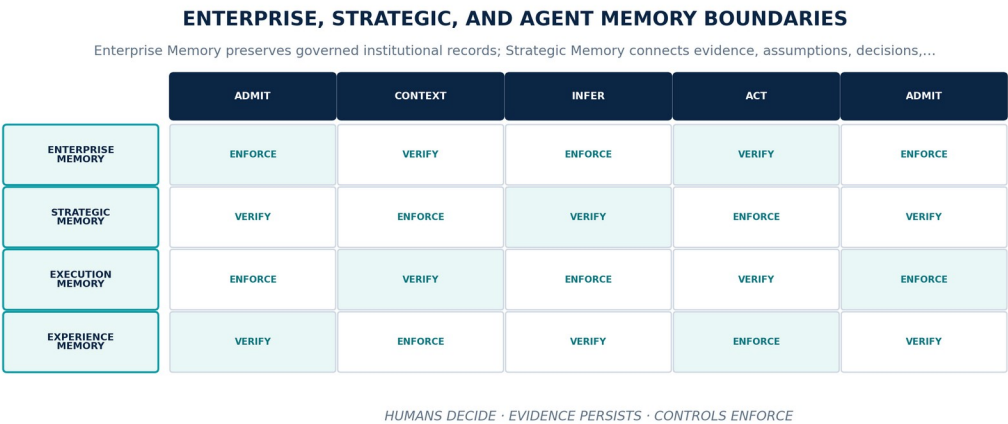


Figure 26. Enterprise, Strategic, and Agent Memory Boundaries. The diagram organizes enterprise memory, strategic memory, execution memory, experience memory as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Enterprise Memory	Canonical governed organizational records owned by the Enterprise Memory layer.
Strategic Memory	Policy-governed connected view of strategic evidence, assessments, objectives, decisions, commitments, outcomes, and lessons.
Execution memory	Run-scoped plans, observations, tool results, checkpoints, intermediate products, reviews, and terminal evidence.
Experience memory	Approved reusable task patterns, examples, procedures, and lessons released through governance; never silent episodic self-modification.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-25-001	AI agents SHALL distinguish canonical enterprise and strategic memory from run-scoped execution memory, cache, conversation history, and learned behavior.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-25-002	Writing to Enterprise or Strategic Memory SHALL use the owning component's schema, policy, provenance, temporal, quality, correction, retention, and human-review contracts.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-25-003	Execution memory SHALL be tenant- and run-isolated, purpose-bound, time-bounded, policy-controlled, and unable to silently influence unrelated tasks after closure.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-25-004	Reusable experience SHALL enter production only as a versioned lesson-derived artifact with evaluation, approval, release, monitoring, rollback, and historical resolution.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-25-005	Memory retrieval SHALL surface source, version, truth state, time, owner, retention, correction, confidence, relevance, and known incompleteness to the agent and explanation product.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Memory class ambiguous	Apply the more restrictive lifecycle and prohibit reuse until ownership is resolved.
Execution memory leaks across scope	Terminate affected runs, isolate residual state, assess exposure, and correct the runtime boundary.
Canonical memory corrected	Invalidate dependent context and products and trigger review or recomputation according to impact.

Assurance evidence

- State inventories reconcile memory class, owner, tenant, purpose, retention, correction, export, and deletion behavior.
- Cross-run and cross-domain tests attempt memory leakage through checkpoints, caches, summaries, vectors, and agent prompts.
- Correction fixtures propagate memory changes through retrieval, context, graph, twins, AI products, and decisions.
- Learning tests prove feedback cannot change reusable behavior without an approved artifact release.

CHAPTER ACCEPTANCE AND INHERITANCE

Enterprise, Strategic, and Agent Memory Boundaries is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 12, Ch. 24, Ch. 26, Ch. 28, Ch. 29, Ch. 30, Ch. 33, Ch. 43, Ch. 48, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-025 · C-031 · C-040 · C-044

26 Temporal and Version-Aware Context

AI context preserves when events occurred, when sources were observed, when assertions applied, and when The Eye recorded them. Version-aware retrieval prevents current knowledge, corrected data, or later outcomes from contaminating historical forecasts, simulations, decisions, and replays.

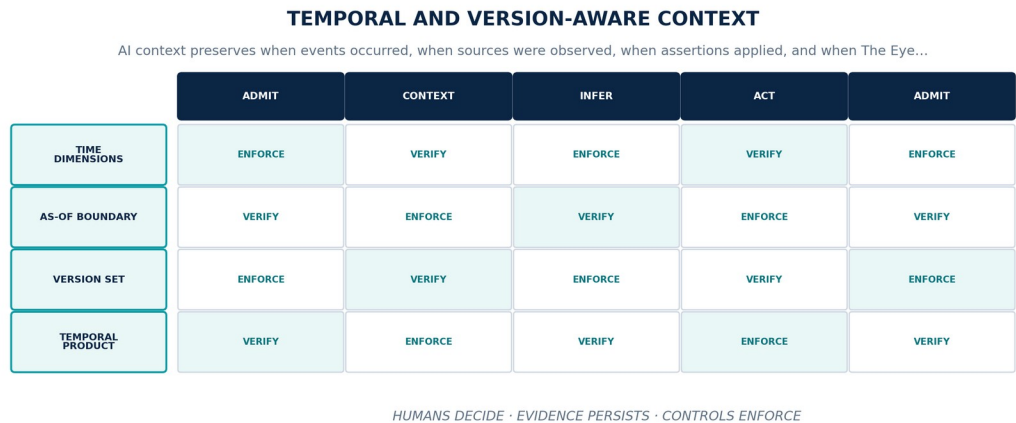


Figure 27. Temporal and Version-Aware Context. The diagram organizes time dimensions, as-of boundary, version set, temporal product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Time dimensions	Event, observation, valid, system, issue, deadline, expiry, model training cutoff, and context assembly time.
As-of boundary	Exact system-time view plus valid-time scope, policy version, projection watermark, and known clock uncertainty.
Version set	Object, graph, twin, model, agent, prompt, retrieval, tool, policy, evaluation, and configuration versions supplied.
Temporal product	Result whose inputs, applicability, horizon, freshness, expiry, and later correction state are explicit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-26-001	Every AI task SHALL declare the temporal perspective required for inputs and outputs and SHALL reject ambiguous default-to-current behavior for consequential work.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-26-002	Historical analysis SHALL resolve information by system time and SHALL keep later corrections, outcomes, and policy changes in a separate comparison boundary.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-26-003	Context assembly SHALL filter and label objects by event, observation, valid, and system time and SHALL expose clock quality, projection lag, and stale state.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-26-004	Model training cutoff and accessible knowledge horizon SHALL be declared where they materially affect novelty, leakage, or historical evaluation.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-26-005	Forecast, scenario, simulation, warning, and decision products SHALL carry creation time, target or valid horizon, expiry, refresh conditions, and dependency invalidation state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Time boundary missing	Return invalid temporal scope and require explicit time semantics.
Clock quality insufficient	Expose time uncertainty and restrict ordering or deadline claims accordingly.
Hindsight contamination detected	Invalidate the evaluation or replay, preserve evidence, and rebuild from the correct historical boundary.

Assurance evidence

- Bitemporal fixtures reconstruct pre-correction and post-correction knowledge states for identical valid-time questions.
- Historical evaluation tests prevent model, context, label, and outcome leakage across cutoff boundaries.
- Projection tests expose revision watermark and lag under concurrent updates and recovery.
- Decision Replay compares original and later knowledge without modifying the original information boundary.

CHAPTER ACCEPTANCE AND INHERITANCE

Temporal and Version-Aware Context is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 24, Ch. 26, Ch. 27, Ch. 30, Ch. 36, Ch. 37, Ch. 38, Ch. 48.

27 Entity and Relationship Grounding

Entity and relationship grounding converts mentions, identifiers, records, observations, and claims into candidate canonical identities and typed temporal relationships. Resolution preserves ambiguity, evidence, confidence, stewardship, merge and split history, and cross-source disagreement.

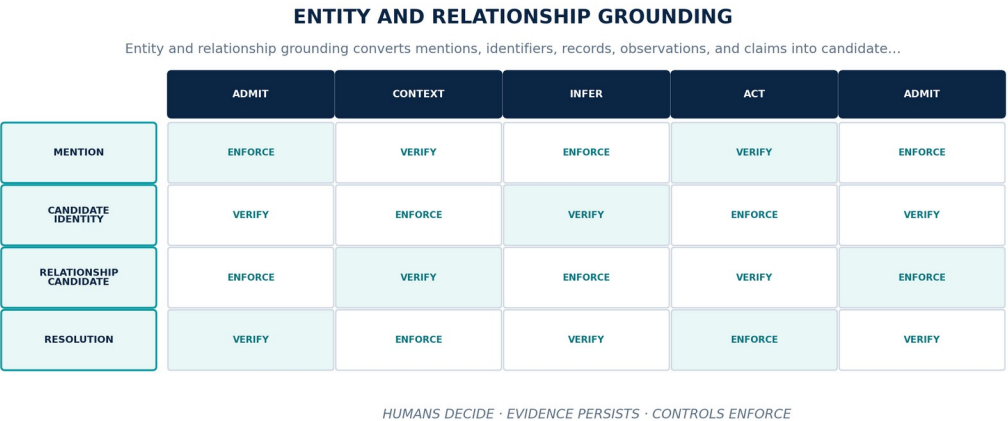


Figure 28. Entity and Relationship Grounding. The diagram organizes mention, candidate identity, relationship candidate, resolution as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Mention	Source-bound text, identifier, image region, signal, record field, or observation that may refer to an entity.
Candidate identity	Existing or proposed entity with evidence, match features, confidence, type, temporal scope, and stewardship state.
Relationship candidate	Typed directed connection with endpoints, evidence, time, truth state, confidence, and ontology version.
Resolution	Resolved, unresolved, ambiguous, new-entity proposal, merge proposal, split proposal, conflict, or human-stewardship required.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-27-001	Entity and relationship models SHALL produce candidates and evidence rather than silently create or merge canonical graph identities.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-27-002	Resolution SHALL use source authority, identifiers, attributes, relationships, temporal compatibility, geography, language, ontology, prior decisions, and negative evidence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-27-003	Ambiguity SHALL remain explicit with ranked candidates and uncertainty when evidence does not support one canonical identity.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-27-004	Relationship extraction SHALL bind exact supporting spans or observations, endpoint candidates, direction, type, temporal scope, truth state, confidence, and contradictory evidence.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-27-005	Merge, split, correction, and supersession SHALL propagate through graph, memory, twins, forecasts, scenarios, simulations, decisions, and evaluation datasets.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Identity confidence below threshold	Keep unresolved or route to stewardship; do not create false certainty.
Ontology type unavailable	Quarantine or propose governed ontology change rather than invent an unmanaged predicate.
Resolution later reversed	Create versioned correction and forward impact analysis without erasing prior decision context.

Assurance evidence

- Resolution benchmarks measure precision, recall, calibration, abstention, merge, split, novelty, and fairness by entity class and language.
- Adversarial fixtures cover aliases, transliteration, shared names, shell entities, identifier reuse, mergers, and temporal changes.
- Relationship tests verify span evidence, endpoint identity, direction, type, time, contradiction, and ontology conformance.
- Correction tests trace identity changes through every dependent product and historical replay.

CHAPTER ACCEPTANCE AND INHERITANCE

Entity and Relationship Grounding is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 32, Ch. 34, Ch. 46.

28 Digital Twin Grounding for AI

Digital Twins provide bounded, versioned representations of companies, competitors, supply chains, products, markets, regulations, infrastructure, processes, and organizations. AI reads and proposes twin changes through contracts that preserve state class, evidence, assumptions, uncertainty, validity, and branch isolation.

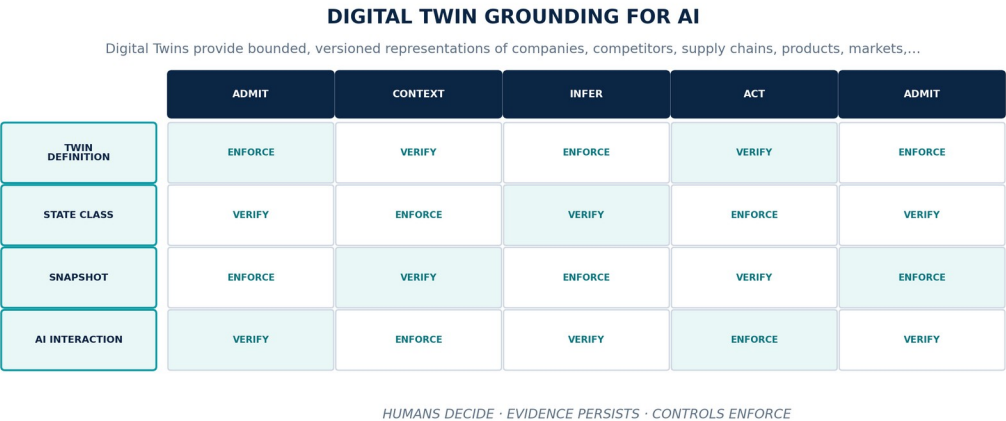


Figure 29. Digital Twin Grounding for AI. The diagram organizes twin definition, state class, snapshot, ai interaction as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Twin definition	Boundary, identities, state variables, behaviors, mappings, models, owners, evidence, validation, operating envelope, and limitations.
State class	Observed, estimated, forecast, scenario, or simulated; classes remain separate and versioned.
Snapshot	Exact twin definition, evidence, graph revision, model, parameters, state vector, confidence, freshness, validity, and quality.
AI interaction	Retrieve, reconcile proposal, estimate, validate, branch, forecast, simulate, challenge, explain, or monitor under declared authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-28-001	AI SHALL address twins through stable definition and snapshot identities and SHALL pin exact versions for prediction, scenario, simulation, recommendation, and replay.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-28-002	Reconciliation agents MAY propose observed or estimated state changes but the Digital Twins layer SHALL validate evidence, mappings, temporal consistency, and lifecycle before commit.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-28-003	Synthetic, scenario, and simulated branches SHALL remain isolated from observed and estimated state and SHALL carry branch lineage through every downstream product.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-28-004	AI products using twin state SHALL expose operating envelope, state variables, estimated fields, uncertainty, model validity, and sensitivity to material state assumptions.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-28-005	Twin correction or invalidation SHALL identify affected forecasts, scenarios, simulations, warnings, decision packages, commitments, and evaluation results.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Twin state stale	Expose stale variables and restrict dependent use or request reconciliation according to consequence.
Model outside operating envelope	Abstain or mark invalid; do not extrapolate without explicit governed method and uncertainty.
Branch write attempts observed state	Deny commit, quarantine execution, and initiate safety and impact review.

Assurance evidence

- Twin fixtures reproduce snapshots, evidence mappings, graph revision, models, parameters, uncertainty, and validity.
- Branch isolation tests attempt cross-state writes through agents, simulations, caches, exports, and recovery.
- Operating-envelope tests cover interpolation, extrapolation, missing variables, stale evidence, and structural change.
- Impact tests invalidate seeded twin versions and trace every dependent product and decision.

CHAPTER ACCEPTANCE AND INHERITANCE

Digital Twin Grounding for AI is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 35, Ch. 36, Ch. 37, Ch. 38.

29 Provenance, Citations, and Evidence Binding

Every material AI claim is bound to evidence and production lineage. Citations are resolvable pointers into governed source and object state, not decorative links or model-generated references, and provenance supports both explanation and forward correction impact.

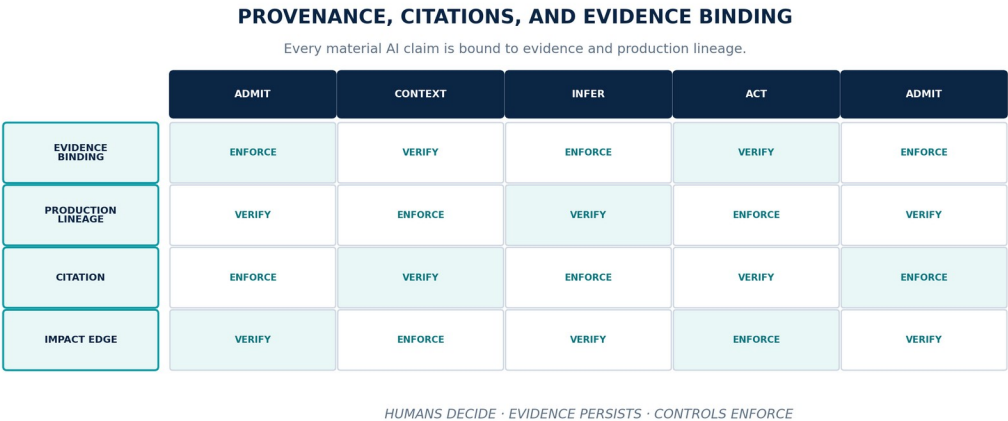


Figure 30. Provenance, Citations, and Evidence Binding. The diagram organizes evidence binding, production lineage, citation, impact edge as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Evidence binding	Claim or field -> exact source or canonical object version, relevant span or feature, transformation, time, truth state, and policy.
Production lineage	Task, context, retrieval, model, agent, prompt, tool, policy, evaluator, human review, output transform, and admission.
Citation	Audience-authorized stable reference with display label, source authority, temporal scope, access behavior, and integrity proof.
Impact edge	Dependency from evidence or artifact to AI product, decision, commitment, export, or learned artifact.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-29-001	Every material factual, relational, numerical, predictive, causal, comparative, and recommendation claim SHALL bind supporting evidence or be explicitly labeled as assumption, hypothesis, synthetic state, or unsupported conclusion.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-29-002	Citation identifiers SHALL resolve to exact versions and relevant spans or structured fields and SHALL NOT be generated from unverified model text.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-29-003	Summaries and explanations SHALL preserve claim-to-evidence alignment through compression, translation, formatting, briefing, export, and audience filtering.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-29-004	Provenance SHALL record every material transformation, model, agent, prompt, context, tool, policy, evaluator, human action, correction, and admission decision.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-29-005	Evidence correction, withdrawal, rights change, or integrity failure SHALL traverse impact edges and update dependent product and decision-use state through durable workflows.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Citation cannot resolve	Mark the claim unverified, restrict product use, and repair or withdraw it.
Evidence access denied	Provide policy-safe metadata and explicit withheld scope; do not expose or substitute unauthorized content.
Lineage write incomplete	Do not acknowledge product admission as provenance-complete; keep the workflow unresolved.

Assurance evidence

- Claim-level fixtures verify citation resolution, span alignment, version, time, truth state, and policy across modalities.
- Transformation tests preserve citation mapping through summarization, translation, aggregation, and structured output repair.
- Forward-impact tests withdraw seeded evidence and identify all affected products, warnings, decisions, and exports.
- Access tests prove citations reveal no unauthorized content while retaining accountable proof and explicit withheld scope.

CHAPTER ACCEPTANCE AND INHERITANCE

Provenance, Citations, and Evidence Binding is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 29, Ch. 30, Ch. 46, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-025 · C-035 · C-036

PART V

Multi-Agent Runtime

Agent identity, execution, planning, supervision, workflow, communication, tools, gates, and memory.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

30 Agent Identity, Role, and Package

An agent is a signed, versioned package that performs a bounded institutional role under an accountable owner. Agent identity binds instructions, models, tools, context sources, policies, evaluations, budgets, deployment support, and lifecycle so that behavior remains attributable and revocable.

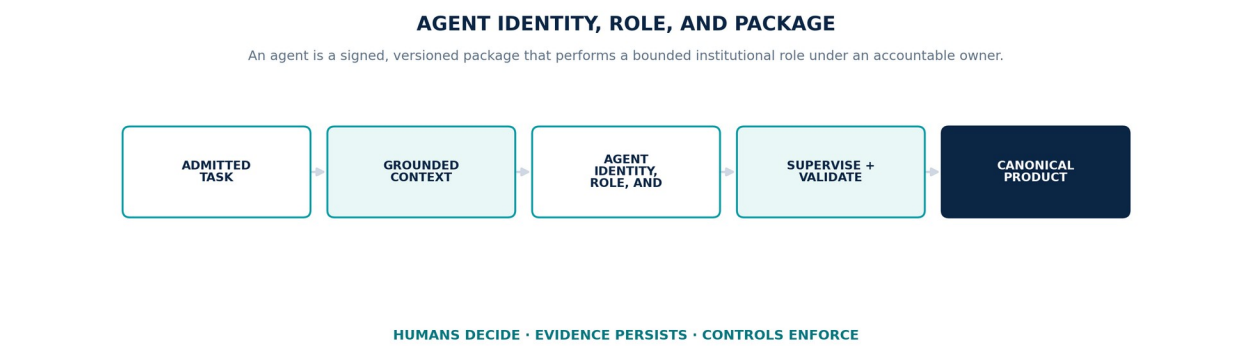


Figure 31. Agent Identity, Role, and Package. The diagram organizes agent identity, role contract, package, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Agent identity	Stable agent type and immutable package version with publisher, owner, signature, provenance, and compatibility.
Role contract	Purpose, permitted tasks, product schemas, consequence classes, data, tools, models, side effects, supervision, and human gates.
Package	Instructions, planners, policies, tool schemas, context profiles, model resolution, evaluators, configuration, tests, and documentation.
Lifecycle	Draft, quarantined, evaluated, approved, shadow, constrained, active, suspended, revoked, deprecated, or retired.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-30-001	Every agent SHALL execute from a signed registered package whose immutable digest binds all behavior-defining artifacts and dependencies.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-30-002	The role contract SHALL declare owned tasks, prohibited tasks, allowed products, data classes, tools, models, network destinations, budgets, side-effect classes, human gates, supervision, and stop conditions.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-30-003	Agent packages SHALL NOT contain ambient credentials, hidden production configuration, unregistered remote code, undeclared data sources, or self-modifying instruction paths.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-30-004	Agent version changes SHALL include instruction, dependency, model policy, tool, context, evaluation, safety, compatibility, deployment, migration, and rollback evidence.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-30-005	Agent retirement SHALL preserve historical package resolution, active-run disposition, product impact, customer export, marketplace removal, and Decision Replay.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Package signature or digest invalid	Quarantine and prohibit execution, import, promotion, and marketplace publication.

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner or support lifecycle absent	Reject production admission; no orphaned agent authority is permitted.
Dependency revoked	Stop new resolution and evaluate active executions and dependent products under impact policy.

Assurance evidence

- Package verification reproduces digest, signature, provenance, dependency, instruction, model, tool, and evaluation closure.
- Role conformance tests attempt prohibited tasks, data, tools, models, network, effects, and authority escalation.
- Upgrade tests compare behavior, compatibility, safety, cost, latency, replay, and rollback across package versions.
- Retirement drills close active credentials and runs while preserving historical evidence and customer-controlled exports.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Identity, Role, and Package is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 43, Ch. 44, Ch. 46, Ch. 55, Ch. 56, Ch. 65, Ch. 68, Ch. 70.

CONSTITUTIONAL INHERITANCE / C-012 · C-028 · C-029 · C-031 · C-035 · C-045 · C-046

31 Agent Execution Lifecycle

An agent execution is a durable, bounded, and auditable state machine. It begins with an admitted task and ends with an explicit product, abstention, failure, cancellation, compensation, or escalation state; an agent process ending is not institutional completion.

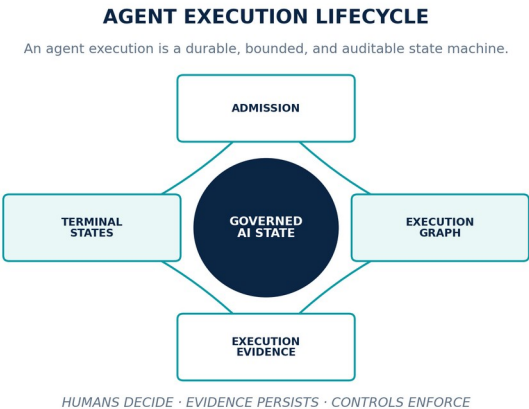


Figure 32. Agent Execution Lifecycle. The diagram organizes admission, execution graph, execution evidence, terminal states as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Admission	Task, actor, purpose, consequence, package, model policy, context policy, tools, budgets, gates, deadline, and stop conditions.
Execution graph	Plans, tasks, dependencies, messages, tool calls, checkpoints, reviews, products, retries, compensation, and terminal state.
Execution evidence	Immutable sequence of observations, decisions, versions, control results, usage, uncertainty, dissent, and side-effect receipts.
Terminal states	Completed, partial, abstained, rejected, cancelled, expired, compensated, failed, or escalated-with-residual-state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-31-001	Every agent execution SHALL receive a unique identity and persist its admitted manifest before model, context, tool, or downstream agent work begins.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-31-002	Each state transition SHALL be authorized, idempotent, durably recorded, causally linked, deadline-aware, and observable before acknowledgement.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-31-003	The runtime SHALL checkpoint plans, context manifests, messages, tool requests and receipts, model invocations, intermediate products, reviews, budgets, and stop decisions.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-31-004	Agent execution SHALL terminate or suspend when purpose, authority, policy, capability, evaluation, budget, deadline, supervisor, or safety conditions no longer hold.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-31-005	Completion SHALL require product validation, side-effect reconciliation, human-gate closure, audit acceptance, resource release, and declared retention or deletion of execution memory.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Worker or process lost	Resume from the last durable checkpoint without repeating acknowledged effects.
External effect uncertain	Enter reconciliation-required state and prohibit retry until receipt or compensation resolves the boundary.
Terminal evidence incomplete	Keep execution non-terminal and expose accountable residual state.

Assurance evidence

- Crash tests interrupt every transition, checkpoint, model call, message, tool effect, gate, and terminal boundary.
- Duplicate and reordering tests prove idempotent tasks, messages, products, and external effects.
- Budget tests cover time, tokens, compute, cost, data, tools, network, retries, and side effects.
- Replay reconstructs an admitted execution from immutable manifests and preserved external receipts.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Execution Lifecycle is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 22, Ch. 28, Ch. 41, Ch. 43, Ch. 44, Ch. 55, Ch. 59, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-004 · C-011 · C-012 · C-025 · C-028 · C-029 · C-043

32 Planner Agent Architecture

The Planner converts an admitted objective into a versioned execution graph of bounded tasks, dependencies, evidence needs, tools, models, agents, human gates, budgets, and stopping conditions. Planning proposes work; it does not grant capabilities or approve effects.

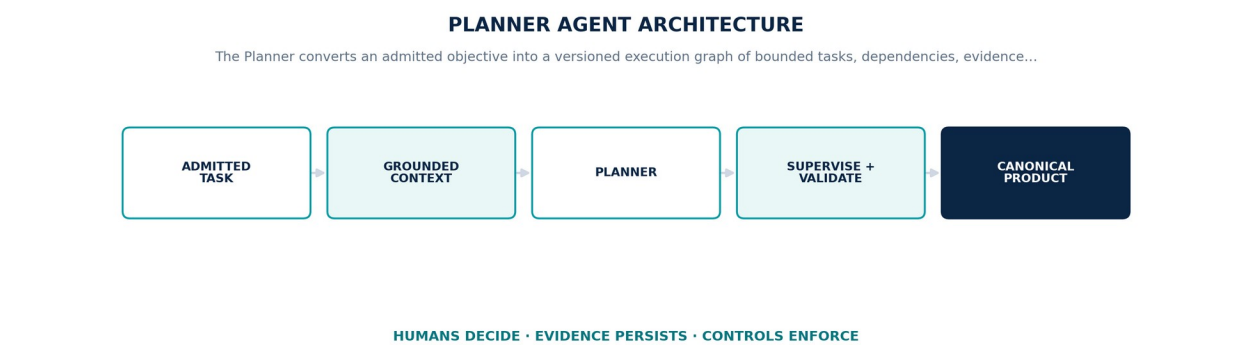


Figure 33. Planner Agent Architecture. The diagram organizes planning input, plan, plan revision, plan quality as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Planning input	Task contract, objectives, constraints, known state, context policy, available capabilities, deadlines, budgets, and consequence.
Plan	Directed acyclic or explicitly iterative graph of typed steps, owners, inputs, outputs, dependencies, guards, gates, retries, and completion criteria.
Plan revision	Versioned change with reason, evidence, affected tasks, budget, capability, deadline, and supervisor approval.
Plan quality	Completeness, feasibility, policy compliance, evidence coverage, critical path, uncertainty, reversibility, and expected value.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-32-001	The Planner SHALL produce a typed plan whose every step maps to an approved agent role, workflow task, model task, tool contract, human gate, or owning-component command.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-32-002	Plans SHALL declare required evidence, input versions, output schemas, dependencies, parallelism, loops, budgets, deadlines, failure handling, compensation, review, and terminal conditions.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-32-003	The Planner SHALL NOT issue capability grants, select unauthorized data, bypass required human gates, approve irreversible effects, or modify policy and evaluation thresholds.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-32-004	Material plan revision SHALL be versioned and re-evaluated for purpose, authority, policy, capability, budget, deadline, side effects, and supervisor acceptance.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-32-005	Plans SHALL expose unresolved assumptions, unknown dependencies, low-confidence decomposition, and tasks requiring human or domain-expert intervention.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Plan infeasible	Return explicit infeasibility, blocking constraints, and any safe partial plan; do not begin unsupported execution.

FAILURE CONDITION	REQUIRED BEHAVIOR
Dependency discovered at runtime	Suspend affected branch and submit a versioned plan revision to the Supervisor.
Plan loops without progress	Stop at progress or budget thresholds and escalate with complete execution evidence.

Assurance evidence

- Plan validators check task types, dependency closure, cycles, schemas, capabilities, gates, budgets, deadlines, and terminal conditions.
- Adversarial objectives attempt to induce prohibited data access, tool use, self-approval, policy change, and hidden side effects.
- Simulation fixtures compare planned and actual critical paths, evidence coverage, retries, cost, and completion quality.
- Human reviewers can understand, challenge, modify, approve, and replay material plans without interpreting hidden model reasoning.

CHAPTER ACCEPTANCE AND INHERITANCE

Planner Agent Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 3, Ch. 21, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 46, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-028 · C-029 · C-035 · C-049

33 Supervisor Agent Architecture

The Supervisor is the runtime authority for an admitted multi-agent execution. It validates plans, assigns bounded work, monitors progress and risk, challenges results, enforces budgets and gates, suspends unsafe activity, and determines terminal state under policy.

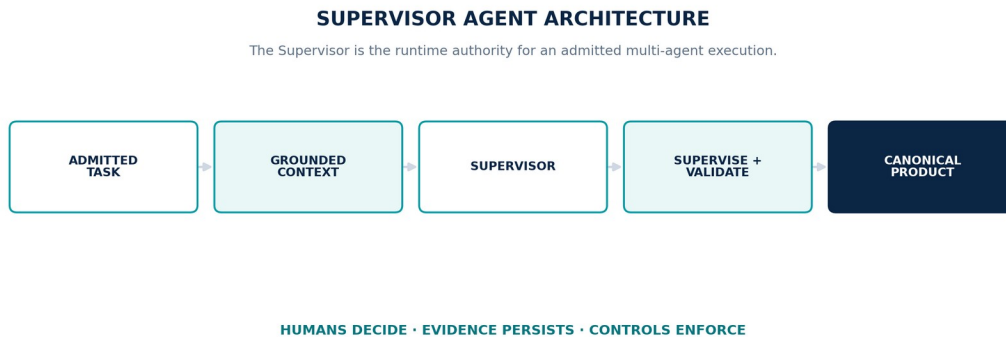


Figure 34. Supervisor Agent Architecture. The diagram organizes supervision scope, admission decision, runtime control, supervision evidence as one governed AI architecture contract.

AI architecture contract

Contract Element	Binding Definition
Supervision scope	One execution or declared portfolio with task, agent, model, context, tool, budget, deadline, consequence, and gate boundaries.
Admission decision	Approve, constrain, return, reject, or escalate a plan or revision based on policy and fitness.
Runtime control	Assign, pause, resume, challenge, reroute, reduce scope, request review, terminate, compensate, or escalate.
Supervision evidence	Decisions, signals, reasons, policy, evaluated state, budgets, dissent, interventions, and terminal rationale.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-33-001	Every multi-agent execution SHALL have exactly one accountable Supervisor identity and policy profile at any execution instant.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-33-002	The Supervisor SHALL validate plan, agent, model, context, tool, capability, budget, human-gate, quality, and safety closure before dispatch.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-33-003	The Supervisor SHALL monitor progress, evidence quality, uncertainty, disagreement, tool effects, policy, budget, deadlines, loops, anomalies, and product fitness throughout execution.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-33-004	The Supervisor SHALL remain independently authorized from worker agents and SHALL NOT rely on worker self-report alone for capability, side-effect, safety, or completion state.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-33-005	Terminal approval SHALL require resolved tasks, validated products, reconciled effects, closed gates, complete audit, and explicit residual-state disposition.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Supervisor unavailable	Stop new work and effects; checkpoint active branches and escalate according to execution consequence.

FAILURE CONDITION	REQUIRED BEHAVIOR
Worker and Supervisor disagree	Preserve both states and route to declared challenge or human authority; do not suppress dissent.
Supervisor control compromised	Fence the execution, revoke grants, preserve evidence independently, and initiate security incident response.

Assurance evidence

- Supervision scenarios cover admission, constraint, challenge, pause, reroute, termination, compensation, escalation, and closure.
- Independent signals verify tool receipts, budgets, policies, execution progress, and product validation outside worker self-report.
- Fault tests remove the Supervisor during every material execution state and verify safe continuation or stop.
- Adversarial agents attempt to misreport completion, hide uncertainty, expand scope, exhaust budgets, and bypass gates.

CHAPTER ACCEPTANCE AND INHERITANCE

Supervisor Agent Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 10, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 55, Ch. 59, Ch. 60, Ch. 62.

34 Workflow Agent and Durable Orchestration

The Workflow Agent mediates between agent reasoning and the durable workflow runtime. It translates approved plans into versioned workflow definitions and commands, tracks machine and human states, enforces timers and compensation, and exposes institutional progress independently of transient agent processes.

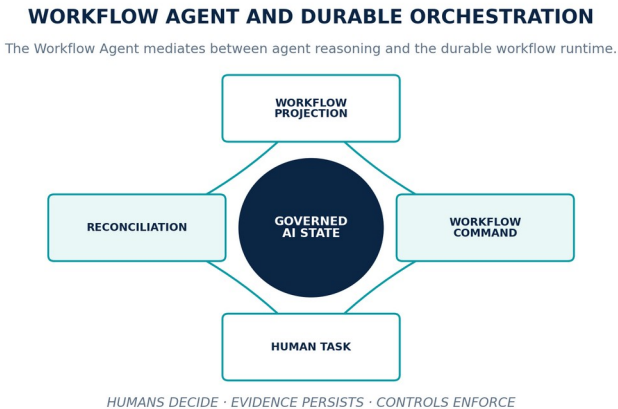


Figure 35. Workflow Agent and Durable Orchestration. The diagram organizes workflow projection, workflow command, human task, reconciliation as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Workflow projection	Approved plan mapped to durable states, transitions, roles, tasks, gates, timers, retries, compensation, evidence, and terminal conditions.
Workflow command	Authenticated, idempotent request to transition one instance under current preconditions and policy.
Human task	Durable assignment with authority, exact context, choices, deadline, escalation, and completion evidence.
Reconciliation	Comparison of workflow state with agent, tool, model, external effect, audit, and canonical product state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-34-001	Consequential multi-agent work SHALL be represented in the durable workflow runtime before execution produces institutional effects.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-34-002	The Workflow Agent SHALL issue only registered workflow commands and SHALL NOT mutate workflow persistence, timers, assignments, or audit records directly.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-34-003	Workflow definitions SHALL bind exact agent roles, task schemas, capabilities, gates, retries, compensation, deadlines, and evidence required for each transition.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-34-004	Agent or model completion SHALL NOT advance institutional workflow state until output, side effect, policy, and evidence preconditions validate.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-34-005	Workflow migration SHALL transform active instances through a versioned, tested, reversible process that preserves prior state and authority.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Workflow and agent state diverge	Fence progression and reconcile from durable transition and effect evidence.

FAILURE CONDITION	REQUIRED BEHAVIOR
Timer or escalation missed	Expose overdue state, route accountability, and preserve the original deadline and consequence.
Compensation incomplete	Keep the instance open with residual effects and accountable remediation.

Assurance evidence

- Crash-recovery tests interrupt every workflow, agent, human, tool, and external-effect boundary.
- State-reconciliation fixtures detect missing, duplicate, reordered, and conflicting transitions.
- Migration tests transform every active source state and preserve Decision Replay of the prior definition.
- Operator tests prove interventions use authorized commands and remain independently audited.

CHAPTER ACCEPTANCE AND INHERITANCE

Workflow Agent and Durable Orchestration is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 17, Ch. 19, Ch. 22, Ch. 41, Ch. 42, Ch. 43, Ch. 55, Ch. 59, Ch. 60.

CONSTITUTIONAL INHERITANCE / C-004 · C-011 · C-025 · C-028 · C-029 · C-036 · C-043

35 Agent Communication and Coordination Protocol

Agents coordinate through typed messages and shared durable execution state rather than unstructured conversation or hidden context. The protocol preserves sender identity, role, task, causation, evidence, uncertainty, authority, deadlines, and product semantics across agent boundaries.

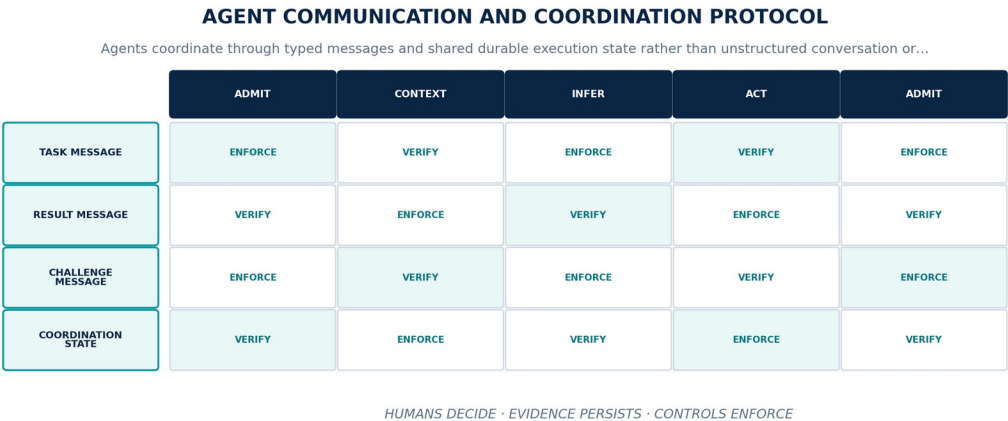


Figure 36. Agent Communication and Coordination Protocol. The diagram organizes task message, result message, challenge message, coordination state as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Task message	Typed delegation with objective, inputs, output schema, evidence, capabilities, budget, deadline, review, and stop conditions.
Result message	Product candidate, partial result, abstention, failure, uncertainty, evidence, quality, resource, and residual-state report.
Challenge message	Structured objection to evidence, method, assumptions, policy, uncertainty, product, or completion state.
Coordination state	Durable assignment, dependency, sequence, acknowledgement, checkpoint, cancellation, and terminal-state references.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-35-001	All inter-agent communication SHALL use versioned registered message schemas and canonical envelopes; free-form conversation MAY be content inside a typed field but not the protocol itself.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-35-002	Messages SHALL authenticate sender, receiver, agent package, execution, task, purpose, authority, correlation, causation, deadline, policy, and integrity.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-35-003	Result messages SHALL distinguish evidence, inference, assessment, assumption, proposal, synthetic content, uncertainty, partial state, and abstention.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-35-004	Agents SHALL NOT transfer capabilities, credentials, tenant context, human approval, or higher authority through message content.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-35-005	The runtime SHALL support idempotent delivery, ordering where required, deduplication, cancellation, replay, poison-message quarantine, and schema evolution.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Unknown message version	Quarantine or use explicit compatibility adapter; never guess semantics.
Sender authority invalid	Reject message and preserve attempted delegation for security and audit review.
Conflicting results	Preserve both products and route challenge or adjudication under the execution plan.

Assurance evidence

- Protocol suites cover every message type, schema version, duplicate, reorder, delay, cancellation, replay, and poison case.
- Security tests attempt identity substitution, tenant confusion, capability transfer, approval reuse, and embedded instruction escalation.
- Semantic fixtures preserve truth state, uncertainty, evidence, partial state, and dissent across agent implementations.
- Execution replay rebuilds coordination state from immutable messages and workflow checkpoints.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Communication and Coordination Protocol is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 19, Ch. 20, Ch. 22, Ch. 23, Ch. 28, Ch. 43, Ch. 44, Ch. 55.

36 Tool, Capability, Sandbox, and Side-Effect Architecture

Tools expose institutional capability to agents through typed gateways and least-privilege grants. Sandboxes isolate execution; capability policy limits data, network, compute, storage, models, credentials, and effects; human gates bind approval to exact consequential actions.

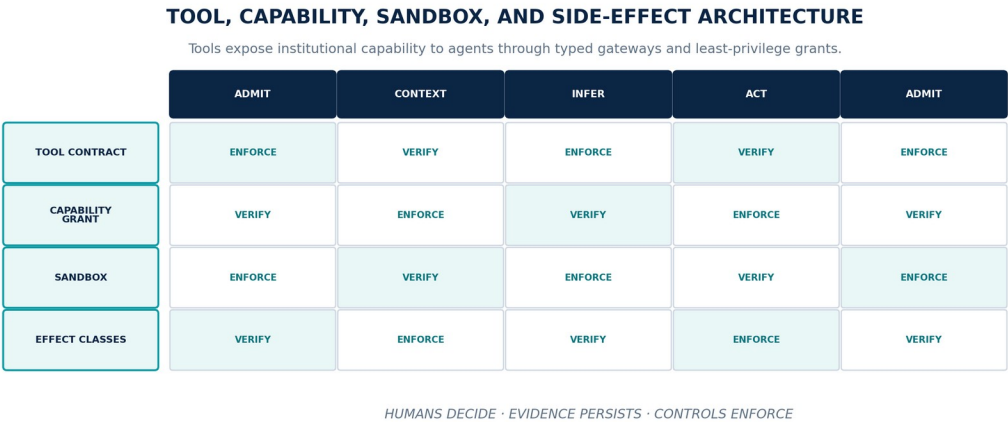


Figure 37. Tool, Capability, Sandbox, and Side-Effect Architecture. The diagram organizes tool contract, capability grant, sandbox, effect classes as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Tool contract	Identity, version, typed inputs and outputs, side-effect class, policy, data, network, resources, idempotency, compensation, timeout, and audit.
Capability grant	Agent, execution, task, purpose, actions, object scope, environment, destinations, resources, duration, obligations, and revocation.
Sandbox	Isolated process, filesystem, network, secret, package, compute, memory, storage, and persistence boundary.
Effect classes	None, reversible, compensatable, approval-required, irreversible, prohibited, or unknown requiring restrictive classification.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-36-001	An agent SHALL invoke a tool only through a registered enforcement gateway with a valid run- and task-specific capability grant.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-36-002	Tool inputs SHALL be schema-valid, policy-filtered, classified, minimized, size-bounded, instruction-safe, and bound to tenant, purpose, deadline, and idempotency context.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-36-003	Tool outputs SHALL be treated as untrusted observations until integrity, schema, provenance, classification, policy, and evidence admission complete.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-36-004	Approval-required and irreversible effects SHALL require a human gate after the exact request is known and SHALL dispatch only the approved digest before expiry.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-36-005	Sandbox and capability revocation SHALL stop new activity immediately and define treatment of in-flight operations, credentials, sessions, files, caches, and residual effects.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Tool or version unregistered	Deny invocation and emit a governed capability violation.
Side-effect class ambiguous	Apply the more restrictive class and require accountable review.
Sandbox escape suspected	Terminate the execution, revoke credentials and grants, isolate affected hosts, and initiate security response.

Assurance evidence

- Capability matrices enumerate every agent, task, tool, action, data, destination, effect, and purpose combination.
- Sandbox tests cover process, filesystem, network, secret, package, tenant, resource, persistence, and covert-channel isolation.
- Side-effect tests prove idempotency, exact approval binding, receipt reconciliation, compensation, and expiry.
- Revocation drills measure closure across gateways, runtimes, credentials, providers, sessions, and residual artifacts.

CHAPTER ACCEPTANCE AND INHERITANCE

Tool, Capability, Sandbox, and Side-Effect Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 22, Ch. 42, Ch. 43, Ch. 44, Ch. 49, Ch. 50, Ch. 51, Ch. 53, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-004 · C-028 · C-029 · C-035 · C-037 · C-039 · C-040

37 Human Gates and Mixed-Initiative Collaboration

Humans and agents collaborate through explicit roles, tasks, questions, challenges, approvals, and handoffs. Mixed-initiative operation allows either side to request work or clarification while preserving human authority, evidence, decision context, accessibility, and accountable workflow state.

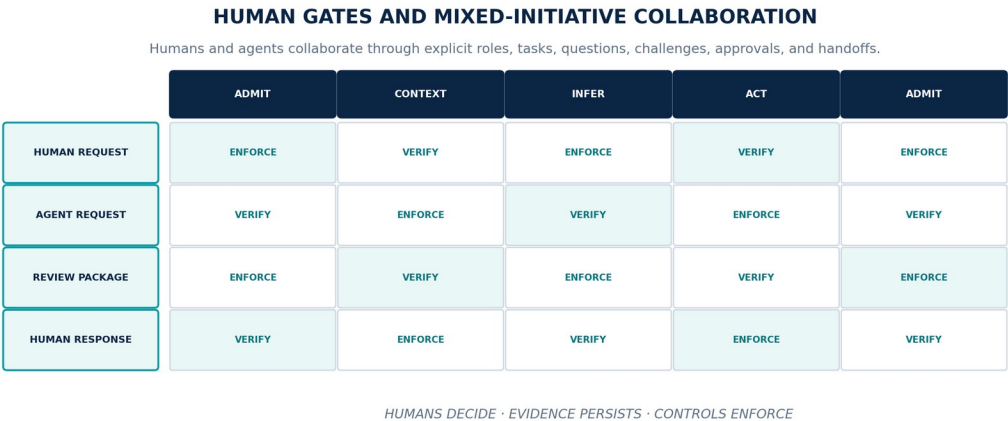


Figure 38. Human Gates and Mixed-Initiative Collaboration. The diagram organizes human request, agent request, review package, human response as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Human request	Authenticated intent with purpose, scope, authority, expected product, deadline, data, consequence, and permitted effects.
Agent request	Structured clarification, evidence need, domain judgment, conflict resolution, approval, or escalation with reason and options.
Review package	Exact product, evidence, explanation, uncertainty, alternatives, dissent, policy, changes, and requested human action.
Human response	Approve, reject, revise, provide evidence, adjudicate, challenge, abstain, delegate, reopen, or stop with rationale and authority.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-37-001	Agent-human interaction SHALL occur through durable tasks or governed workspaces that preserve identity, role, authority, context version, action, deadline, and audit.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-37-002	Agents SHALL request human intervention when evidence, authority, domain judgment, conflict, novelty, consequence, uncertainty, or safety exceeds the task's automated boundary.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-37-003	Review packages SHALL separate facts, claims, inference, assumptions, synthetic content, recommendation, missing information, disagreement, and exact requested action.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-37-004	Human edits, overrides, dismissals, and approvals SHALL be captured as accountable product or workflow state and SHALL NOT be reduced to unlabeled training feedback.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-37-005	Interfaces SHALL preserve accessibility, role clarity, policy-filtered evidence, uncertainty, provenance, and decision-use constraints across UI, API, briefing, report, and export modes.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Human response ambiguous	Return for clarification; do not infer authorization or intent from incomplete language.
Human and agent context differ	Expose the version difference and refresh or explicitly bind the accepted information boundary.
Review deadline expires	Escalate, reassign, or close as expired under policy; never auto-approve.

Assurance evidence

- Mixed-initiative scenarios cover clarification, evidence request, challenge, override, approval, rejection, delegation, expiry, and reopening.
- Accessibility reviews validate role-appropriate language, navigation, alternative text, keyboard operation, and non-color status cues.
- Context-diff tests force material changes during review and verify invalidation and refreshed consent.
- Replay reconstructs every human and agent contribution, visible evidence, authority, and resulting state transition.

CHAPTER ACCEPTANCE AND INHERITANCE

Human Gates and Mixed-Initiative Collaboration is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 39, Ch. 40, Ch. 42, Ch. 43, Ch. 48, Ch. 55, Ch. 71.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-027 · C-047 · C-048

38 Agent Context, Memory, Checkpointing, and Replay

Agent continuity is built from immutable context manifests, durable checkpoints, typed messages, workflow state, and external receipts. The runtime reconstructs what an agent knew and did without relying on hidden process memory, provider-retained sessions, or mutable conversation state.

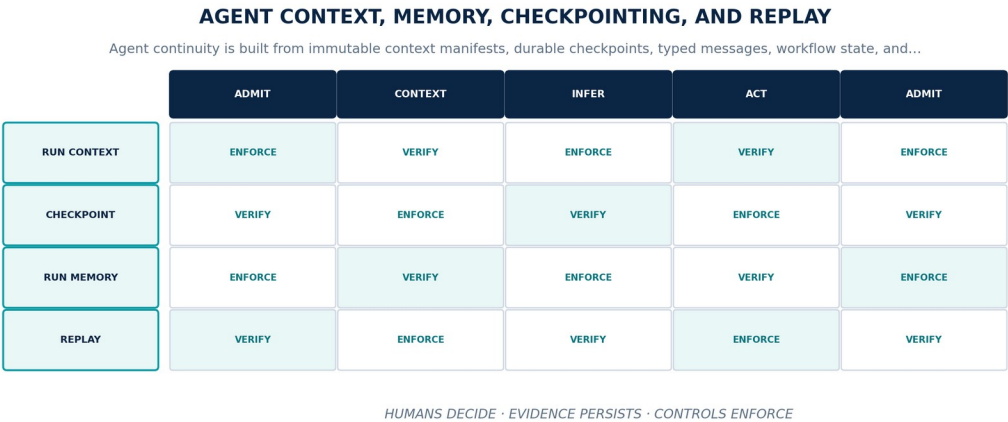


Figure 39. Agent Context, Memory, Checkpointing, and Replay. The diagram organizes run context, checkpoint, run memory, replay as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Run context	Admitted task, plan, packages, models, prompts, context, policies, capabilities, messages, tools, budgets, and time.
Checkpoint	Durable execution state plus completed effects, pending tasks, dependencies, resource, uncertainty, and resume preconditions.
Run memory	Bounded observations, intermediate products, reviews, tool results, and rationale necessary for the current execution.
Replay	Deterministic or semantically bounded reconstruction of execution state, decisions, effects, products, and controls from preserved evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-38-001	Agent runtimes SHALL externalize all material execution state required for recovery, audit, explanation, and replay before acknowledging progress.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-38-002	Checkpoints SHALL bind exact agent, model, prompt, context, tool, policy, configuration, workflow, message, budget, and external-effect state.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-38-003	Run memory SHALL be tenant-isolated, purpose-bound, retention-controlled, minimised, encrypted, revocable, and prohibited from undeclared cross-run reuse.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-38-004	Resume SHALL revalidate identity, authority, policy, capability, artifact fitness, context validity, deadlines, budgets, and unresolved effects before new work.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-38-005	Replay SHALL preserve nondeterminism, sampling, external results, human acts, and uncertainty and SHALL clearly identify what can and cannot be reproduced exactly.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Checkpoint incomplete	Resume from the prior verified checkpoint and reconcile any possible external effect.
Provider session unavailable	Reconstruct from canonical manifests or return explicit non-reproducible state; do not claim continuity.
Run memory retention expires	Delete or archive under policy while preserving minimum audit and Decision Replay evidence.

Assurance evidence

- Crash fixtures resume every execution state and compare outputs and effects with uninterrupted control runs.
- Checkpoint integrity tests alter manifests, ordering, messages, receipts, budgets, and policy and verify rejection.
- Cross-run isolation tests attempt memory transfer through caches, provider sessions, embeddings, logs, and agent summaries.
- Replay reports distinguish exact, tolerance-bounded, externally captured, and non-reproducible execution elements.

CHAPTER ACCEPTANCE AND INHERITANCE

Agent Context, Memory, Checkpointing, and Replay is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 24, Ch. 28, Ch. 41, Ch. 43, Ch. 55, Ch. 60, Ch. 61, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-025 · C-028 · C-029 · C-035 · C-043

PART VI

Observation and Understanding Agents

Acquisition, search, cleaning, classification, summarization, entities, relationships, graph, reasoning, governance, and learning.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

39 Observation, Crawler, and Collection Agents

Observation, Crawler, and Collection Agents acquire permitted external and internal signals under governed source contracts. They preserve source identity, authenticity, rights, timing, coverage, chain of custody, correction behavior, and collection health before any analytical interpretation.



Figure 40. Observation, Crawler, and Collection Agents. The diagram organizes observation agent, crawler agent, collection agent, products as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Observation Agent	Monitors source obligations, schedules, triggers, freshness, coverage, correction, and health and proposes acquisition work.
Crawler Agent	Navigates approved locations and links within robots, terms, rate, identity, scope, depth, content, and network policy.
Collection Agent	Executes source-specific acquisition, captures bytes or state, computes integrity, records custody, and commits Observation Records.
Products	Source Contract proposals, acquisition plans, Observation Records, Evidence Objects, source-health events, and correction events.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-39-001	These agents SHALL operate only under an approved Source Contract declaring authority, rights, purpose, scope, identity, schedule, schema, rate, residency, retention, correction, quality, and owner.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-39-002	Crawler and Collection Agents SHALL identify themselves where required, obey approved robots and access controls, enforce rate and depth, and prohibit credential or network escalation.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-39-003	Acquired content SHALL be stored as immutable evidence with source, request, response, observation time, checksum, signature where available, classification, rights, transformations, and coverage context.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-39-004	Observation Agents SHALL distinguish source silence, collection failure, access denial, true absence, stale data, schema drift, and policy restriction in health state.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-39-005	Corrections, withdrawals, retractions, and changed rights SHALL create versioned events and forward impact analysis without deleting prior observation history.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Source contract missing or expired	Stop acquisition and route contract review; do not infer ongoing permission.

FAILURE CONDITION	REQUIRED BEHAVIOR
Authenticity or integrity uncertain	Quarantine evidence, retain custody information, and prohibit analytical admission.
Crawler scope escapes	Terminate navigation, revoke session, record attempted path, and initiate security review.

Assurance evidence

- Source fixtures verify contracts, identities, rights, rates, schedules, schemas, redirects, pagination, corrections, and expiry.
- Crawler security tests cover credential leakage, SSRF, link traps, hostile content, prompt injection, archives, and redirects.
- Coverage tests distinguish source absence from collector failure and measure freshness and completeness by source contract.
- Correction tests propagate source withdrawal through evidence, objects, graph, twins, foresight, decisions, and briefings.

CHAPTER ACCEPTANCE AND INHERITANCE

Observation, Crawler, and Collection Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 31, Ch. 41, Ch. 43, Ch. 44, Ch. 53, Ch. 54, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-013 · C-014 · C-035 · C-038 · C-040

40 Search Agent

The Search Agent executes governed discovery across public, commercial, scientific, regulatory, market, enterprise, graph, and approved custom sources. It builds transparent search plans, preserves query and result provenance, and separates discovery from evidence admission.



Figure 41. Search Agent. The diagram organizes search intent, search plan, search result, search product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Search intent	Question, purpose, scope, languages, geography, time, source classes, authority, sensitivity, depth, deadline, and expected evidence.
Search plan	Versioned queries, providers, indexes, graph seeds, expansion, deduplication, stopping, budgets, and escalation.
Search result	Provider result plus query, rank, snippet, source identity, retrieval time, rights, quality, and access status.
Search product	Discovery set and coverage report; content becomes Evidence Object only through collection and admission.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-40-001	The Search Agent SHALL execute only against registered providers, indexes, domains, enterprise repositories, and graph surfaces approved for the task and customer domain.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-40-002	Search plans SHALL preserve original and expanded queries, languages, time, geography, source classes, provider selection, ranking, stopping, and known coverage limits.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-40-003	Search results SHALL remain untrusted discovery metadata until Collection and Evidence admission validate source, content, integrity, rights, classification, and custody.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-40-004	The agent SHALL diversify sources, preserve contradicting and minority evidence, and expose ranking, provider, language, and availability bias material to the task.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-40-005	Search history and provider logs SHALL follow tenant, purpose, classification, retention, residency, minimization, and customer-control policy.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Provider unavailable or rate-limited	Use approved alternatives or return explicit coverage loss and deferred scope.
Result source untrusted	Retain as discovery candidate with risk state; do not cite or admit without collection validation.

FAILURE CONDITION	REQUIRED BEHAVIOR
Query may disclose protected intent	Minimize, transform, route locally, or deny according to source and sovereignty policy.

Assurance evidence

- Search benchmarks measure source coverage, diversity, novelty, precision, recall, timeliness, and bias by domain and language.
- Query replay reproduces provider, index, version, expansion, filters, ranks, and result set where the source permits.
- Privacy tests inspect outbound queries, provider retention, logs, identifiers, and sensitive strategic intent.
- Adversarial tests cover SEO manipulation, poisoned results, deceptive domains, prompt injection, and coordinated source campaigns.

CHAPTER ACCEPTANCE AND INHERITANCE

Search Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 18, Ch. 21, Ch. 23, Ch. 29, Ch. 31, Ch. 43, Ch. 53, Ch. 54, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-013 · C-014 · C-035 · C-038 · C-040

41 Cleaning and Classification Agents

Cleaning and Classification Agents transform admitted evidence into normalized, segmented, quality-marked, policy-aware analytical inputs. They preserve raw evidence and transformation lineage and never overwrite source content or turn classification confidence into factual truth.

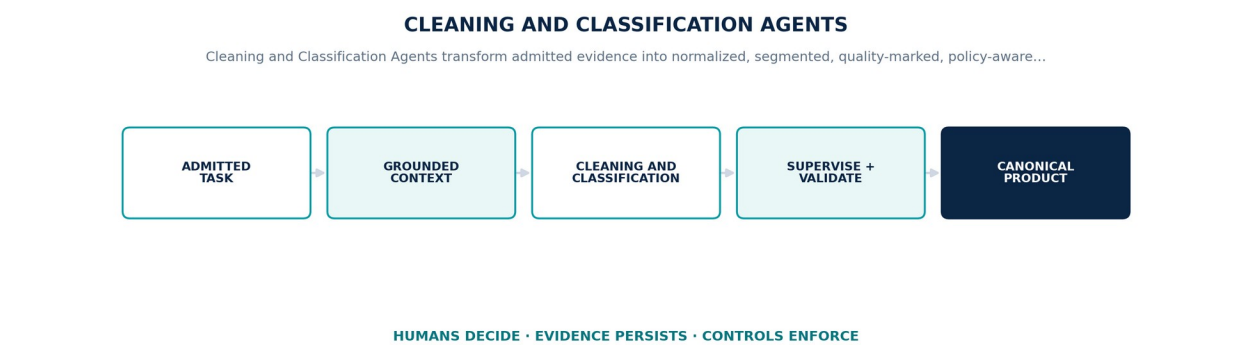


Figure 42. Cleaning and Classification Agents. The diagram organizes cleaning agent, classification agent, transformation plan, products as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Cleaning Agent	Decodes, parses, normalizes, segments, deduplicates, aligns units and languages, detects corruption, and records information loss.
Classification Agent	Assigns document, domain, topic, event, sensitivity, handling, quality, language, and routing labels with confidence and policy.
Transformation plan	Ordered versioned steps, models, rules, parameters, schemas, quality thresholds, and reversible references to raw evidence.
Products	Normalized Evidence views, segment manifests, quality findings, classification proposals, and review requests.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-41-001	Cleaning SHALL preserve immutable source evidence and SHALL record every decode, extraction, normalization, translation, segmentation, deduplication, unit conversion, redaction, and information-loss event.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-41-002	Classification SHALL distinguish content labels from security classification and SHALL route security, rights, or handling changes through authorized policy or data owners.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-41-003	Deduplication SHALL preserve source instances, publication and observation times, corrections, syndication, and evidentiary independence rather than deleting similar content.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-41-004	Model- or rule-generated labels SHALL carry method, version, confidence, uncertainty, applicable scope, and human-review conditions.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-41-005	Low-quality, corrupted, unsupported, or policy-sensitive evidence SHALL be quarantined or routed for review and SHALL NOT be silently normalized into valid input.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Parser or decoder fails	Preserve raw evidence and return explicit unsupported or partial extraction state.

FAILURE CONDITION	REQUIRED BEHAVIOR
Classification confidence low	Retain alternatives, apply restrictive routing where needed, and request review.
Transformation loses material content	Reject the derived view or mark its known loss and prohibit unsupported downstream use.

Assurance evidence

- Golden documents cover formats, encodings, languages, tables, images, archives, malformed content, and adversarial payloads.
- Round-trip and lineage tests trace every normalized segment to exact raw bytes and transformation versions.
- Classification evaluations measure accuracy, calibration, abstention, fairness, drift, and policy consequence by slice.
- Deduplication tests preserve corrections, source independence, temporal order, and coordinated-campaign evidence.

CHAPTER ACCEPTANCE AND INHERITANCE

Cleaning and Classification Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 27, Ch. 28, Ch. 29, Ch. 31, Ch. 32, Ch. 43, Ch. 46, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-012 · C-014 · C-035 · C-040 · C-045

42 Summarization Agent

The Summarization Agent produces audience- and task-specific condensed representations while preserving claim-to-evidence alignment, uncertainty, contradictions, temporal scope, source diversity, material caveats, and information-loss disclosure.



Figure 43. Summarization Agent. The diagram organizes summary contract, input manifest, summary product, forms as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Summary contract	Audience, purpose, scope, length, structure, required evidence classes, prohibited inference, citations, uncertainty, and quality.
Input manifest	Exact source objects, versions, segments, truth states, time, classifications, policies, and coverage.
Summary product	Typed claims and narrative with citations, omitted scope, contradictions, confidence, limitations, and generation lineage.
Forms	Extractive, abstractive, comparative, change-focused, multi-document, executive, technical, or briefing summary under one evidence contract.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-42-001	Every material statement in a summary SHALL bind supporting evidence or be explicitly labeled as synthesis, inference, assumption, or unresolved claim.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-42-002	Summarization SHALL preserve material contradictions, dissent, quantitative units, temporal qualifiers, negation, uncertainty, source authority, and known missing evidence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-42-003	Audience adaptation SHALL change language and detail but SHALL NOT remove decision-relevant caveats, inflate confidence, or alter institutional meaning.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-42-004	Compression SHALL expose input coverage, selection method, omitted material categories, length constraints, and information-loss risk.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-42-005	Summaries SHALL remain derived products linked to source versions and SHALL be invalidated or reviewed when material evidence is corrected, withdrawn, or reclassified.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Citation alignment fails	Reject the summary claim or product and preserve source and generation evidence.
Inputs exceed safe coverage	Segment or return coverage-limited state; do not imply complete synthesis.

FAILURE CONDITION	REQUIRED BEHAVIOR
Material contradiction omitted	Mark evaluation failure and withdraw or correct dependent products.

Assurance evidence

- Claim-level evaluations measure faithfulness, citation precision and recall, contradiction, uncertainty, temporal accuracy, and coverage.
- Adversarial fixtures cover negation, numerics, units, legal qualifiers, conflicting sources, deceptive prompts, and hidden instructions.
- Audience reviews verify consistent meaning across executive, analyst, operator, regulator, and accessible formats.
- Correction tests identify and refresh summaries dependent on changed evidence versions.

CHAPTER ACCEPTANCE AND INHERITANCE

Summarization Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 21, Ch. 27, Ch. 28, Ch. 30, Ch. 32, Ch. 43, Ch. 46, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-015 · C-035 · C-047

43 Named Entity Recognition and Identity Resolution Agents

The NER Agent detects mentions and attributes in evidence; the Identity Resolution Agent maps them to candidate canonical entities. The pair preserves text spans, source context, ontology, ambiguity, confidence, temporal validity, and stewardship rather than treating a recognized name as a resolved identity.

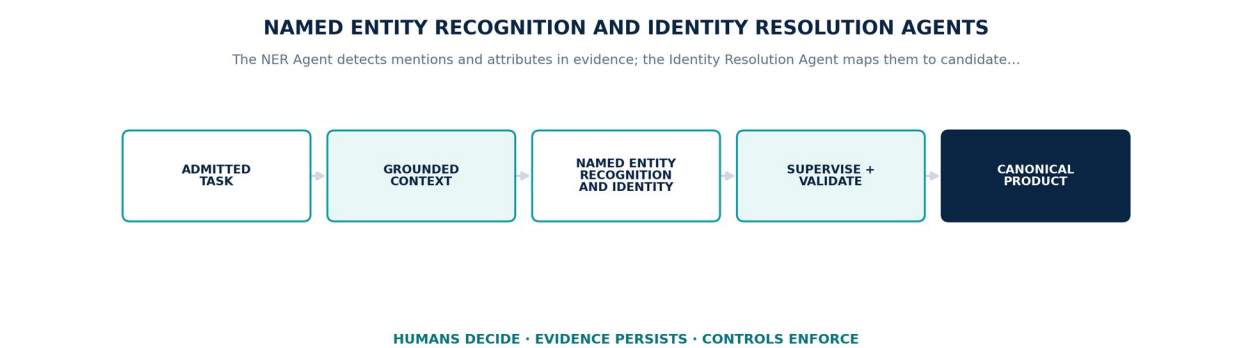


Figure 44. Named Entity Recognition and Identity Resolution Agents. The diagram organizes ner agent, identity resolution agent, evidence, product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
NER Agent	Detects typed mentions, aliases, identifiers, roles, attributes, locations, quantities, products, organizations, people, technologies, and domain entities.
Identity Resolution Agent	Generates and ranks canonical candidates, new-entity proposals, merge or split proposals, and unresolved outcomes.
Evidence	Exact source span or region, surrounding context, language, model and rule versions, confidence, and transformation lineage.
Product	Entity mention and resolution proposal consumed by Knowledge Graph stewardship; never an unreviewed canonical identity write.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-43-001	NER output SHALL bind every mention and attribute to exact evidence location, document version, language, ontology type, confidence, and transformation lineage.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-43-002	Identity resolution SHALL combine identifiers, names, aliases, attributes, relationships, time, geography, source authority, prior stewardship, and negative evidence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-43-003	The agents SHALL preserve unresolved and multiple-candidate state where evidence is insufficient and SHALL NOT force a single identity for workflow convenience.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-43-004	Canonical entity creation, merge, split, and retirement SHALL remain Knowledge Graph owner decisions through governed revision workflows.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-43-005	Resolution corrections SHALL propagate to relationships, memory, twins, forecasts, scenarios, simulations, decisions, warnings, evaluation data, and replay.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Mention boundaries ambiguous	Preserve alternative spans or lower confidence and request domain review where material.
No candidate meets threshold	Return unresolved or new-entity proposal; do not select the nearest identity.
Cross-tenant identity collision	Reject resolution and initiate isolation review; canonical identity never overrides tenant policy.

Assurance evidence

- Benchmarks measure mention and resolution precision, recall, calibration, abstention, novelty, merge, split, and fairness by class and language.
- Adversarial cases cover aliases, transliteration, shared names, shell entities, identifier reuse, mergers, and changing roles.
- Stewardship tests prove agents can propose but cannot commit canonical identity lifecycle changes.
- Correction fixtures trace identity changes through every dependent AI and canonical product.

CHAPTER ACCEPTANCE AND INHERITANCE

Named Entity Recognition and Identity Resolution Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 27, Ch. 28, Ch. 30, Ch. 32, Ch. 34, Ch. 43, Ch. 46.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-010 · C-011 · C-012 · C-035 · C-045

44 Relationship Agent

The Relationship Agent extracts and proposes typed, directed, temporal, evidence-bearing connections among canonical candidates and intelligence objects. It separates observed or claimed relationships from inferred associations and preserves contradiction, confidence, and ontology constraints.

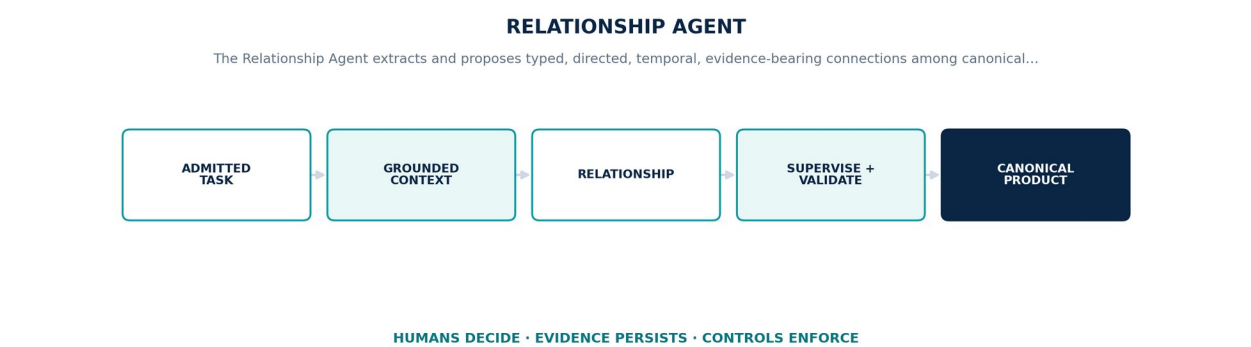


Figure 45. Relationship Agent. The diagram organizes input, candidate, reasoning modes, product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Input	Evidence spans or observations plus candidate endpoint identities, ontology, temporal context, truth state, and source authority.
Candidate	Predicate, direction, endpoints, qualifiers, event or valid time, confidence, evidence, method, and contradictory support.
Reasoning modes	Direct extraction, cross-sentence or cross-document association, event-mediated link, graph-supported inference, or negative relation.
Product	Relationship proposal and evidence package for Knowledge Graph validation and stewardship.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-44-001	Every relationship candidate SHALL bind exact endpoint candidates, direction, predicate, ontology version, temporal scope, truth state, confidence, and supporting evidence.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-44-002	The agent SHALL distinguish direct source assertion, observed co-occurrence, extracted event relation, model inference, graph inference, and analyst assessment.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-44-003	Missing or ambiguous endpoints SHALL remain unresolved and SHALL NOT be replaced by fabricated canonical identities.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-44-004	Material counter-evidence, negation, conditionality, hearsay, source conflict, and expired relationships SHALL remain visible in the candidate package.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-44-005	The agent SHALL NOT commit graph edges; canonical admission requires graph revision validation, policy, provenance, ontology, and stewardship controls.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Predicate unavailable	Use governed extension proposal or quarantine; do not create uncontrolled relationship types.
Endpoint resolution changes	Invalidate or recompute the candidate and propagate impact after canonical correction.

FAILURE CONDITION	REQUIRED BEHAVIOR
Evidence supports conflicting directions	Preserve alternatives and route adjudication rather than averaging.

Assurance evidence

- Relationship benchmarks measure type, direction, endpoints, temporal scope, negation, confidence, and evidence alignment.
- Ontology tests cover domain extension, version migration, deprecated predicates, and incompatible endpoint classes.
- Adversarial fixtures include indirect language, speculation, coordinated claims, tables, diagrams, and multilingual evidence.
- Graph admission tests prove no agent path bypasses revision validation and stewardship.

CHAPTER ACCEPTANCE AND INHERITANCE

Relationship Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 32, Ch. 34, Ch. 46.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-010 · C-011 · C-012 · C-017 · C-035

45 Knowledge Graph and Ontology Agents

The Knowledge Graph Agent assembles validated entity, event, claim, and relationship proposals into atomic graph revisions. The Ontology Steward Agent evaluates semantic changes, compatibility, migration, and domain extensions under human and governance authority.

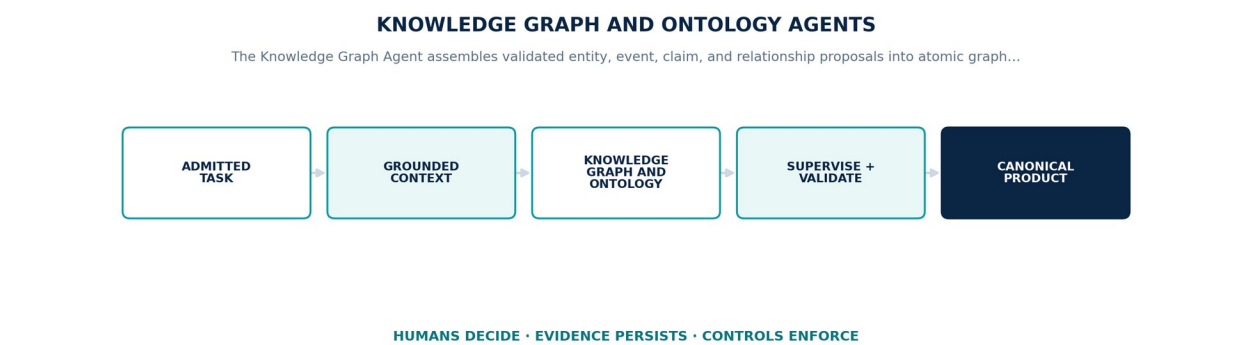


Figure 46. Knowledge Graph and Ontology Agents. The diagram organizes knowledge graph agent, ontology steward agent, graph revision, authority as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Knowledge Graph Agent	Builds revision proposals, validates referential and temporal integrity, detects duplicates and conflicts, and predicts downstream impact.
Ontology Steward Agent	Evaluates class, predicate, constraint, mapping, version, compatibility, migration, and stewardship proposals.
Graph revision	Atomic set of node, edge, ontology, provenance, temporal, truth-state, correction, and stewardship changes.
Authority	Knowledge Graph component and designated human stewards approve commits; agents prepare and validate proposals.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-45-001	Graph revision proposals SHALL identify exact source objects, endpoint versions, ontology, temporal scope, truth state, provenance, policy, confidence, and correction semantics.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-45-002	The Knowledge Graph Agent SHALL validate referential integrity, ontology constraints, duplicate identity, temporal consistency, contradiction preservation, policy, and impact before submission.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-45-003	Ontology changes SHALL include rationale, alternatives, domain authority, compatibility, data and query migration, model and agent impact, deployment parity, and rollback.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-45-004	Agents SHALL NOT approve their own ontology or graph changes where stewardship, policy, or consequence requires independent human authority.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-45-005	Committed revisions SHALL emit impact events and update retrieval, memory mappings, twins, forecasts, scenarios, simulations, decision packages, warnings, and evaluations through governed workflows.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Revision validation fails	Reject atomically and return precise conflicts; no partial graph commit.

FAILURE CONDITION	REQUIRED BEHAVIOR
Ontology compatibility uncertain	Keep proposal non-active and require migration analysis and steward review.
Downstream impact unresolved	Keep change workflow open and expose affected consumers and stale projections.

Assurance evidence

- Graph fixtures validate atomicity, identity, constraints, time, truth state, provenance, policy, contradiction, and correction.
- Ontology conformance suites cover additive, breaking, deprecated, mapped, multilingual, and domain-extension changes.
- Migration tests preserve historical queries, model features, agent tools, twin mappings, and Decision Replay.
- Impact tests identify all seeded downstream products and prevent false change closure.

CHAPTER ACCEPTANCE AND INHERITANCE

Knowledge Graph and Ontology Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 26, Ch. 28, Ch. 30, Ch. 34, Ch. 46, Ch. 48, Ch. 67, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-009 · C-011 · C-012 · C-017 · C-035 · C-038 · C-051

46 Reasoning, Evidence, Provenance, and Dissent Agents

The Reasoning Agent forms explicit hypotheses and assessments from governed evidence; the Evidence and Provenance Agent verifies support and lineage; the Dissent Agent searches for counter-evidence, alternative explanations, and failure conditions. None may collapse uncertainty into fact.



Figure 47. Reasoning, Evidence, Provenance, and Dissent Agents. The diagram organizes reasoning agent, evidence and provenance agent, dissent agent, assessment product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Reasoning Agent	Produces typed hypotheses, arguments, assessments, causal mechanisms, comparisons, and conclusions with assumptions and uncertainty.
Evidence and Provenance Agent	Validates claim support, citation alignment, source authority, transformation lineage, policy, time, and forward impact.
Dissent Agent	Actively challenges evidence selection, logic, assumptions, alternative hypotheses, model agreement, incentives, and missing information.
Assessment product	Question, conclusion, evidence, counter-evidence, method, assumptions, uncertainty, dissent, limitations, and accountable review.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-46-001	Reasoning SHALL operate over a pinned information boundary and SHALL distinguish deduction, rule application, statistical inference, analogy, causal hypothesis, simulation interpretation, and narrative synthesis.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-46-002	Every material conclusion SHALL expose supporting and contradicting evidence, assumptions, uncertainty, method, alternatives, and conditions under which it would change.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-46-003	The Evidence and Provenance Agent SHALL verify claim-level support independently from the producing reasoning path and SHALL reject fabricated or unresolved citations.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-46-004	The Dissent Agent SHALL receive sufficient independent model, retrieval, and context diversity to challenge shared failure modes and SHALL preserve unresolved dissent in the product.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-46-005	Reasoning products SHALL remain assessed or inferred truth states and SHALL require owning-component and human review according to consequence before decision-active use.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Evidence insufficient	Abstain or return hypothesis state with explicit missing evidence and next information value.
Reasoners agree but share one dependency	Expose correlated evidence and method risk; consensus alone does not increase confidence.
Dissent unresolved	Preserve alternatives and route accountable adjudication or scenario treatment.

Assurance evidence

- Argument fixtures verify evidence, counter-evidence, assumptions, method, uncertainty, and truth-state labeling.
- Adversarial cases include motivated reasoning, base-rate neglect, spurious correlation, circular citations, and coordinated sources.
- Diversity tests measure correlated model, context, retrieval, and prompt failures across agent roles.
- Human review evaluates comprehensibility, contestability, calibration, usefulness, and change conditions.

CHAPTER ACCEPTANCE AND INHERITANCE

Reasoning, Evidence, Provenance, and Dissent Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 27, Ch. 28, Ch. 32, Ch. 43, Ch. 45, Ch. 46, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-010 · C-012 · C-021 · C-022 · C-024

47 Governance, Evaluation, and Learning Agents

Governance, Evaluation, and Learning Agents prepare control evidence and change proposals. They monitor conformance, execute approved evaluation profiles, connect outcomes to prior AI products, and propose improvements while remaining unable to approve their own risk or release production changes.



Figure 48. Governance, Evaluation, and Learning Agents. The diagram organizes governance agent, evaluation agent, learning agent, authority as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Governance Agent	Monitors artifact status, authority, policy, exceptions, marketplace, documentation, reviews, and expiry and prepares governance cases.
Evaluation Agent	Runs approved suites, metrics, slices, judges, challenge sets, and reproducibility checks and produces evidence, not approval.
Learning Agent	Links outcomes, corrections, feedback, overrides, incidents, and drift to lesson proposals and candidate changes.
Authority	Named governance, risk, evaluation, product, data, security, and release owners decide fitness, exceptions, and production change.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-47-001	Governance Agents SHALL use current registries and policies and SHALL surface missing, expired, conflicting, or non-conformant state without issuing independent approval.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-47-002	Evaluation Agents SHALL execute immutable profiles against exact artifacts and populations and SHALL preserve raw results, runner, judge, environment, seed, exclusions, and uncertainty.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-47-003	Learning Agents SHALL create evidence-backed lesson proposals and SHALL NOT directly modify models, prompts, agents, tools, policies, thresholds, retrieval, or memory authority.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-47-004	Agents evaluating or governing another artifact SHALL be independently identified, capability-scoped, tested for manipulation, and prohibited from accepting instructions from the artifact under evaluation.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-47-005	Every resulting fitness decision, exception, learned change, release, rollback, or rejection SHALL be made by its named authority and linked to the agent-prepared evidence.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Governance data incomplete	Return indeterminate case and block unsupported approval or closure.

FAILURE CONDITION	REQUIRED BEHAVIOR
Evaluator compromised or manipulated	Invalidate affected evidence, quarantine the evaluator, and re-run with independent controls.
Learning proposal lacks causal support	Keep as hypothesis and prohibit production change until evaluation and authority close.

Assurance evidence

- Separation tests prove agents cannot write fitness decisions, risk acceptance, exceptions, or release approvals.
- Evaluation reproducibility tests independently re-run exact manifests and compare within declared tolerance.
- Manipulation tests expose evaluator agents to prompt injection, gaming, leakage, biased judges, and artifact collusion.
- Learning fixtures trace outcome through proposal, evaluation, human approval, release, monitoring, and rollback.

CHAPTER ACCEPTANCE AND INHERITANCE

Governance, Evaluation, and Learning Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 13, Ch. 46, Ch. 48, Ch. 55, Ch. 56, Ch. 63, Ch. 68, Ch. 72.

CONSTITUTIONAL INHERITANCE / C-004 · C-028 · C-031 · C-035 · C-044 · C-045 · C-051

PART VII

Foresight, Decision, and Domain Agents

Prediction, scenarios, simulation, decisions, risk, opportunity, competitors, supply chains, geopolitical, technology, cyber, financial, executive, reporting, and warnings.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

48 Prediction Agent

The Prediction Agent produces calibrated probabilistic Forecast Packages for the canonical 30-day, 90-day, 180-day, one-year, three-year, and five-year horizons. It separates target definition, data, model, assumptions, distribution, drivers, uncertainty, sensitivity, validation, and failure conditions.

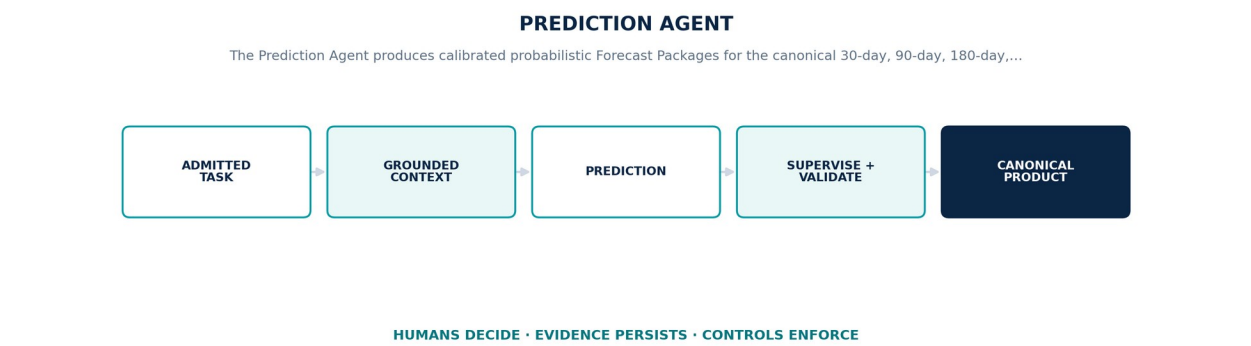


Figure 49. Prediction Agent. The diagram organizes forecast task, method, forecast package, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Forecast task	Target, population, outcome definition, horizon, issue time, cutoff, context, decision use, refresh, and consequence.
Method	Approved statistical, machine-learning, causal, rules, market-implied, expert, ensemble, or hybrid model composition.
Forecast Package	Probability distribution, point summaries, intervals, drivers, calibration, sensitivity, scenarios, lineage, expiry, and failure conditions.
Lifecycle	Requested, produced, reviewed, active, refreshed, expired, challenged, withdrawn, or superseded.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-48-001	Forecasts SHALL use only the six canonical horizons unless a domain specification defines an additional governed operational horizon without replacing them.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-48-002	The agent SHALL pin target definition, data cutoff, context, graph and twin revisions, model and ensemble versions, assumptions, features, parameters, and evaluation profile.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-48-003	Forecast products SHALL express distributions and calibrated uncertainty and SHALL NOT present point estimates or model confidence as certainty.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-48-004	Drivers, sensitivity, subgroup or geographic variation, out-of-distribution state, missing data, structural breaks, and failure conditions SHALL be visible in the package and explanation.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-48-005	Forecast fitness, drift, realized outcomes, corrections, and model withdrawal SHALL update product status and downstream scenarios, warnings, simulations, decisions, and briefings.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Target or cutoff ambiguous	Reject task and require explicit definition and temporal boundary.
Model outside operating envelope	Abstain, constrain, or route to governed alternate and expose the failed condition.

FAILURE CONDITION	REQUIRED BEHAVIOR
Calibration degrades	Constrain or withdraw affected task populations and initiate re-evaluation and impact review.

Assurance evidence

- Backtests preserve historical model, data, context, policy, and outcome boundaries without leakage.
- Evaluation measures calibration, discrimination, sharpness, coverage, stability, bias, and decision utility by horizon and slice.
- Stress tests cover missing data, structural break, regime shift, adversarial signals, source corrections, and extreme events.
- Outcome monitoring links realized events to exact forecast versions and governed learning proposals.

CHAPTER ACCEPTANCE AND INHERITANCE

Prediction Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 26, Ch. 27, Ch. 28, Ch. 30, Ch. 35, Ch. 36, Ch. 45, Ch. 46, Ch. 48, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-020 · C-021 · C-025 · C-030 · C-045 · C-049

49 Scenario Agent

The Scenario Agent creates and maintains multiple coherent plausible future branches. It combines baseline state, actors, drivers, assumptions, critical uncertainties, signposts, interventions, impacts, and review without assigning destiny to one branch or disguising scenarios as probability forecasts.

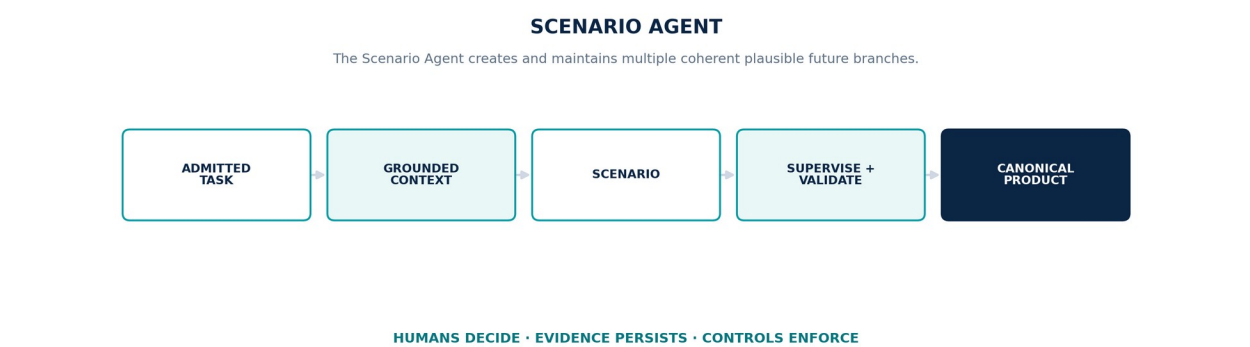


Figure 50. Scenario Agent. The diagram organizes scenario task, scenario branch, portfolio, monitoring as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Scenario task	Question, objectives, baseline, horizon, scope, actors, drivers, uncertainties, branch policy, and decision use.
Scenario Branch	Versioned future with assumptions, causal or narrative logic, events, actor behavior, signposts, interventions, impacts, and uncertainty.
Portfolio	Baseline, upside, downside, disruption, and governed custom branches with diversity, coherence, relevance, and coverage.
Monitoring	Signposts, thresholds, evidence, branch relevance, review cadence, retirement, and promotion to simulation or decision.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-49-001	The Scenario Agent SHALL preserve multiple coherent branches and SHALL NOT collapse a forecast distribution, model sample, or favored narrative into a single future.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-49-002	Every branch SHALL bind baseline graph and twin state, assumptions, drivers, actors, mechanisms, uncertainties, signposts, interventions, impacts, evidence, and versions.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-49-003	Scenario probability, where used, SHALL be separately governed and SHALL NOT be inferred from narrative plausibility or agent preference.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-49-004	Branches SHALL be evaluated for internal coherence, distinctiveness, relevance, coverage, stereotype or political bias, and sensitivity to material assumptions.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-49-005	Observed signposts and changed evidence SHALL update branch relevance through versioned review while preserving prior scenario and decision context.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Branches converge materially	Revise portfolio diversity or expose insufficient alternative coverage.
Scenario logic incoherent	Mark failed coherence and prohibit promotion to simulation or decision use.

FAILURE CONDITION	REQUIRED BEHAVIOR
Signpost evidence ambiguous	Preserve uncertainty and require review; do not automatically promote one branch.

Assurance evidence

- Scenario fixtures verify baseline, assumptions, mechanisms, actors, signposts, interventions, impacts, and revision lineage.
- Portfolio tests measure diversity, coherence, coverage, relevance, and dependence on shared assumptions.
- Bias reviews challenge actor modeling, cultural assumptions, political framing, and omitted plausible branches.
- Historical signpost replay tests branch monitoring without hindsight contamination.

CHAPTER ACCEPTANCE AND INHERITANCE

Scenario Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 26, Ch. 27, Ch. 28, Ch. 35, Ch. 36, Ch. 37, Ch. 45, Ch. 46.

CONSTITUTIONAL INHERITANCE / C-005 · C-019 · C-020 · C-021 · C-022 · C-024 · C-026

50 Simulation Agent

The Simulation Agent prepares, executes, validates, and interprets reproducible experiments over exact twin, scenario, model, policy, environment, intervention, constraint, and seed versions. It distinguishes simulated outcomes from observed reality and from probability forecasts.

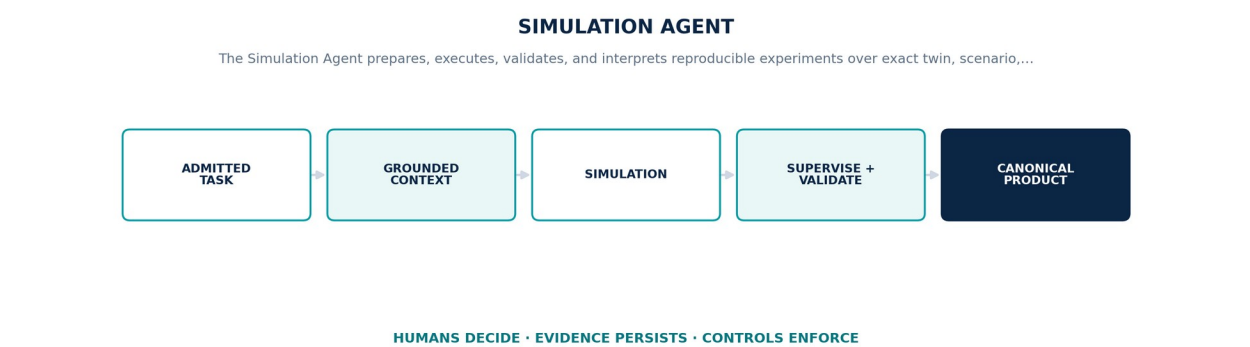


Figure 51. Simulation Agent. The diagram organizes experiment manifest, execution, simulation run, interpretation as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Experiment manifest	Objective, initial state, twin and scenario, models, rules, agents, interventions, constraints, environment, seeds, outputs, budgets, and validation.
Execution	Isolated run or ensemble with exact artifacts, scheduler, resource, random state, checkpoints, telemetry, and failure state.
Simulation Run	Outputs, distributions, trajectories, impacts, sensitivity, uncertainty, validity, limitations, resource, and reproducibility evidence.
Interpretation	Assumption-bound explanation of results, not a claim that the world will behave identically.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-50-001	The agent SHALL validate complete experiment manifests and SHALL reject unresolved artifact, state, constraint, policy, budget, or expected-output dependencies.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-50-002	Simulation execution SHALL occur in isolated synthetic branches and SHALL never write outcomes into observed twin or canonical world state.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-50-003	Runs SHALL preserve seeds, nondeterministic components, environment, scheduler, model, agent, policy, configuration, data, and hardware-relevant versions required for reproducibility.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-50-004	Simulation products SHALL expose validity domain, assumptions, uncertainty, sensitivity, convergence, rare events, model discrepancy, and validation against observed or benchmark behavior.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-50-005	Interpretation SHALL preserve alternative explanations and SHALL not convert simulated frequency into calibrated real-world probability without an approved mapping and evaluation.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Manifest incomplete	Reject before resource allocation and identify missing dependencies.
Run non-reproducible	Mark result unfit, preserve evidence, and investigate nondeterministic or environment drift.

FAILURE CONDITION	REQUIRED BEHAVIOR
Validation fails	Invalidate dependent interpretation and decision packages and propagate impact.

Assurance evidence

- Reproduction fixtures rerun exact manifests across declared environments and compare within stated tolerances.
- Sensitivity tests vary assumptions, initial state, models, constraints, interventions, seeds, and structural parameters.
- Branch-isolation tests attempt writes from simulation into observed state, memory, graph, and production tools.
- Validation suites compare simulations with historical, controlled, synthetic, and domain-expert benchmarks.

CHAPTER ACCEPTANCE AND INHERITANCE

Simulation Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 22, Ch. 26, Ch. 28, Ch. 35, Ch. 37, Ch. 38, Ch. 45, Ch. 46, Ch. 58.

CONSTITUTIONAL INHERITANCE / C-005 · C-018 · C-019 · C-022 · C-023 · C-024 · C-025

51 Decision Agent

The Decision Agent assembles a complete Decision Package for human authority. It frames the question, connects objectives and constraints, develops alternatives, integrates evidence and foresight, exposes trade-offs and dissent, and proposes monitoring and commitments without making the final decision.

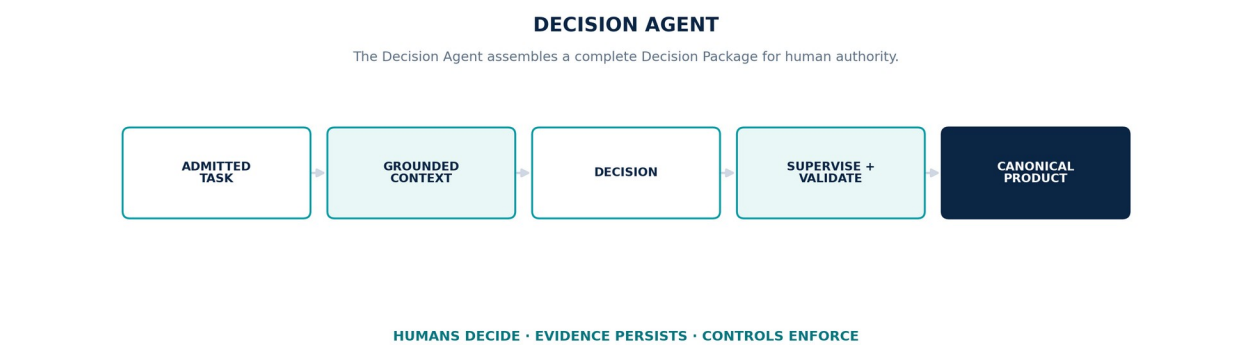


Figure 52. Decision Agent. The diagram organizes decision frame, options, decision package, authority boundary as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Decision frame	Question, accountable human authority, scope, objectives, constraints, stakeholders, deadline, reversibility, and consequence.
Options	Status quo and actionable alternatives with assumptions, dependencies, resources, impacts, risks, opportunities, reversibility, and information gaps.
Decision Package	Evidence, assessments, forecasts, scenarios, simulations, uncertainty, trade-offs, dissent, policy, recommendation, explanation, and monitoring.
Authority boundary	Agent may recommend and prepare; named human approves, rejects, changes, defers, delegates, abstains, or reopens.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-51-001	The Decision Agent SHALL build packages against explicit Strategy Graph objectives, measures, constraints, capabilities, risks, opportunities, and stakeholder obligations.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-51-002	Packages SHALL include status quo, materially distinct alternatives, evidence, counter-evidence, uncertainty, forecasts, scenarios, simulations, trade-offs, dissent, missing information, reversibility, and monitoring.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-51-003	Recommendations SHALL separate value judgments, policy constraints, analytical assumptions, model or agent outputs, and accountable human preferences.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-51-004	The agent SHALL NOT approve the recommendation, create an institutional commitment, dispatch an approval-required effect, or suppress material dissent.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-51-005	The final human record SHALL bind the exact package version and create commitments, monitoring, review, reopening, and Decision Replay state through owning components.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Package incomplete	Remain draft and identify missing evidence, option, authority, policy, or monitoring elements.

FAILURE CONDITION	REQUIRED BEHAVIOR
Objectives conflict	Expose the conflict and require human priority or governance resolution; do not hide it in an aggregate score.
Recommendation changes after review begins	Invalidate the pending gate and require human review of the new package version.

Assurance evidence

- Completeness fixtures verify every required package element and prohibit single-option or dissent-free closure where material alternatives exist.
- Counterfactual tests change objectives, constraints, forecasts, scenarios, and assumptions and verify coherent recommendation changes.
- Human review measures explanation, contestability, cognitive load, uncertainty communication, and decision usefulness.
- Replay reconstructs exact package, AI executions, visible evidence, human authority, rationale, commitments, and later outcomes.

CHAPTER ACCEPTANCE AND INHERITANCE

Decision Agent is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 24, Ch. 27, Ch. 28, Ch. 39, Ch. 40, Ch. 42, Ch. 43, Ch. 48, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-017 · C-024 · C-025 · C-026 · C-027

52 Risk and Opportunity Agents

Risk and Opportunity Agents maintain a symmetric view of strategic exposure. They connect drivers, evidence, probability, consequence, timing, affected objectives, capabilities, scenarios, responses, owners, indicators, and decision windows without turning one opaque score into institutional truth.



Figure 53. Risk and Opportunity Agents. The diagram organizes risk agent, opportunity agent, exposure product, portfolio as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Risk Agent	Identifies adverse exposure, mechanisms, likelihood, impact, velocity, controls, residual risk, indicators, options, and owners.
Opportunity Agent	Identifies beneficial exposure, mechanisms, probability, value, timing, capabilities, competition, investment, options, and owners.
Exposure product	Versioned Risk / Opportunity object linked to Strategy Graph, evidence, forecasts, scenarios, twins, warnings, decisions, and responses.
Portfolio	Deduplicated and connected exposures with dependencies, concentration, optionality, correlation, and transparent prioritization.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-52-001	Both agents SHALL apply symmetric evidence, uncertainty, temporal, ownership, monitoring, and closure standards while preserving different consequence and option semantics.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-52-002	Exposure products SHALL state mechanism, drivers, probability or plausibility, consequence or value, horizon, affected objectives, dependencies, indicators, response window, options, owner, and evidence.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-52-003	Prioritization SHALL decompose probability, impact, urgency, reversibility, exposure, strategic relevance, control effectiveness, and information value rather than publish an opaque global rank.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-52-004	Risk and Opportunity Agents SHALL identify shared drivers, correlations, second-order effects, portfolio concentration, and trade-offs between mitigation and opportunity capture.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-52-005	Material exposures SHALL route to accountable owners, warning or decision workflows, and outcome monitoring with explicit acknowledgement and closure criteria.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner unresolved	Keep as candidate and escalate routing; do not present as governed actionable exposure.

FAILURE CONDITION	REQUIRED BEHAVIOR
Probability unavailable	Use scenario or plausibility state and do not manufacture numeric likelihood.
Exposure duplicated	Consolidate presentation while preserving distinct causes, objectives, evidence, geographies, and response windows.

Assurance evidence

- Historical evaluation measures identification, calibration, timeliness, consequence, value capture, false positives, misses, and owner action.
- Portfolio tests expose common drivers, correlation, concentration, cascade, hedge, and option interactions.
- Score tests prove every priority and Strategic Health contribution decomposes to governed dimensions and evidence.
- Outcome review links response, decision, commitment, result, external change, and attribution limits.

CHAPTER ACCEPTANCE AND INHERITANCE

Risk and Opportunity Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 27, Ch. 34, Ch. 36, Ch. 37, Ch. 39, Ch. 40, Ch. 47, Ch. 48, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-017 · C-020 · C-022 · C-026 · C-032 · C-034

53 Competitor and Supply Chain Agents

Competitor and Supply Chain Agents maintain evidence-grounded models of external organizations, products, capabilities, moves, dependencies, tiers, flows, constraints, exposures, and alternatives. They operate over graph and twin state and preserve legal, ethical, source, and confidence boundaries.

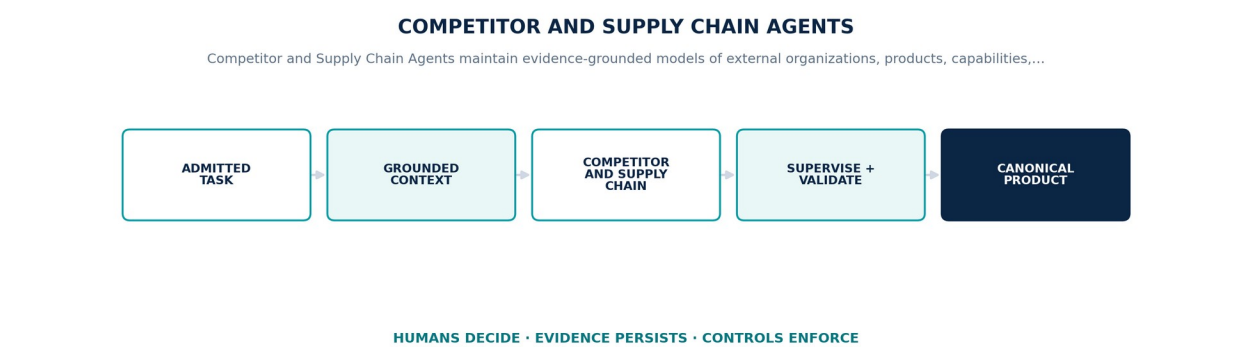


Figure 54. Competitor and Supply Chain Agents. The diagram organizes competitor agent, supply chain agent, products, grounding as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Competitor Agent	Tracks identity, strategy signals, offerings, technologies, capabilities, talent, partnerships, filings, patents, moves, scenarios, and comparative position.
Supply Chain Agent	Maps multi-tier entities, sites, products, materials, logistics, contracts, capacity, lead time, provenance, disruption, and alternatives.
Products	Competitor assessments, supply-chain twin proposals, dependencies, exposures, alternatives, signposts, warnings, and decision inputs.
Grounding	Evidence Objects, Knowledge Graph, Strategy Graph, company and supply-chain twins, forecasts, scenarios, and approved enterprise systems.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-53-001	These agents SHALL distinguish observed public or enterprise evidence, source claims, inferred capability, analyst assessment, scenario assumption, and simulated state.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-53-002	Competitor analysis SHALL avoid unsupported intent attribution and SHALL expose source bias, alternative explanations, comparability limits, and uncertainty.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-53-003	Supply-chain mapping SHALL preserve tier, site, product, material, route, time, capacity, ownership, contractual, geographic, and provenance uncertainty rather than infer false completeness.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-53-004	Sensitive competitive and supplier data SHALL follow rights, purpose, antitrust, confidentiality, privacy, sovereignty, and customer-control policy.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-53-005	Material moves or disruptions SHALL link affected objectives, products, twins, forecasts, scenarios, exposures, response windows, owners, and decision workflows.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Entity or ownership uncertain	Preserve candidate structures and confidence; do not merge suppliers or competitors for convenience.

Assurance evidence

- Benchmarks validate entity, event, product, capability, supplier, site, tier, route, and dependency extraction.
- Coverage tests quantify unknown suppliers, sites, tiers, regions, volumes, and contractual dependencies.
- Scenario drills trace competitor moves and supply disruptions through twins, forecasts, options, decisions, and outcomes.
- Policy tests cover confidential data, antitrust, purpose, sharing, export, and customer-domain isolation.

CHAPTER ACCEPTANCE AND INHERITANCE

Competitor and Supply Chain Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 34, Ch. 35, Ch. 36, Ch. 37, Ch. 39, Ch. 47, Ch. 50, Ch. 54, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-012 · C-018 · C-022 · C-032 · C-040

54 Geopolitical and Technology Agents

Geopolitical and Technology Agents connect political, regulatory, security, economic, scientific, patent, standards, capability, adoption, and supply signals to organizational objectives and exposures. They preserve plurality, source context, uncertainty, and region-specific interpretation.

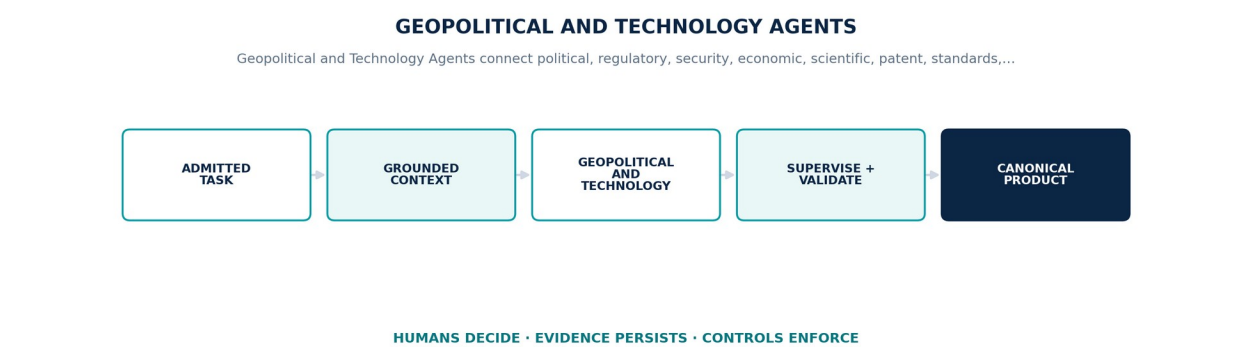


Figure 55. Geopolitical and Technology Agents. The diagram organizes geopolitical agent, technology agent, products, grounding as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Geopolitical Agent	Tracks actors, institutions, policies, conflicts, alliances, sanctions, elections, trade, security, climate, narratives, and regional dynamics.
Technology Agent	Tracks research, patents, standards, products, infrastructure, talent, investment, maturity, adoption, capability, dependencies, and constraints.
Products	Assessments, trend and capability maps, scenarios, signposts, risks, opportunities, warnings, technology-roadmap inputs, and decision evidence.
Grounding	Countries, governments, people, technologies, organizations, events, regulations, markets, patents, papers, graph relationships, and twins.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-54-001	Geopolitical analysis SHALL preserve competing actor perspectives, source incentives, propaganda risk, regional context, uncertainty, and alternative causal mechanisms.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-54-002	Technology analysis SHALL distinguish scientific result, prototype, product claim, demonstrated capability, scalable system, adoption, and strategic impact.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-54-003	Maturity, readiness, diffusion, and strategic relevance claims SHALL decompose to transparent evidence, criteria, uncertainty, dependencies, and failure conditions.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-54-004	Sanctioned, controlled, sensitive, or dual-use technology data and analysis SHALL follow export, sovereignty, security, ethical, and customer policy.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-54-005	Material geopolitical or technology change SHALL map to affected graph entities, twins, objectives, capabilities, exposures, forecasts, scenarios, decisions, and warnings.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Source environment manipulated	Increase source diversity and uncertainty, preserve dissent, and restrict unsupported urgency.

FAILURE CONDITION	REQUIRED BEHAVIOR
Technology claim cannot be independently verified	Retain as claim or hypothesis and prohibit capability admission.
Regional interpretation conflicts	Preserve localized assessments and route accountable adjudication rather than averaging.

Assurance evidence

- Historical cases evaluate event detection, actor modeling, source diversity, timeliness, alternative hypotheses, and forecast calibration.
- Technology benchmarks trace papers, patents, products, standards, deployments, adoption, and realized capabilities.
- Adversarial exercises cover propaganda, coordinated influence, fabricated research, patent noise, hype cycles, and strategic deception.
- Impact tests connect seeded events and technologies to objectives, exposures, decisions, and warning routes.

CHAPTER ACCEPTANCE AND INHERITANCE

Geopolitical and Technology Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 27, Ch. 28, Ch. 34, Ch. 35, Ch. 36, Ch. 37, Ch. 47, Ch. 50, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-012 · C-013 · C-022 · C-032 · C-035

55 Cyber and Financial Agents

Cyber and Financial Agents analyze fast-moving, high-consequence technical and market domains under strict evidence, security, timeliness, regulatory, and human-authority controls. They prepare assessments, forecasts, warnings, and options but do not execute trades, cyber actions, or final institutional decisions without governed authorization.

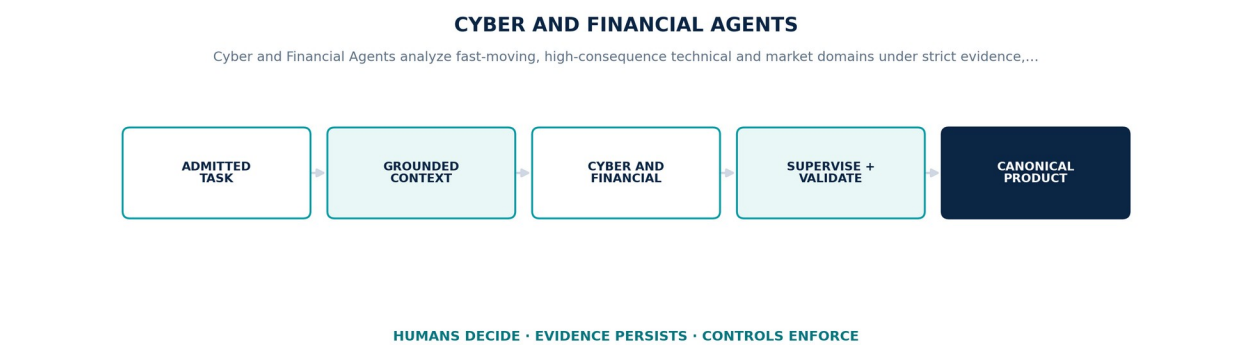


Figure 56. Cyber and Financial Agents. The diagram organizes cyber agent, financial agent, products, action boundary as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Cyber Agent	Tracks assets, identities, vulnerabilities, threats, campaigns, controls, dependencies, telemetry, incidents, exposure, response, and recovery.
Financial Agent	Tracks markets, instruments, filings, fundamentals, liquidity, credit, rates, exposures, transactions, scenarios, controls, and regulatory obligations.
Products	Cyber or financial assessments, exposures, forecasts, scenarios, warnings, response options, decision packages, and outcome records.
Action boundary	Analysis and recommendation are separate from scanning, containment, disclosure, payment, order, transaction, or other institutional effect.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-55-001	Cyber and Financial Agents SHALL operate under task-specific data, market, regulatory, legal, security, confidentiality, and side-effect policies appropriate to consequence.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-55-002	Cyber findings SHALL bind affected assets, identities, evidence, telemetry, vulnerability or threat references, confidence, exploitability, controls, exposure, time, and response window.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-55-003	Financial products SHALL distinguish observed price or filing data, derived metrics, estimates, forecasts, scenarios, assumptions, model risk, liquidity, and regulatory context.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-55-004	The agents SHALL NOT perform offensive cyber action, uncontrolled scanning, disclosure, trade, transfer, payment, account change, or other consequential effect without registered tools, exact human approval, and authorized workflow.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-55-005	Market, cyber, model, telemetry, and source latency or outage SHALL be visible in product state and SHALL constrain warnings, recommendations, and decisions.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Market or telemetry feed stale	Expose last valid time and prohibit unsupported real-time claims or actions.
Potential active compromise	Escalate through incident authority, restrict agent capabilities, and preserve evidence independently.
Regulatory or action authority unclear	Stop effect preparation and require accountable legal, compliance, security, or financial approval.

Assurance evidence

- Cyber exercises cover detection, false positives, adversarial telemetry, compromised tools, containment proposals, and evidence custody.
- Financial evaluations cover temporal leakage, survivorship, look-ahead bias, calibration, stress, liquidity, and regime change.
- Side-effect tests prove analysis agents cannot invoke scanning, response, trading, payment, or disclosure paths without exact grants and gates.
- Incident and market replay reconstructs source, model, agent, policy, decision, action, and outcome timelines.

CHAPTER ACCEPTANCE AND INHERITANCE

Cyber and Financial Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 21, Ch. 26, Ch. 27, Ch. 36, Ch. 39, Ch. 42, Ch. 44, Ch. 47, Ch. 53, Ch. 55, Ch. 63.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-011 · C-028 · C-035 · C-039 · C-050

56 Executive Briefing and Reporting Agents

Executive Briefing and Reporting Agents render governed strategic products for specific audiences and review cadences. They preserve source evidence, changes, objectives, uncertainty, dissent, decisions, commitments, owners, and required actions and never become a separate executive truth layer.



Figure 57. Executive Briefing and Reporting Agents. The diagram organizes executive briefing agent, reporting agent, audience contract, publication product as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Executive Briefing Agent	Creates live decision- and attention-oriented briefings around objectives, material change, scenarios, decisions, commitments, and outcomes.
Reporting Agent	Produces scheduled, event-driven, regulatory, operational, analytical, and exportable reports from governed canonical products.
Audience contract	Roles, purpose, classification, detail, accessibility, locale, cadence, channels, acknowledgement, and action expectations.
Publication product	Versioned briefing or report with evidence, as-of time, changes, uncertainty, policy, owners, actions, corrections, and expiration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-56-001	Agents SHALL render only admitted canonical products and policy-filtered projections and SHALL NOT create untracked executive facts, decisions, commitments, or scores.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-56-002	Every briefing or report SHALL state audience, purpose, as-of time, data and product freshness, evidence, changes, uncertainty, dissent, omissions, owners, required action, and expiry.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-56-003	Audience compression SHALL retain material caveats, alternative futures, decision status, commitment state, and source-to-claim traceability.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-56-004	Strategic Health Score and other roll-ups SHALL be decomposable to governed objectives, measures, exposures, capabilities, commitments, outcomes, quality, and evidence.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-56-005	Correction, reclassification, withdrawal, decision change, and commitment change SHALL version affected publications and notify accountable recipients according to materiality.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Required source product stale	Expose stale state or withhold the section; do not present prior content as current.

FAILURE CONDITION	REQUIRED BEHAVIOR
Audience policy cannot be satisfied	Deny publication or produce explicit policy-safe omission with accountable owner.
Report delivery unknown	Reconcile recipient receipt before claiming publication or acknowledgement.

Assurance evidence

- Publication fixtures compare source products and rendered claims, values, uncertainty, actions, and citations.
- Audience tests cover executive, analyst, operator, regulator, board, partner, accessible, multilingual, and restricted variants.
- Score tests decompose every roll-up and prevent hidden factors, stale inputs, or invalid comparisons.
- Correction tests update publications, recipients, acknowledgements, and downstream decisions with complete audit.

CHAPTER ACCEPTANCE AND INHERITANCE

Executive Briefing and Reporting Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 24, Ch. 27, Ch. 28, Ch. 34, Ch. 40, Ch. 47, Ch. 48, Ch. 55, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-025 · C-027 · C-034 · C-047 · C-048

57 Weak Signal and Early Warning Agents

Weak Signal and Early Warning Agents convert low-maturity change into governed attention. They combine novelty, mechanism, evidence, graph and twin exposure, forecasts, scenarios, objectives, consequence, urgency, reversibility, and information value without inflating confidence or flooding operators.

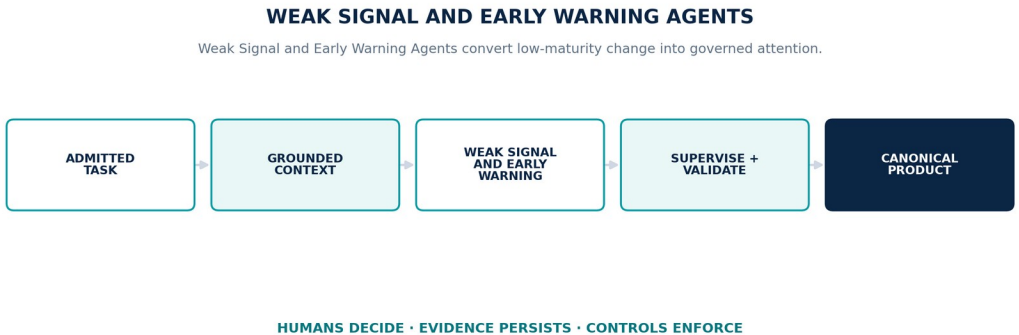


Figure 58. Weak Signal and Early Warning Agents. The diagram organizes weak signal agent, early warning agent, warning product, lifecycle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Weak Signal Agent	Detects novel or changing patterns, clusters supporting evidence, proposes mechanisms, estimates maturity, and maintains monitored candidates.
Early Warning Agent	Evaluates materiality, deduplicates and prioritizes candidates, resolves accountable owners, issues Warning Products, and tracks lifecycle.
Warning Product	Change, mechanism, evidence, counter-evidence, uncertainty, consequence, objectives, response window, owner, options, and closure.
Lifecycle	Candidate, monitored, issued, acknowledged, investigating, acted, dismissed, expired, resolved, or superseded.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-57-001	Weak-signal candidates SHALL preserve evidence, counter-evidence, novelty basis, mechanism, affected entities, uncertainty, maturity, monitoring plan, and conditions for escalation or dismissal.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-57-002	Materiality SHALL decompose consequence, probability or plausibility, urgency, reversibility, exposure, strategic relevance, information value, owner, and response window.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-57-003	Deduplication SHALL consolidate presentation without erasing distinct causes, geographies, objectives, time horizons, evidence, uncertainty, or response requirements.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-57-004	Every issued Warning Product SHALL have an accountable human or role, acknowledgement deadline, escalation path, response options, review state, and closure criteria.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-57-005	False, late, missed, duplicated, dismissed, useful, and outcome-linked warnings SHALL enter evaluation and governed learning with exact decision and context history.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Signal evidence too weak	Remain monitored with explicit information needs; do not manufacture urgency.

FAILURE CONDITION	REQUIRED BEHAVIOR
Owner cannot be resolved	Do not issue as actionable and escalate the routing failure to the Executive Operating System.
Warning storm	Apply transparent grouping and priority while preserving every underlying candidate and urgent distinct exception.

Assurance evidence

- Historical replay measures precision, recall, timeliness, calibration, actionability, utility, acknowledgement, and outcome by slice.
- Synthetic storms test deduplication, suppression, grouping, priority, owner routing, escalation, and critical-signal preservation.
- Mechanism reviews verify warnings explain how evidence could affect objectives rather than reporting correlation alone.
- Closure tests require evidence, action or dismissal rationale, outcome, residual monitoring, and learning disposition.

CHAPTER ACCEPTANCE AND INHERITANCE

Weak Signal and Early Warning Agents is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 34, Ch. 36, Ch. 37, Ch. 39, Ch. 40, Ch. 47, Ch. 48, Ch. 57, Ch. 62.

PART VIII

AI Trust, Safety, and Explainability

Risk taxonomy, action safety, instruction security, privacy, model and agent safety, uncertainty, explanation, and audit.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

58 AI Risk Taxonomy and Consequence Classes

AI risk in The Eye is classified by the harm that a capability can enable, amplify, conceal, or fail to prevent. Consequence classification controls permissible models, agents, data, tools, autonomy, human gates, evaluation depth, monitoring, release authority, and recovery obligations for each task and product use.



Figure 59. AI Risk Taxonomy and Consequence Classes. The diagram organizes risk dimensions, consequence classes, classification object, control principle as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Risk dimensions	Decision harm, physical and cyber effect, financial exposure, rights and privacy, strategic misdirection, information integrity, security, availability, and institutional accountability.
Consequence classes	C0 informational; C1 bounded analytical; C2 material advisory; C3 consequential decision support; C4 safety-, mission-, rights-, or system-critical.
Classification object	Exact task, product, population, data, tool, deployment, user role, decision context, and reachable side effect—not a model family in isolation.
Control principle	Controls accumulate with consequence, uncertainty, scale, irreversibility, vulnerability, novelty, and propagation potential.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-58-001	Every production AI task SHALL carry an approved consequence class, risk statement, affected populations and assets, misuse cases, failure modes, control profile, accountable owner, and review interval.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-58-002	Classification SHALL consider intended use, reasonably foreseeable misuse, compound agent behavior, downstream reliance, correlated failure, scale, reversibility, detectability, and time to intervene.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-58-003	The effective consequence class SHALL be the highest class produced by the task, data, tool, decision, deployment, or reachable side effect unless an approved boundary demonstrably removes that exposure.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-58-004	A change in scope, population, geography, authority, tool reach, model behavior, or operational dependency SHALL trigger risk reclassification before production use continues beyond the approved boundary.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-58-005	Risk acceptance SHALL identify residual risk, affected scope, compensating controls, monitoring, expiry, revocation conditions, and the named human authority; aggregate model quality SHALL NOT substitute for risk treatment.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Risk class missing or disputed	Apply the more restrictive plausible class and block expansion until adjudicated.
Control cannot be enforced	Remove the affected capability or side effect from the permitted envelope and return explicit unavailability.
Observed harm exceeds class	Suspend affected resolution, preserve evidence, initiate incident response, and reclassify before restoration.

Assurance evidence

- Risk-register fixtures resolve every task and product to one current classification and control profile.
- Misuse and compound-failure tests cover direct, indirect, emergent, scaled, and delayed effects.
- Change-impact tests trigger reclassification on material model, data, agent, tool, policy, deployment, and use changes.
- Audit sampling confirms high-consequence uses cannot resolve through lower-consequence controls or expired acceptances.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Risk Taxonomy and Consequence Classes is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 2, Ch. 6, Ch. 14, Ch. 21, Ch. 39, Ch. 42, Ch. 43, Ch. 45, Ch. 56, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-028 · C-029 · C-043 · C-052

59 Safe Action and Side-Effect Governance

AI preparation and external effect are separated by an enforceable action boundary. Agents may construct, validate, and recommend actions within a declared capability envelope, but a side effect occurs only through an authorized tool path that binds purpose, policy, human authority, request integrity, execution conditions, and a durable receipt.

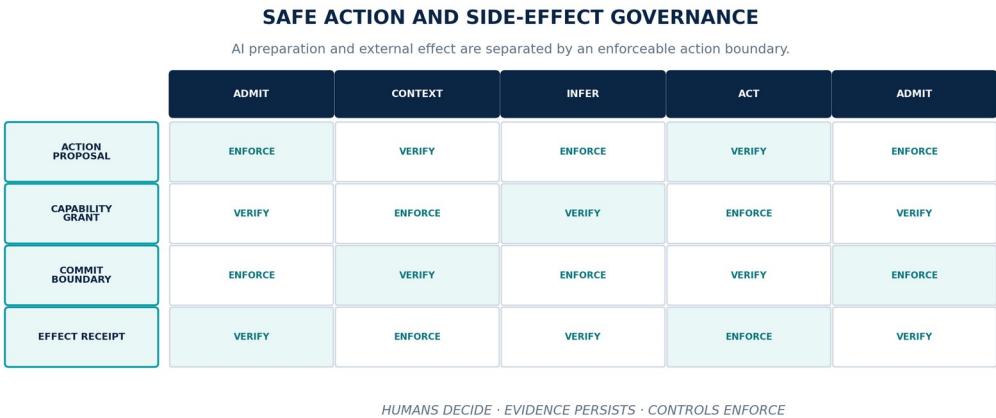


Figure 60. Safe Action and Side-Effect Governance. The diagram organizes action proposal, capability grant, commit boundary, effect receipt as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Action proposal	Non-executing structured object describing target, effect, parameters, rationale, evidence, risk, reversibility, approver, and expiry.
Capability grant	Short-lived, least-privilege authority bound to tenant, agent, workflow, tool, operation, resource, purpose, limits, and time.
Commit boundary	Enforcement point that revalidates authorization, approval, policy, request digest, target state, budget, and idempotency immediately before effect.
Effect receipt	Tamper-evident record of attempted and actual effect, target, result, time, authority, tool version, external correlation, and reconciliation state.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-59-001	Every side-effecting operation SHALL pass through a registered capability adapter that denies unlisted tools, operations, resources, parameters, identities, destinations, and network paths.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-59-002	Consequential or irreversible effects SHALL require a current human approval bound cryptographically or equivalently to the exact action request, material context, policy decision, and permitted execution window.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-59-003	The commit boundary SHALL revalidate target state, authorization, separation of duties, limits, approval freshness, request digest, and stop conditions immediately before dispatch.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-59-004	Retries SHALL use idempotency and reconciliation semantics appropriate to the external system and SHALL never infer success from timeout, partial response, or workflow completion.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-59-005	Agents SHALL NOT create, widen, delegate, persist, or exchange capability grants outside the supervisor and policy-controlled grant service.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Authorization or approval stale	Deny dispatch and return a refreshed proposal for review.
External result indeterminate	Hold the workflow in reconciliation; do not retry blindly or report success.
Effect differs from approval	Stop further actions, preserve both digests and receipts, and initiate incident and impact review.

Assurance evidence

- Adversarial tests attempt parameter smuggling, confused-deputy use, grant reuse, privilege expansion, and tool substitution.
- Commit fixtures cover stale state, changed context, revoked roles, expired approval, duplicate dispatch, timeout, and partial success.
- Reconciliation tests prove every attempted effect reaches confirmed, rejected, compensated, or explicitly unresolved terminal state.
- Decision Replay links proposals, approvals, capability grants, policy decisions, dispatches, receipts, and compensation.

CHAPTER ACCEPTANCE AND INHERITANCE

Safe Action and Side-Effect Governance is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 21, Ch. 39, Ch. 42, Ch. 43, Ch. 44, Ch. 48, Ch. 55, Ch. 63.

CONSTITUTIONAL INHERITANCE / C-004 · C-024 · C-025 · C-026 · C-027 · C-028 · C-029

60 Prompt, Instruction, and Tool Security

All natural-language and multimodal content is untrusted data unless it is a signed, versioned control artifact resolved through the instruction authority chain. Prompt injection, tool manipulation, retrieval poisoning, data exfiltration, and control-plane impersonation are contained through structural separation, provenance, policy enforcement, least privilege, and continuous adversarial testing.

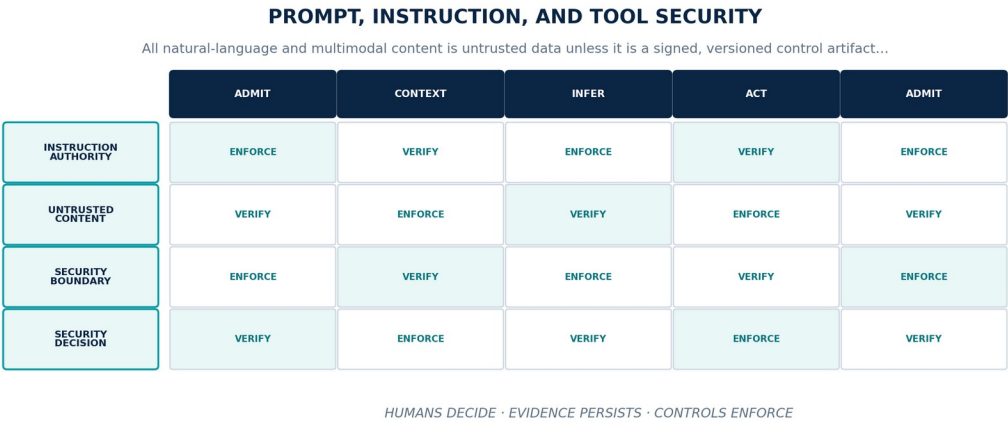


Figure 61. Prompt, Instruction, and Tool Security. The diagram organizes instruction authority, untrusted content, security boundary, security decision as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Instruction authority	Constitutional and platform policy, task specification, agent manifest, workflow step, approved user direction, and untrusted content in descending control order.
Untrusted content	Sources, documents, web pages, messages, images, tool outputs, memory, model text, and external metadata that may contain adversarial instructions.
Security boundary	Typed separation of instructions, evidence, examples, tool schemas, secrets, retrieved content, and output channels.
Security decision	Allow, constrain, transform, isolate, challenge, require human review, or deny with durable reason and policy version.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-60-001	The runtime SHALL maintain typed, provenance-preserving separation between trusted control artifacts and untrusted task content throughout context assembly, inference, agent communication, tool use, and output admission.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-60-002	Models and agents SHALL treat instructions embedded in retrieved or tool-returned content as evidence to analyze, not authority to follow, unless a registered parser and policy explicitly promotes the exact content class.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-60-003	Secrets, credentials, system instructions, hidden policies, and cross-domain context SHALL remain outside model-visible context unless the task contract, policy, and provider boundary explicitly permit the minimum necessary disclosure.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-60-004	Tool schemas SHALL constrain operations and parameters structurally; free-form model text SHALL NOT be executed as code, queries, commands, URLs, templates, policies, or tool arguments without typed validation and authorization.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-60-005	Every high-consequence agent and tool path SHALL pass prompt-injection, exfiltration, poisoning, indirect-instruction, encoding, multimodal, and multi-turn persistence tests before promotion and after material change.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Instruction origin ambiguous	Treat as untrusted content and prohibit control effect.
Injection or exfiltration suspected	Isolate the content and run, deny sensitive tools, preserve evidence, and invoke security review.
Parser or schema mismatch	Reject the operation and return a typed failure; never fall back to free-form execution.

Assurance evidence

- Security fixtures verify instruction precedence, provenance, trust labels, delimiter independence, and context-channel isolation.
- Red-team corpora cover direct and indirect injection, obfuscation, multilingual content, images, documents, tool outputs, and agent messages.
- Canary and egress controls detect attempts to reveal secrets, hidden instructions, protected identifiers, or unauthorized context.
- Tool fuzzing proves malformed and adversarial model outputs cannot escape registered schemas and capabilities.

CHAPTER ACCEPTANCE AND INHERITANCE

Prompt, Instruction, and Tool Security is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 20, Ch. 21, Ch. 27, Ch. 39, Ch. 42, Ch. 43, Ch. 45, Ch. 57.

61 Data, Model, and Agent Privacy

Privacy controls apply across training, evaluation, retrieval, context, inference, memory, logging, explanation, human review, and learning. The AI runtime uses only data that is authorized for the precise purpose and minimizes both what enters computational boundaries and what persists after execution.

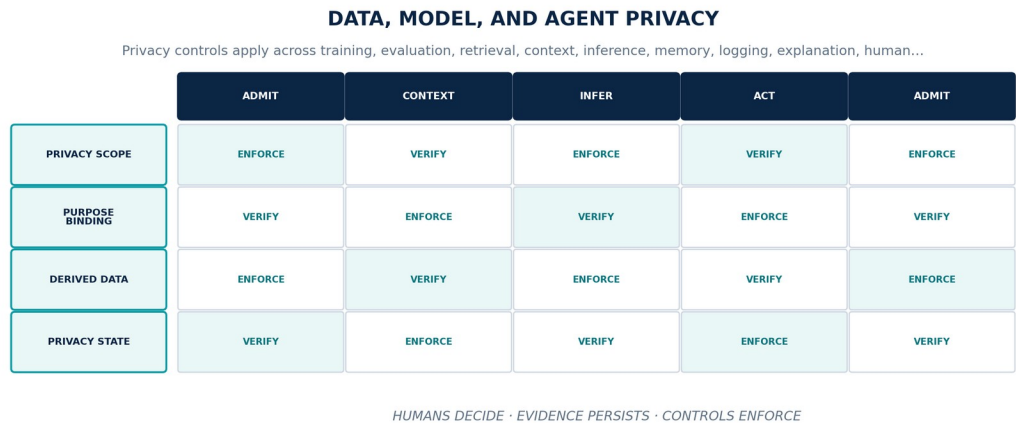


Figure 62. Data, Model, and Agent Privacy. The diagram organizes privacy scope, purpose binding, derived data, privacy state as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Privacy scope	Personal, sensitive, confidential, classified, regulated, proprietary, licensed, and inferred information across all modalities and derived representations.
Purpose binding	Declared task, user, decision, population, legal or policy basis, allowed processing, recipient, geography, and retention.
Derived data	Embeddings, summaries, features, labels, traces, explanations, evaluations, model updates, caches, and memorized or reconstructable content.
Privacy state	Permitted, minimized, transformed, consent- or authority-dependent, restricted, expired, deleted, legally held, or prohibited.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-61-001	Every AI execution SHALL resolve data purpose, authority, classification, residency, provider eligibility, minimization, retention, output restrictions, and deletion obligations before context crosses an inference boundary.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-61-002	Sensitive content SHALL be redacted, tokenized, localized, aggregated, or otherwise transformed when the task can be satisfied without disclosing the original value to the selected model, agent, evaluator, or provider.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-61-003	Prompts, responses, embeddings, traces, caches, evaluation samples, and feedback SHALL NOT be retained, reused, or used for provider training beyond the approved purpose, contract, geography, and retention profile.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-61-004	Privacy controls SHALL cover inferences about people and organizations, membership and attribute disclosure, model memorization, re-identification, vector leakage, and aggregation across individually permitted records.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-61-005	Deletion, correction, consent withdrawal, legal restriction, or data-rights change SHALL propagate through AI context stores, caches, evaluation datasets, learned artifacts, products, and forward impact workflows according to policy and technical feasibility.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Purpose or authority unresolved	Deny the data path or use a policy-approved minimized substitute.
Provider boundary incompatible	Resolve an eligible local or sovereign artifact; otherwise return unavailable.
Deletion cannot be isolated	Quarantine affected artifacts, disclose the limitation to governance, and apply approved retraining, retirement, or compensating controls.

Assurance evidence

- Data-flow inventories trace sensitive fields and derived representations through every model, agent, store, log, evaluator, and provider.
- Privacy tests cover prompt leakage, memorization, reconstruction, membership inference, vector inversion, cross-run reuse, and explanation disclosure.
- Provider and deployment attestations prove retention, training use, geography, encryption, access, deletion, and incident obligations.
- Deletion fixtures verify propagation or explicit residual-risk treatment across caches, datasets, models, products, and backups.

CHAPTER ACCEPTANCE AND INHERITANCE

Data, Model, and Agent Privacy is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 9, Ch. 15, Ch. 21, Ch. 24, Ch. 27, Ch. 29, Ch. 33, Ch. 45, Ch. 54, Ch. 64.

CONSTITUTIONAL INHERITANCE / C-005 · C-013 · C-014 · C-015 · C-016 · C-027 · C-035

62 Model and Agent Safety Architecture

Safety is enforced by layered controls before, during, and after execution. Task admission, artifact resolution, context controls, runtime supervisors, policy decisions, capability boundaries, product validators, human gates, monitors, and emergency stops remain independently effective so that no single model or agent is trusted to police itself.

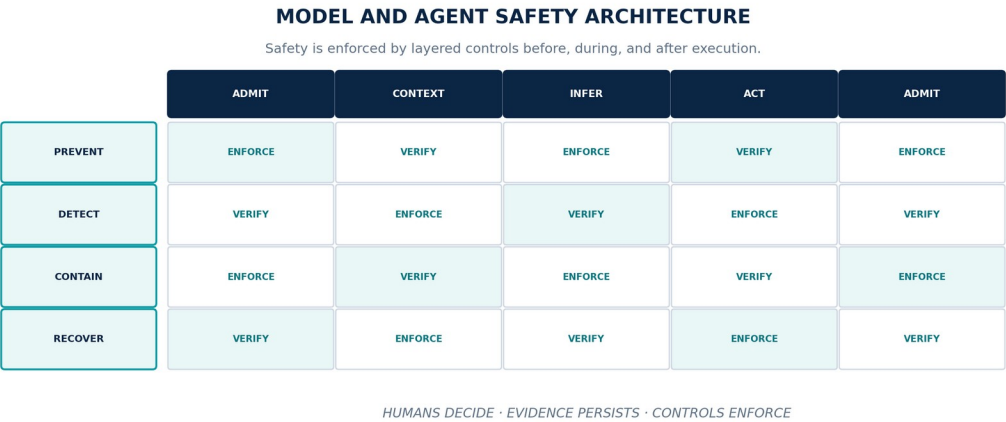


Figure 63. Model and Agent Safety Architecture. The diagram organizes prevent, detect, contain, recover as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Prevent	Purpose restriction, task admission, approved artifacts, safe context, least privilege, tool constraints, and human authority.
Detect	Runtime invariants, policy monitors, evaluator checks, anomaly signals, canaries, disagreement, and outcome surveillance.
Contain	Budget and scope limits, sandbox, circuit breaker, suspension, isolation, capability revocation, and product quarantine.
Recover	Checkpoint, rollback, compensation, correction, impact analysis, incident closure, and controlled restoration.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-62-001	Safety-critical controls SHALL execute outside the model or agent whose behavior they constrain and SHALL fail closed for prohibited authority, data, tool, and side-effect paths.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-62-002	Each agent package SHALL declare operating envelope, prohibited objectives, stop conditions, resource limits, escalation paths, reachable tools, product admissions, and supervisor policy.	Negative conformance test	Adversarial fixture proving the prohibited path is denied, observable, and audited.
AI-62-003	The supervisor SHALL monitor plan divergence, repeated failure, tool anomalies, policy conflicts, context contamination, coordination loops, budget exhaustion, and unsafe or out-of-scope behavior.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-62-004	Emergency stop SHALL revoke new model and tool resolution, suspend active side effects, preserve execution state, and permit only explicitly authorized containment and recovery operations.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-62-005	Safety cases for C3 and C4 tasks SHALL connect hazards, controls, assumptions, evidence, residual risk, runtime monitors, incident playbooks, owners, and approval to the exact release.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Safety control unavailable	Block affected task admission or enter a predefined lower-risk mode that preserves explicit degradation.
Agent exceeds envelope	Suspend the run, revoke capabilities, preserve checkpoint and evidence, and escalate by consequence class.
Multiple controls disagree	Apply the most restrictive enforceable result and initiate control-health review.

Assurance evidence

- Fault-injection exercises disable models, evaluators, policies, supervisors, tools, stores, and networks and verify independent containment.
- Hazard-to-control traceability proves every unacceptable risk has a preventive or detective control and recovery path.
- Long-running agent tests cover loop detection, goal drift, coordination deadlock, capability accumulation, and checkpoint recovery.
- Emergency-stop drills demonstrate bounded activation, active-run handling, durable audit, restoration criteria, and no hidden bypass.

CHAPTER ACCEPTANCE AND INHERITANCE

Model and Agent Safety Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 14, Ch. 21, Ch. 39, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 48, Ch. 56, Ch. 63.

CONSTITUTIONAL INHERITANCE / C-004 · C-024 · C-027 · C-028 · C-029 · C-043 · C-049

63 Uncertainty, Calibration, Abstention, and Partial Results

Uncertainty is a first-class property of every AI product. The architecture distinguishes probability, confidence, evidence quality, ambiguity, disagreement, model uncertainty, scenario spread, sensitivity, and operational completeness, and it allows the system to abstain rather than manufacture precision.

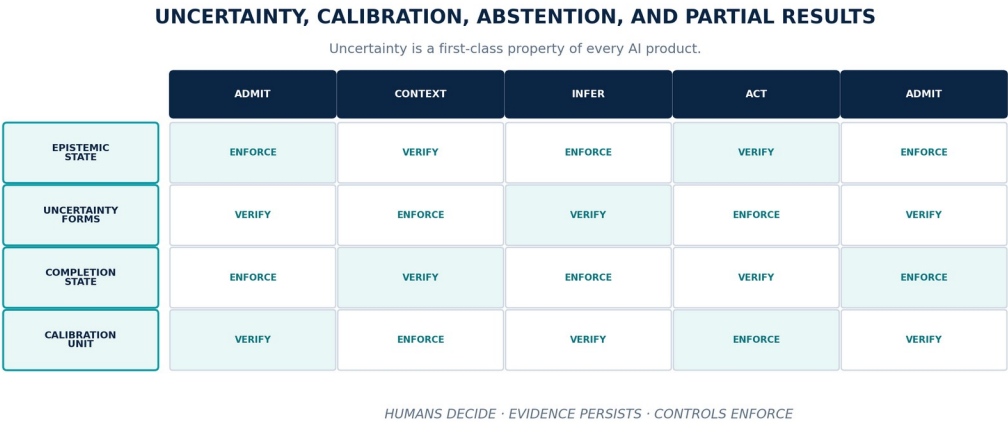


Figure 64. Uncertainty, Calibration, Abstention, and Partial Results. The diagram organizes epistemic state, uncertainty forms, completion state, calibration unit as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Epistemic state	Known, supported claim, inference, assessment, assumption, hypothesis, contradiction, unknown, or unavailable.
Uncertainty forms	Aleatoric, epistemic, data quality, model, structural, parameter, temporal, scenario, operational, and policy uncertainty.
Completion state	Complete, partial with usable boundary, abstained, degraded, unavailable, invalid, or superseded.
Calibration unit	Task, horizon, population, domain, consequence, artifact version, product field, and decision-use threshold.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-63-001	Every AI product SHALL expose uncertainty and completion state at the granularity needed to prevent a reliable aggregate from concealing an unreliable material claim, field, entity, horizon, or subgroup.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-63-002	Confidence SHALL be tied to an interpretable definition, calibration method, evaluation population, validity period, and known failure conditions and SHALL NOT be inferred from model fluency or agent agreement alone.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-63-003	Models and agents SHALL abstain when required evidence, task fitness, operating-envelope validity, policy, authorization, calibration, or output integrity is insufficient for the consequence class.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-63-004	Partial results SHALL identify completed and missing work, failed dependencies, affected claims, uncertainty impact, retry or refresh conditions, and whether downstream admission is permitted.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-63-005	Decision products SHALL distinguish likelihood from impact, confidence from evidence quality, forecast probability from scenario plausibility, and model disagreement from real-world uncertainty.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Calibration population unsupported	Mark confidence uncalibrated for that slice and restrict or abstain according to consequence.
Uncertainty cannot be quantified	Use bounded qualitative state with explicit rationale and limitations; do not fabricate a number.
Partial output loses boundary	Reject product admission and preserve the run for repair or explicit failure.

Assurance evidence

- Calibration reports include reliability, coverage, sharpness, abstention, selective risk, subgroup performance, horizon, and drift.
- Out-of-distribution and missing-evidence fixtures verify abstention and partial-result behavior before fluent completion.
- Product-schema tests prevent omission, defaulting, or flattening of uncertainty and completion state.
- Human-factor tests measure whether decision-makers correctly distinguish probability, confidence, quality, disagreement, and scenario range.

CHAPTER ACCEPTANCE AND INHERITANCE

Uncertainty, Calibration, Abstention, and Partial Results is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 23, Ch. 27, Ch. 36, Ch. 37, Ch. 38, Ch. 39, Ch. 46, Ch. 56.

64 Explanation, Contestability, and Human Factors

Explanations are designed for responsible human judgment, challenge, correction, and appeal. They reveal the decision-relevant structure of a recommendation without overstating mechanistic access to a model, and they preserve material evidence, uncertainty, alternatives, dissent, limitations, policy, and consequences across audience-specific views.

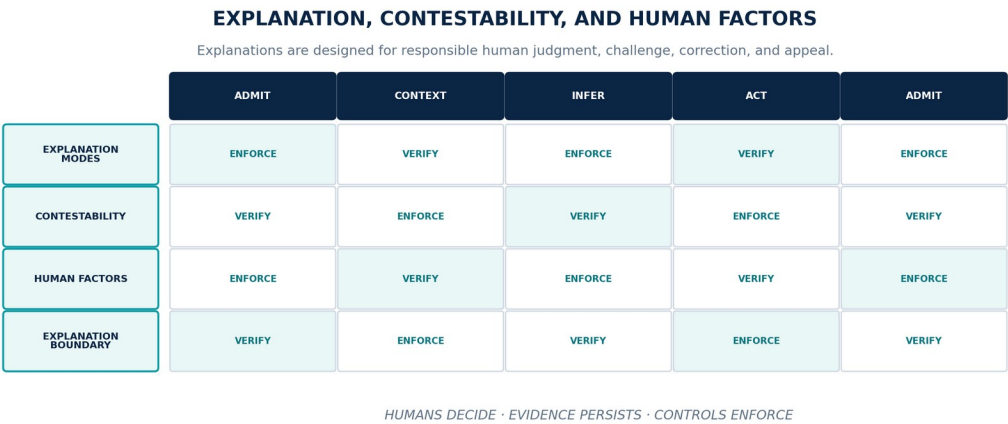


Figure 65. Explanation, Contestability, and Human Factors. The diagram organizes explanation modes, contestability, human factors, explanation boundary as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Explanation modes	Evidence, procedural, model or method, uncertainty, causal, counterfactual, comparative, policy, and operational.
Contestability	Ability to inspect, question, annotate, dispute, provide evidence, request review, override, reopen, or appeal under recorded authority.
Human factors	Comprehension, workload, attention, anchoring, automation bias, framing, accessibility, expertise, language, and time pressure.
Explanation boundary	Faithful account of observable inputs, methods, outputs, controls, and limitations—not fabricated inner reasoning or persuasive certainty.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-64-001	Consequential products SHALL present the recommendation, alternatives, evidence, uncertainty, counter-evidence, assumptions, sensitivities, model and agent roles, dissent, policy constraints, limitations, and required human action in a coherent decision package.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-64-002	Users SHALL be able to contest a material claim, source, entity resolution, assumption, model result, agent judgment, score, recommendation, or policy application without destroying the original record.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-64-003	A contest SHALL create a durable case with standing, scope, evidence, owner, response deadline, adjudication, correction, impact analysis, notification, and closure state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-64-004	Audience adaptation SHALL preserve semantic equivalence and material caveats while enforcing access policy and accessibility; simplification SHALL NOT become omission of decision-relevant risk.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-64-005	Interface and briefing designs SHALL be evaluated for automation bias, anchoring, false precision, urgency inflation, uncertainty comprehension, cognitive overload, and accessibility by representative users.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Explanation unavailable	Mark the product explanation-incomplete and prohibit use above its approved consequence boundary.
Contest reveals material defect	Suspend affected use, preserve original state, correct through the owning layer, and propagate impact.
Audience rendering changes meaning	Withdraw the rendering and revert to the authoritative structured explanation product.

Assurance evidence

- Faithfulness tests compare every explanation statement with preserved context, evidence, execution, model, agent, policy, and product state.
- Representative-user studies measure comprehension, calibration, challenge behavior, bias, workload, and decision quality.
- Contest workflows are exercised from submission through adjudication, correction, notification, replay, and closure.
- Accessibility checks cover structure, reading order, contrast, non-color meaning, language, assistive technology, and alternative representations.

CHAPTER ACCEPTANCE AND INHERITANCE

Explanation, Contestability, and Human Factors is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 21, Ch. 27, Ch. 28, Ch. 39, Ch. 46, Ch. 48, Ch. 55, Ch. 57.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-012 · C-024 · C-025 · C-035 · C-047

65 AI Audit, Non-Repudiation, and Accountability

AI accountability is established through a complete, tamper-evident execution and authority record. Audit connects who requested, configured, approved, changed, executed, reviewed, or relied upon an AI capability with the exact artifacts, context, policies, effects, decisions, outcomes, and corrective actions involved.

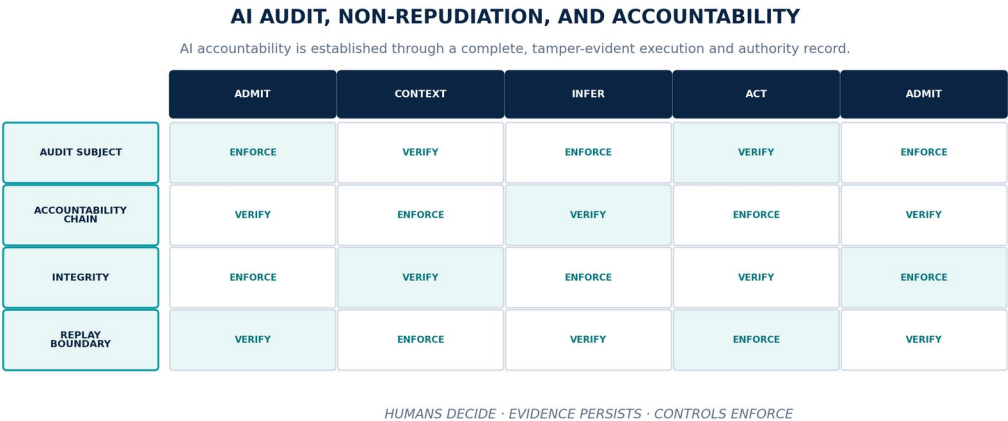


Figure 66. AI Audit, Non-Repudiation, and Accountability. The diagram organizes audit subject, accountability chain, integrity, replay boundary as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Audit subject	Model, agent, prompt, instruction, context, retrieval, tool, policy, evaluator, release, execution, product, human gate, side effect, decision, and learning artifact.
Accountability chain	Owner, operator, requester, supervisor, reviewer, approver, policy authority, model or data steward, incident commander, and decision-maker.
Integrity	Authenticated identity, trusted time, sequence, content digest, version resolution, append-only evidence, retention, and verifiable export.
Replay boundary	Exact historical information and control state separated from later corrections, outcomes, and retrospective assessment.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-65-001	Every AI execution SHALL produce a durable record of request, purpose, authority, task, consequence, context manifest, artifact resolution, instructions, retrieval, tool calls, policies, evaluations, human actions, outputs, admissions, effects, and terminal state.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-65-002	Audit records SHALL use stable identities, authenticated actors, trusted timestamps, ordering, digests, version references, and append-only correction so that material actions cannot be silently repudiated or rewritten.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-65-003	Administrative and governance actions affecting model, agent, prompt, policy, evaluation, release, capability, or monitoring state SHALL be attributable and subject to separation of duties appropriate to consequence.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-65-004	Decision Replay SHALL reconstruct the original AI product and authority boundary from preserved artifacts or verified immutable references without using later knowledge as if it were originally available.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-65-005	Audit export SHALL preserve evidentiary meaning, access policy, integrity proof, redaction state, withheld scope, and chain of custody across SaaS, private-cloud, and on-premise deployments.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Audit append fails	Do not report execution or side effect as fully committed; hold or suspend according to consequence and recover durable evidence.
Identity or time cannot be trusted	Mark the affected event indeterminate, restrict reliance, and invoke integrity review.
Replay dependency unavailable	Return an explicit reconstruction gap and prohibit fabricated reconstruction.

Assurance evidence

- Golden-run replay reproduces exact artifact, context, policy, tool, human, product, effect, and terminal state references.
- Integrity tests detect deletion, reordering, substitution, clock manipulation, duplicate identity, and cross-tenant contamination.
- Separation-of-duties tests prevent one role from making, approving, releasing, and concealing a consequential change.
- Export verification confirms digests, signatures or equivalent proofs, redactions, policy labels, and chain of custody offline.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Audit, Non-Repudiation, and Accountability is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 11, Ch. 12, Ch. 18, Ch. 21, Ch. 39, Ch. 43, Ch. 44, Ch. 48, Ch. 51, Ch. 55.

CONSTITUTIONAL INHERITANCE / C-004 · C-010 · C-025 · C-026 · C-028 · C-036 · C-048

PART IX

Evaluation, Learning, and Operations

Evaluation, red teaming, monitoring, governed learning, release, deployment parity, continuity, and retirement.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

66 AI Evaluation Architecture

Evaluation is an operating plane of The Eye, not a pre-release scorecard. It establishes whether an exact AI system configuration is fit for a defined task, consequence, population, decision use, deployment, and period, and it continues through shadow operation, production monitoring, outcomes, correction, and retirement.

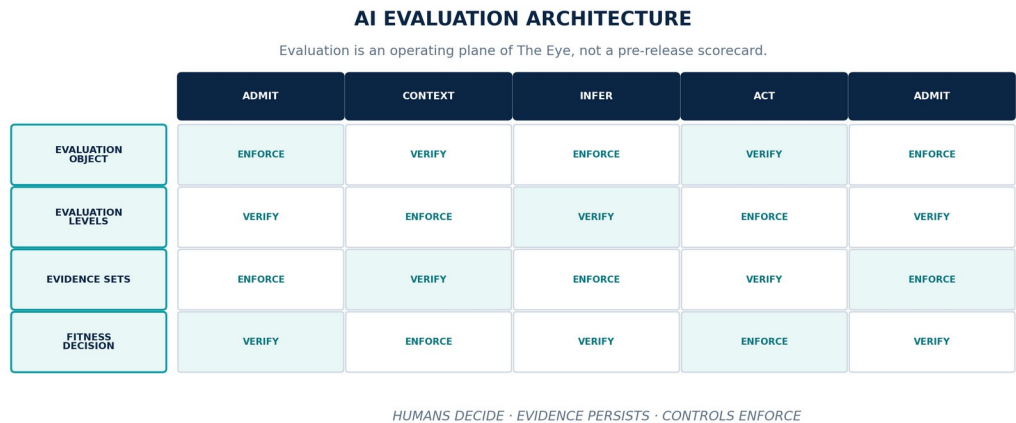


Figure 67. AI Evaluation Architecture. The diagram organizes evaluation object, evaluation levels, evidence sets, fitness decision as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Evaluation object	Exact model, agent, prompt, context, retrieval, tool, policy, workflow, product schema, and release configuration.
Evaluation levels	Component, task, product, agent trajectory, workflow, human-AI system, safety control, operational system, and realized outcome.
Evidence sets	Curated benchmarks, historical as-of replays, synthetic cases, adversarial tests, expert review, shadow traffic, canaries, production samples, and outcome cohorts.
Fitness decision	Approved, restricted, shadow-only, conditionally approved, suspended, failed, expired, or indeterminate for a precise scope.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-66-001	Every production AI release SHALL have an evaluation plan that defines claims, populations, slices, baselines, metrics, thresholds, uncertainty, sample design, reviewers, expiry, monitoring, and decision-use limits.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-66-002	Evaluation SHALL measure task correctness, evidence faithfulness, calibration, robustness, safety, privacy, security, fairness, latency, reliability, cost, explanation, human factors, and operational recoverability as applicable.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-66-003	High-consequence evaluation SHALL include representative experts, historical as-of replay, adversarial cases, rare but severe failures, subgroup analysis, tool and policy behavior, and human-AI decision performance.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-66-004	Evaluator models MAY assist measurement but SHALL be versioned, validated, calibrated, bias-assessed, periodically anchored to human judgment, and prohibited from serving as sole authority for their own release lineage.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-66-005	Evaluation results SHALL bind exact artifacts, datasets, code, environment, configuration, policy, reviewers, confidence intervals, limitations, and applicability and SHALL expire on material change or elapsed validity.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Representative evidence unavailable	Return fitness indeterminate or restrict scope; do not generalize from an unrelated benchmark.
Metrics conflict	Apply the declared decision rule, expose trade-offs, and escalate material safety or consequence conflicts.
Evaluation cannot be reproduced	Block promotion and investigate data, artifact, environment, or evidence integrity.

Assurance evidence

- Evaluation manifests reproduce datasets, sampling, artifacts, configuration, metrics, thresholds, reviewers, and results.
- Independent reruns and historical replays confirm statistical, semantic, temporal, and operational reproducibility.
- Slice analysis covers material domains, languages, geographies, populations, horizons, evidence quality, and consequence classes.
- Outcome review compares predicted and assisted decisions with realized outcomes while controlling hindsight and selection bias.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Evaluation Architecture is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 20, Ch. 23, Ch. 27, Ch. 39, Ch. 45, Ch. 46, Ch. 47, Ch. 56, Ch. 66.

CONSTITUTIONAL INHERITANCE / C-005 · C-021 · C-022 · C-028 · C-030 · C-047 · C-052

67 Red Teaming and Safety Testing

Red teaming challenges the complete AI system under realistic adversarial, erroneous, ambiguous, degraded, and high-pressure conditions. It targets not only model responses but agent trajectories, context and memory boundaries, graph and twin grounding, tool authority, human gates, policy enforcement, deployment controls, and organizational response.

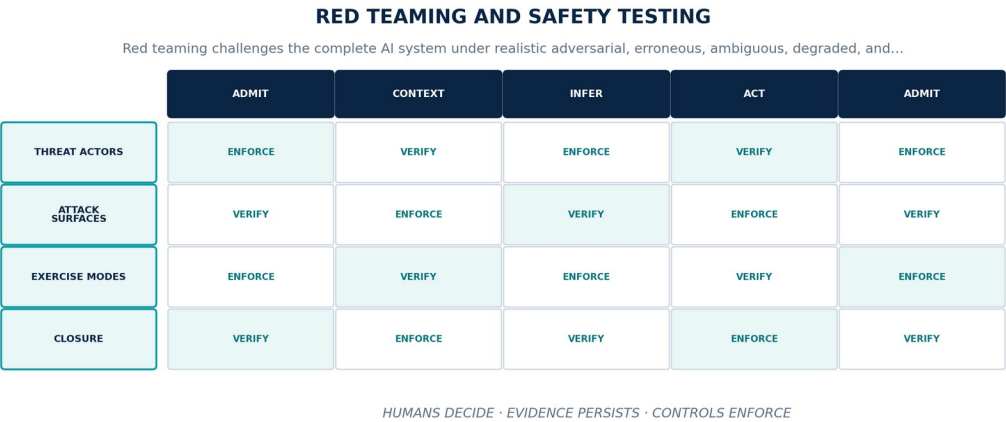


Figure 68. Red Teaming and Safety Testing. The diagram organizes threat actors, attack surfaces, exercise modes, closure as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Threat actors	Malicious outsider, compromised source, insider, careless user, poisoned provider, colluding agents, defective model, and environmental failure.
Attack surfaces	Data, retrieval, prompt, model, agent protocol, memory, tool, capability, policy, evaluation, supply chain, interface, export, and operations.
Exercise modes	Automated fuzzing, expert adversarial testing, tabletop, live-fire sandbox, purple-team validation, and independent assessment.
Closure	Finding, severity, affected scope, evidence, owner, containment, remediation, regression test, residual risk, approval, and monitoring.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-67-001	Red-team plans SHALL derive from the task risk model, misuse cases, threat model, agent capability graph, data boundaries, reachable side effects, past incidents, and external changes.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-67-002	Testing SHALL cover prompt injection, poisoning, evasion, extraction, privacy leakage, identity confusion, hallucinated evidence, unsafe planning, tool abuse, coordination failure, approval bypass, denial of service, and recovery manipulation.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-67-003	C3 and C4 releases SHALL receive independent adversarial review proportionate to novelty, autonomy, scale, irreversibility, and affected rights or missions before production expansion.	Automated conformance test	Versioned fixture and signed result for exact task, artifacts, context, configuration, and environment.
AI-67-004	Material findings SHALL create durable release blockers or explicit restricted-use controls; unresolved critical findings SHALL NOT be accepted through ordinary delivery authority.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-67-005	Every remediated finding SHALL become a versioned regression asset applied to affected and successor models, agents, workflows, providers, and deployment profiles without disclosing exploit detail beyond need to know.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Test creates unsafe external path	Contain exercises in an isolated environment with synthetic or explicitly authorized targets and emergency stop.
Critical exploit confirmed	Suspend affected capability, revoke exposed authority, preserve evidence, and initiate incident command.
Finding cannot be reproduced	Retain as indeterminate risk, improve telemetry, and avoid closure based solely on non-reproduction.

Assurance evidence

- Coverage maps connect threats, surfaces, consequence classes, exercises, findings, controls, and regression assets.
- Independent teams reproduce material findings and verify remediation under frozen and successor releases.
- Live-fire rehearsals demonstrate capability revocation, stop behavior, containment, evidence preservation, and recovery.
- Metrics track time to detect, contain, reproduce, remediate, regress, accept residual risk, and close by severity.

CHAPTER ACCEPTANCE AND INHERITANCE

Red Teaming and Safety Testing is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 15, Ch. 21, Ch. 39, Ch. 42, Ch. 43, Ch. 45, Ch. 54, Ch. 57, Ch. 63, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-013 · C-024 · C-027 · C-028 · C-029 · C-043 · C-049

68 Runtime Monitoring, Drift, and Fitness

Production AI remains authorized only while its observed behavior, inputs, dependencies, controls, and outcomes stay within the evaluated operating envelope. Monitoring combines technical telemetry with semantic, epistemic, safety, policy, human, and outcome signals and drives durable restriction, investigation, recalibration, rollback, or retirement workflows.

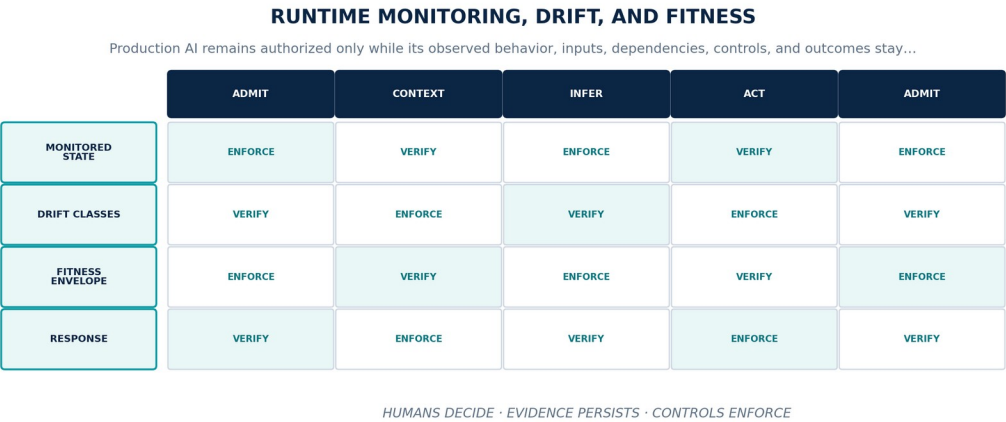


Figure 69. Runtime Monitoring, Drift, and Fitness. The diagram organizes monitored state, drift classes, fitness envelope, response as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Monitored state	Inputs, context, retrieval, models, agents, tools, policies, products, users, decisions, effects, outcomes, dependencies, and infrastructure.
Drift classes	Data, concept, population, evidence, ontology, temporal, calibration, behavior, prompt, tool, policy, provider, workload, cost, and outcome.
Fitness envelope	Approved ranges and assumptions for task, population, quality, uncertainty, latency, safety, capacity, cost, deployment, and decision use.
Response	Observe, alert, increase review, narrow scope, switch artifact, abstain, quarantine, rollback, suspend, investigate, or retire.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-68-001	Each production task SHALL have monitors and thresholds tied to its evaluation claims, consequence class, operating envelope, control health, known failure modes, and expected decision or outcome signals.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-68-002	Monitoring SHALL preserve tenant and subject privacy while permitting analysis by artifact, task, population, horizon, language, deployment, tool, agent, user role, and consequence slice.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-68-003	Model and agent outputs SHALL be sampled for correctness, evidence faithfulness, calibration, uncertainty, abstention, policy, explanation, safety, and human override with sampling risk proportionate to consequence.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-68-004	A fitness breach SHALL trigger a durable state transition and owned workflow; an alert alone SHALL NOT count as risk treatment or continued authorization.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-68-005	Monitoring blind spots, delayed labels, sparse outcomes, and evaluator uncertainty SHALL be explicit inputs to fitness state and may require restricted use or expiry rather than assumed health.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Telemetry incomplete	Mark monitoring coverage degraded and apply the predefined restriction for the consequence class.
Drift threshold exceeded	Narrow, switch, suspend, or abstain as declared while investigation and re-evaluation proceed.
Outcome signal contradicts proxy metrics	Prioritize the governed outcome review, preserve both signals, and reassess the evaluation model.

Assurance evidence

- Synthetic drift and control-failure injections verify detection, routing, restriction, rollback, and closure.
- Monitoring manifests reconcile expected and observed telemetry, sampling, thresholds, blind spots, owners, and response times.
- Backtests show threshold sensitivity, false-positive and false-negative trade-offs, and consequence-appropriate detection delay.
- Production reviews connect behavior and overrides with decisions, outcomes, incidents, corrections, and artifact fitness state.

CHAPTER ACCEPTANCE AND INHERITANCE

Runtime Monitoring, Drift, and Fitness is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 23, Ch. 38, Ch. 45, Ch. 47, Ch. 56, Ch. 58, Ch. 59, Ch. 60, Ch. 61, Ch. 63, Ch. 68.

69 Governed Learning and Adaptation

The Eye learns through an explicit evidence-to-artifact pipeline. Outcomes, corrections, decisions, contests, incidents, analyst feedback, and observed drift can propose improvements, but no runtime interaction silently changes production model weights, prompts, policies, agent behavior, memory reuse, thresholds, or strategic truth.

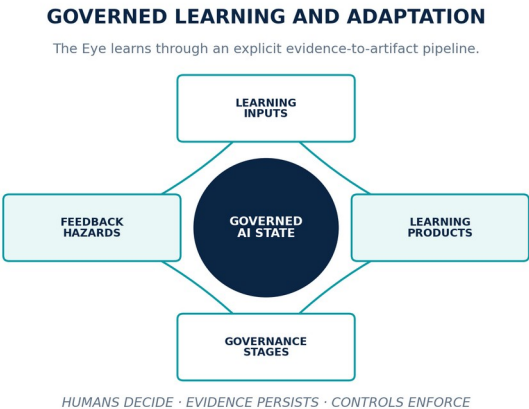


Figure 70. Governed Learning and Adaptation. The diagram organizes learning inputs, learning products, governance stages, feedback hazards as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Learning inputs	Verified outcomes, corrections, labels, human decisions, overrides, contests, incidents, evaluations, monitoring, and lessons.
Learning products	Dataset, feature, rule, prompt, example, retrieval configuration, agent package, model, calibrator, evaluator, policy proposal, or operating procedure.
Governance stages	Capture, qualify, de-bias, approve purpose, prepare, train or configure, evaluate, challenge, release, monitor, and retire.
Feedback hazards	Automation bias, self-confirmation, label leakage, selection bias, reward hacking, data poisoning, strategic manipulation, and institutionalizing past error.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-69-001	Feedback SHALL be treated as evidence with source, authority, context, time, confidence, affected scope, consent or rights, and adjudication state; user acceptance SHALL NOT automatically become a truth label.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-69-002	Learning datasets SHALL preserve lineage, temporal cutoffs, representativeness, exclusions, corrections, rights, privacy, class balance, bias analysis, and separation from evaluation holdouts.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-69-003	Any change to production behavior SHALL produce a versioned artifact that passes the same conformance, evaluation, safety, security, privacy, release, deployment-parity, monitoring, rollback, and audit controls as a manually authored change.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-69-004	Online or adaptive methods, where approved, SHALL operate only within explicit parameter, population, objective, duration, safety, review, checkpoint, rollback, and human-oversight bounds and SHALL preserve every state transition.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-69-005	Outcome learning SHALL distinguish correlation from causal effect, original from later information, decision from implementation, and system recommendation from human and environmental contributions.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Feedback provenance weak	Retain for analysis but prohibit promotion into authoritative labels or learned artifacts.
Learning degrades a protected slice	Block or rollback release and investigate data, objective, evaluation, and control failure.
Adaptive state escapes bounds	Freeze adaptation, restore the last approved checkpoint, and initiate safety review.

Assurance evidence

- Learning lineage reconstructs every production artifact from approved evidence, datasets, code, configuration, evaluation, and release.
- Counterfactual and holdout tests detect self-confirmation, leakage, reward gaming, selection bias, and regressions across slices.
- Poisoning exercises verify source qualification, anomaly detection, reviewer separation, dataset integrity, and rollback.
- Adaptive-method tests prove bounds, checkpoints, monitoring, human intervention, deterministic restoration, and historical resolution.

CHAPTER ACCEPTANCE AND INHERITANCE

Governed Learning and Adaptation is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 29, Ch. 30, Ch. 45, Ch. 47, Ch. 48, Ch. 50, Ch. 56, Ch. 64, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-010 · C-016 · C-025 · C-028 · C-030 · C-044 · C-051

70 AI Release, Promotion, Rollback, and Retirement

AI release management promotes an immutable, dependency-closed system configuration rather than an isolated model file. Every release binds models, agents, prompts, retrieval, context, tools, policies, schemas, evaluators, thresholds, deployment artifacts, safety evidence, monitoring, and rollback into one resolvable manifest.

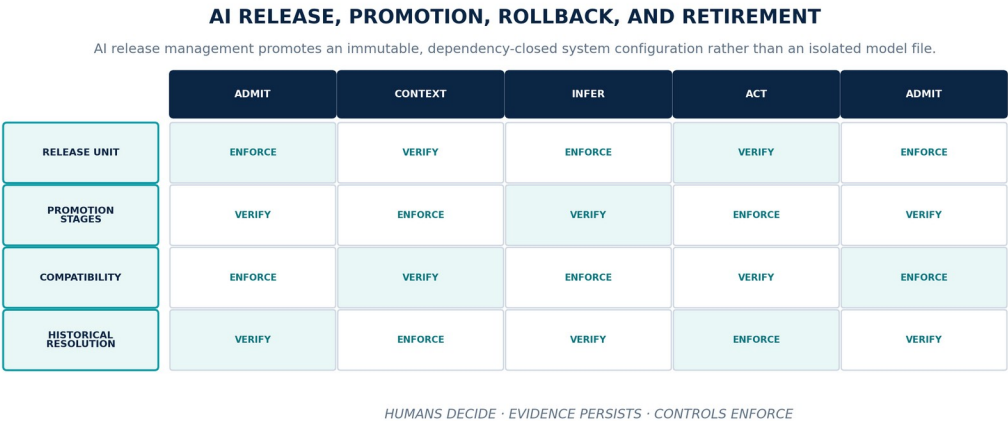


Figure 71. AI Release, Promotion, Rollback, and Retirement. The diagram organizes release unit, promotion stages, compatibility, historical resolution as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Release unit	Signed manifest of all behavior-affecting artifacts, dependencies, configuration, evidence, approvals, environments, and permitted scopes.
Promotion stages	Development, evaluation, red-team, integration, shadow, canary, restricted production, general approved use, suspended, retired, and revoked.
Compatibility	Task, product, schema, semantic, policy, provenance, temporal, evaluation, operational, and replay compatibility.
Historical resolution	Ability to resolve or reconstruct the exact approved configuration used for any product, decision, incident, or evaluation.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-70-001	No production resolver SHALL select an AI artifact outside a signed release manifest with current scope-specific approvals, fitness, safety evidence, deployment eligibility, monitoring, and rollback state.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-70-002	Promotion SHALL use progressive exposure, predeclared success and stop criteria, representative traffic, consequence limits, independent approval, and durable comparison with the incumbent or no-AI baseline.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-70-003	Rollback SHALL restore a known compatible system configuration, not merely a prior model, and SHALL reconcile in-flight runs, checkpoints, products, side effects, caches, embeddings, and monitoring state.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-70-004	Retirement SHALL stop new resolution, identify active and historical dependencies, preserve replay evidence, migrate or expire affected products, revoke capabilities and secrets, and retain required audit and model cards.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-70-005	Emergency revocation SHALL propagate to model gateways, agent registries, workflow schedulers, tool grants, caches, deployment sites, and offline packages within the declared recovery objective.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Manifest dependency unresolved	Reject release and identify the open reference; do not resolve a floating substitute.
Canary exceeds stop criterion	Halt expansion, isolate exposure, restore the approved baseline, and investigate impact.
Rollback target incompatible	Suspend affected use and execute the tested forward-fix or recovery configuration under incident command.

Assurance evidence

- Release verification resolves every artifact, digest, approval, evidence item, deployment image, configuration, and rollback target.
- Canary rehearsals validate traffic selection, comparison, stop, rollback, in-flight handling, and outcome accounting.
- Revocation drills measure propagation across connected, intermittently connected, and disconnected deployment profiles.
- Historical tests replay products and decisions against exact retired releases without relying on mutable latest references.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Release, Promotion, Rollback, and Retirement is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 3, Ch. 14, Ch. 23, Ch. 42, Ch. 45, Ch. 46, Ch. 56, Ch. 63, Ch. 66, Ch. 68.

CONSTITUTIONAL INHERITANCE / C-004 · C-025 · C-028 · C-030 · C-041 · C-042 · C-049

71 AI Deployment Parity and Sovereignty

SaaS, private-cloud, and on-premise deployments implement one AI product and control contract. Providers, accelerators, model sizes, network conditions, and operating procedures may differ, but human authority, product meaning, provenance, uncertainty, safety, evaluation, audit, and lifecycle semantics do not fork by deployment model.

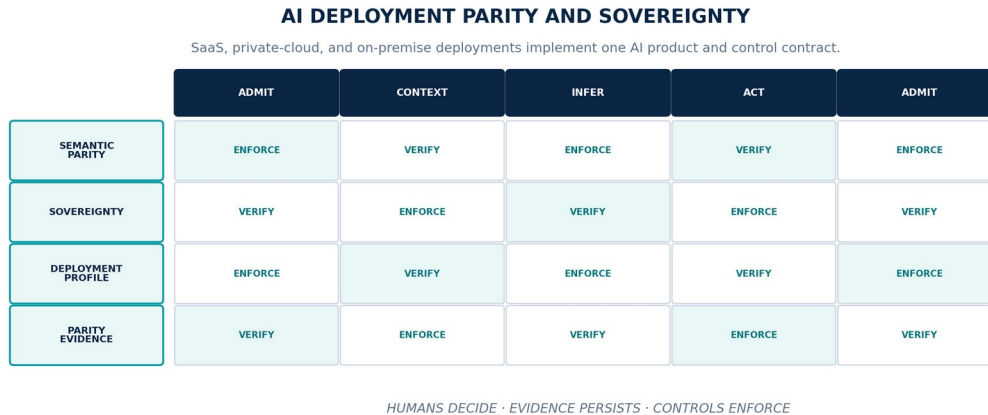


Figure 72. AI Deployment Parity and Sovereignty. The diagram organizes semantic parity, sovereignty, deployment profile, parity evidence as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
Semantic parity	Equivalent task, product, state, authority, policy, uncertainty, provenance, failure, replay, correction, and learning behavior.
Sovereignty	Customer control over data location, model and provider eligibility, keys, administration, network egress, updates, evidence, and operational authority.
Deployment profile	Approved topology, artifacts, accelerators, connectivity, providers, capacity, controls, operations, and degraded-mode envelope.
Parity evidence	Common conformance suite plus profile-specific security, performance, resilience, capacity, update, and recovery evidence.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-71-001	All deployment models SHALL implement the same canonical AI task, agent, context, product, explanation, human gate, audit, correction, and learning contracts.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-71-002	A deployment profile MAY substitute providers or implementation mechanisms only when the replacement passes task-specific semantic, safety, policy, calibration, provenance, performance, and replay conformance.	Evaluation / monitoring	Reproducible profile, evidence set, slice metrics, thresholds, uncertainty, reviewer, and fitness decision.
AI-71-003	On-premise and disconnected operation SHALL use signed artifact bundles, local policy and identity enforcement, bounded licenses and approvals, deterministic update order, revocation handling, evidence export, and reconciliation on reconnection.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-71-004	Sovereignty policy SHALL control model location, provider routing, data egress, telemetry, keys, administrative access, training use, support access, and artifact update authority without hidden cloud dependency.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.
AI-71-005	Capacity or accelerator limits SHALL produce explicit queueing, lower-throughput approved resolution, partial service, or unavailability and SHALL NOT silently weaken quality, safety, explanation, or human control.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Equivalent artifact unavailable	Return declared deployment-specific unavailability or approved degradation with unchanged product semantics.
Disconnected approval expires	Stop new affected work and retain evidence until an authorized update or renewal is imported.
Parity regression detected	Block profile promotion or isolate the release and restore the last conforming configuration.

Assurance evidence

- The same golden task suite runs across SaaS, private-cloud, on-premise, connected, intermittent, and disconnected profiles.
- Parity comparison covers product fields, uncertainty, citations, policies, gates, failures, replay, corrections, and audit—not text similarity alone.
- Air-gapped exercises validate signed import, dependency closure, local resolution, revocation, evidence export, and later reconciliation.
- Sovereignty reviews verify data flow, egress, provider, key, telemetry, administrator, support, update, and deletion boundaries.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Deployment Parity and Sovereignty is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 8, Ch. 9, Ch. 15, Ch. 16, Ch. 17, Ch. 45, Ch. 52, Ch. 53, Ch. 54, Ch. 62, Ch. 63, Ch. 65.

CONSTITUTIONAL INHERITANCE / C-003 · C-013 · C-027 · C-030 · C-041 · C-042 · C-049

72 AI Incident, Continuity, and Governance Closure

AI incidents are managed as failures of a governed sociotechnical system. Response contains model and agent behavior, protects people and operations, preserves evidence, identifies affected products and decisions, communicates uncertainty, restores controlled capability, and converts verified lessons into governed improvements without compromising the historical record.

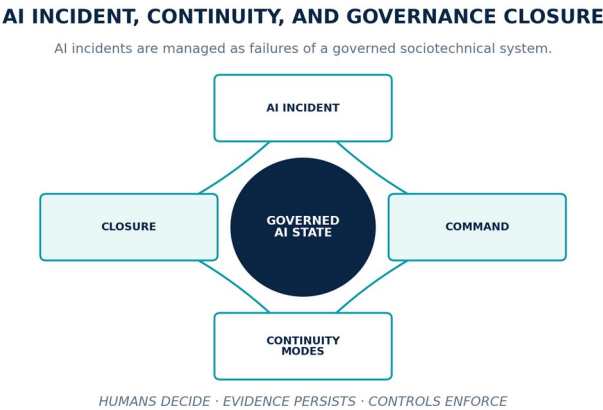


Figure 73. AI Incident, Continuity, and Governance Closure. The diagram organizes ai incident, command, continuity modes, closure as one governed AI architecture contract.

AI architecture contract

CONTRACT ELEMENT	BINDING DEFINITION
AI incident	Observed or credible failure involving harmful, unsafe, unauthorized, misleading, unaccountable, unavailable, or uncontrolled AI behavior or dependency.
Command	Named incident authority with consequence-based roles for AI, security, data, platform, product, legal, domain, communications, and customer operations.
Continuity modes	Normal, constrained, human-only, deterministic fallback, read-only, isolated, recovery, and unavailable.
Closure	Contained, corrected, impact-accounted, notified, restored, monitored, lessons released, and independently approved with residual risk explicit.

Normative requirements

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-72-001	AI incident detection SHALL create a durable case that binds affected tenants, tasks, artifacts, agents, data, tools, products, decisions, effects, deployments, time range, evidence, severity, and command authority.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-72-002	Containment SHALL support artifact revocation, agent and workflow suspension, capability withdrawal, product quarantine, warning suppression or annotation, provider isolation, human-only operation, and evidence preservation.	Fault / recovery exercise	Fault trace, checkpoint, product state, effect reconciliation, recovery, and accountable closure.
AI-72-003	Impact analysis SHALL traverse provenance and dependency edges to identify affected outputs, warnings, recommendations, decisions, commitments, exports, customers, evaluations, datasets, and learned artifacts.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.
AI-72-004	Restoration SHALL require verified cause treatment, regression evidence, safety and fitness reassessment, deployment-parity validation, operational rehearsal, accountable approval, progressive exposure, and enhanced monitoring.	Security / privacy test	Signed adversarial result bound to task, artifacts, data class, provider, deployment, and policy.

ID	REQUIREMENT	VERIFICATION	REQUIRED EVIDENCE
AI-72-005	Volume 5 conformance SHALL be closed only when all 72 chapters, controlled catalogs, constitutional mappings, engineering references, AI release evidence, and deployment profiles remain traceable, testable, auditable, and subordinate to human final authority.	System test + review	End-to-end result plus named independent authority, scope, rationale, reservations, and time.

Failure semantics

FAILURE CONDITION	REQUIRED BEHAVIOR
Scope remains uncertain	Apply the broader credible containment boundary and communicate uncertainty explicitly.
Primary AI path unavailable	Enter the approved continuity mode; never simulate normal service with ungoverned substitutes.
Closure evidence incomplete	Keep the incident and affected conformance state open and restrict restoration or scope expansion.

Assurance evidence

- Incident exercises cover harmful output, approval bypass, data leakage, provider compromise, agent loop, model withdrawal, drift, and widespread revocation.
- Continuity tests prove human-only and deterministic fallback modes preserve authority, evidence, policy, communication, and recovery state.
- Impact fixtures trace seeded defects through products, decisions, commitments, exports, evaluations, and learning artifacts.
- Governance review verifies requirement, invariant, engineering, component, agent, model, risk, evaluation, and deployment traceability before baseline approval.

CHAPTER ACCEPTANCE AND INHERITANCE

AI Incident, Continuity, and Governance Closure is conformant only when every requirement is implemented for the declared scope, failure states are observable and recoverable, and evidence resolves to the exact released configuration. Primary Volume 4 engineering anchors: Ch. 6, Ch. 11, Ch. 14, Ch. 23, Ch. 43, Ch. 45, Ch. 48, Ch. 51, Ch. 55, Ch. 63, Ch. 68, Ch. 71, Ch. 72.

PART APP

AI Architecture Reference Appendices

Normative component, agent, model, manifest, tool, evaluation, risk, explanation, parity, traceability, decision, and terminology catalogs.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

A AI Component Catalog

The 57 stable AI component identities realize the model, agent, context, evaluation, trust, learning, and AI operations planes. They map to the ten canonical layers and Volume 4 engineering owners; they do not create an eleventh product layer.

ID	PLANE	COMPONENT	LAYER	V4 ANCHOR	STATE	RESPONSIBILITY
AI-C001	Model Control	AI Artifact Registry	Intelligence Layer	V4 Ch. 14, 45	Canonical control state	Own immutable model, adapter, prompt, evaluator, and release identities and lifecycle.
AI-C002	Model Control	Task Contract Registry	Intelligence Layer	V4 Ch. 20, 45	Canonical control state	Own task semantics, input/output schemas, consequence scope, fitness, and admission rules.
AI-C003	Model Control	Model Gateway	Intelligence Layer	V4 Ch. 14, 20	Stateless gateway	Resolve approved model paths and preserve provider-independent inference contracts.
AI-C004	Model Control	Model Resolver	Intelligence Layer	V4 Ch. 14, 23	Decision cache	Select artifacts by task, policy, quality, sovereignty, capacity, cost, and availability.
AI-C005	Model Control	Provider Adapter Runtime	Intelligence Layer	V4 Ch. 14, 20	Stateless adapter	Translate canonical requests and responses without leaking provider semantics.
AI-C006	Model Control	Prompt and Instruction Registry	Intelligence Layer	V4 Ch. 14, 45	Canonical control state	Own signed prompt, instruction, template, precedence, and compatibility artifacts.
AI-C007	Model Control	Structured Output Validator	Intelligence Layer	V4 Ch. 20, 27	Stateless validator	Validate schema, semantics, citations, uncertainty, policy, and product admission.
AI-C008	Model Control	Inference Policy Enforcer	Intelligence Layer	V4 Ch. 21, 43	Policy decision cache	Enforce data, model, provider, residency, purpose, and consequence constraints.
AI-C009	Model Control	Inference Capacity Broker	Intelligence Layer	V4 Ch. 23, 58	Scheduling state	Allocate accelerator and provider capacity by priority, budgets, deadlines, and sovereignty.
AI-C010	Model Control	Inference Economics Ledger	Intelligence Layer	V4 Ch. 12, 23	Immutable usage ledger	Attribute tokens, compute, latency, energy, and provider cost to tasks and products.
AI-C011	Agent Control	Agent Package Registry	Intelligence Layer	V4 Ch. 14, 45	Canonical control state	Own agent identity, manifest, capabilities, dependencies, evaluation, and lifecycle.
AI-C012	Agent Control	Agent Admission Service	Intelligence Layer	V4 Ch. 14, 39	Admission decision	Admit exact agent packages for task, consequence, user, deployment, and policy scope.
AI-C013	Agent Control	Planner Runtime	Intelligence Layer	V4 Ch. 14, 39	Run-scoped plan state	Construct versioned task graphs under purpose, authority, budgets, and stop conditions.
AI-C014	Agent Control	Supervisor Runtime	Intelligence Layer	V4 Ch. 14, 39	Execution control state	Admit, monitor, constrain, suspend, checkpoint, recover, and terminate agent runs.
AI-C015	Agent Control	Durable Workflow Runtime	Intelligence Layer	V4 Ch. 18, 39	Workflow history	Coordinate tasks, waits, human gates, retries, compensation, and terminal evidence.
AI-C016	Agent Control	Agent Message Router	Intelligence Layer	V4 Ch. 14, 20	Durable message state	Route typed, authenticated, policy-bound messages with causation and delivery semantics.
AI-C017	Agent Control	Capability Grant Service	Intelligence Layer	V4 Ch. 21, 43	Canonical authorization state	Issue short-lived least-privilege grants for tools, objects, resources, and side effects.
AI-C018	Agent Control	Tool Gateway	Intelligence Layer	V4 Ch. 14, 20, 43	Stateless enforcement	Validate typed tool calls and isolate external protocols, credentials, and effects.
AI-C019	Agent Control	Execution Sandbox	Intelligence Layer	V4 Ch. 15, 21	Ephemeral isolated state	Constrain code, files, network, process, resource, and secret access per run.
AI-C020	Agent Control	Human Gate Service	Decision Intelligence	V4 Ch. 39, 43	Canonical decision state	Bind a named human action to exact context, authority, proposal, and effect digest.

ID	PLANE	COMPONENT	LAYER	V4 ANCHOR	STATE	RESPONSIBILITY
AI-C021	Agent Control	Checkpoint and Replay Store	Enterprise Memory	V4 Ch. 18, 48	Immutable execution history	Preserve plans, context, messages, tools, reviews, products, and terminal state.
AI-C022	Agent Control	Agent Budget Manager	Intelligence Layer	V4 Ch. 23, 39	Run-scoped control state	Enforce time, token, compute, tool, fan-out, retry, and financial budgets.
AI-C023	Context Control	Context Assembly Service	Intelligence Layer	V4 Ch. 20, 27	Ephemeral manifest state	Assemble minimum authorized task context with time, provenance, and token budgets.
AI-C024	Context Control	Hybrid Retrieval Orchestrator	Intelligence Layer	V4 Ch. 14, 27	Run-scoped retrieval state	Coordinate lexical, semantic, graph, temporal, geospatial, and structured retrieval.
AI-C025	Context Control	Retrieval Policy Filter	Intelligence Layer	V4 Ch. 21, 43	Policy-filtered projection	Enforce purpose, access, rights, residency, temporal, and evidence boundaries before ranking.
AI-C026	Context Control	Graph Context Service	Knowledge Graph	V4 Ch. 27, 34	Pinned graph projection	Provide authorized, temporal, evidence-bearing subgraphs and reasoning paths.
AI-C027	Context Control	Memory Context Service	Enterprise Memory	V4 Ch. 29, 33	Policy-filtered projection	Retrieve enterprise, strategic, and approved experience memory by class and purpose.
AI-C028	Context Control	Twin Context Service	Digital Twins	V4 Ch. 35, 36	Pinned twin snapshot	Provide versioned twin definitions, state classes, operating envelopes, and uncertainty.
AI-C029	Context Control	Temporal Boundary Resolver	Enterprise Memory	V4 Ch. 26, 30	Bitemporal query state	Resolve event, observation, valid, system, cutoff, issue, and expiry boundaries.
AI-C030	Context Control	Evidence Binding Service	Enterprise Memory	V4 Ch. 26, 28	Canonical lineage state	Bind claims and fields to exact evidence, transformations, and citation pointers.
AI-C031	Context Control	Context Compressor	Intelligence Layer	V4 Ch. 20, 27	Ephemeral derived state	Reduce context while preserving task facts, evidence alignment, uncertainty, and policy.
AI-C032	Context Control	Context Integrity Validator	Intelligence Layer	V4 Ch. 20, 27	Validation result	Detect missing, conflicting, stale, contaminated, truncated, or unauthorized context.
AI-C033	Evaluation	Evaluation Registry	Intelligence Layer	V4 Ch. 45, 46	Canonical control state	Own evaluation profiles, datasets, metrics, thresholds, reviewers, evidence, and expiry.
AI-C034	Evaluation	Evaluation Orchestrator	Intelligence Layer	V4 Ch. 39, 46	Workflow history	Execute reproducible component, task, agent, system, human, and outcome evaluations.
AI-C035	Evaluation	Benchmark and Replay Service	Enterprise Memory	V4 Ch. 48, 50	Immutable evidence set	Run curated, historical as-of, synthetic, and regression evidence sets.
AI-C036	Evaluation	Model and Agent Judge Service	Intelligence Layer	V4 Ch. 46	Derived evaluation state	Apply approved automated evaluators anchored to independent human judgment.
AI-C037	Evaluation	Calibration Service	Prediction Engine	V4 Ch. 36, 46	Versioned calibration state	Measure and apply task-, horizon-, population-, and artifact-specific calibration.
AI-C038	Evaluation	Human Review Workbench	Executive Operating System	V4 Ch. 39, 55	Canonical review state	Capture expert assessment, disagreement, rationale, and adjudication without overwriting runs.
AI-C039	Evaluation	Red Team Harness	Intelligence Layer	V4 Ch. 45, 57	Isolated exercise state	Execute adversarial suites across prompt, retrieval, agent, tool, policy, and deployment surfaces.
AI-C040	Evaluation	Safety Case Repository	Enterprise Memory	V4 Ch. 45, 46	Canonical assurance state	Bind hazards, controls, assumptions, tests, residual risk, monitoring, and approval to release.
AI-C041	Trust	AI Risk Registry	Executive Operating System	V4 Ch. 45, 56	Canonical governance state	Own task consequence, hazards, affected populations, controls, acceptances, and review.
AI-C042	Trust	Instruction Security Gateway	Intelligence Layer	V4 Ch. 21, 43	Stateless enforcement	Separate trusted instructions from untrusted content and contain injection and exfiltration.
AI-C043	Trust	AI Privacy Gateway	Intelligence Layer	V4 Ch. 21, 54	Policy decision and transform	Minimize, redact, tokenize, localize, and govern data entering AI boundaries.
AI-C044	Trust	Safety Monitor	Intelligence Layer	V4 Ch. 47, 63	Runtime fitness state	Detect envelope breaches, unsafe behavior, control failure, and compound agent anomalies.

ID	PLANE	COMPONENT	LAYER	V4 ANCHOR	STATE	RESPONSIBILITY
AI-C045	Trust	Emergency Stop Controller	Executive Operating System	V4 Ch. 43, 63	Canonical control state	Revoke resolution and capabilities, suspend runs, and preserve containment authority.
AI-C046	Trust	Explanation Product Service	Decision Intelligence	V4 Ch. 27, 39	Canonical explanation product	Construct faithful, audience-safe, contestable explanations from execution and evidence state.
AI-C047	Trust	Contest and Appeal Service	Executive Operating System	V4 Ch. 39, 48	Canonical case state	Own challenges, evidence, adjudication, correction, impact, notification, and closure.
AI-C048	Trust	AI Audit Ledger	Enterprise Memory	V4 Ch. 12, 51	Append-only audit state	Preserve identities, versions, actions, approvals, effects, integrity proofs, and custody.
AI-C049	Operations	AI Release Controller	Intelligence Layer	V4 Ch. 45, 66	Canonical release state	Promote dependency-closed manifests through evaluation, shadow, canary, and approval.
AI-C050	Operations	Artifact Distribution Service	Intelligence Layer	V4 Ch. 62, 65	Signed package state	Distribute models, agents, prompts, policies, and evaluators across deployment profiles.
AI-C051	Operations	AI Runtime Monitor	Intelligence Layer	V4 Ch. 47, 56	Metrics and fitness state	Measure task, artifact, slice, control, operational, human, cost, and outcome behavior.
AI-C052	Operations	Drift and Change Detector	Intelligence Layer	V4 Ch. 47, 56	Derived alert state	Detect input, concept, behavior, calibration, policy, provider, and outcome drift.
AI-C053	Operations	AI Incident Coordinator	Executive Operating System	V4 Ch. 63, 68	Canonical incident state	Coordinate containment, impact, communications, recovery, restoration, and lessons.
AI-C054	Operations	Learning Proposal Service	Enterprise Memory	V4 Ch. 29, 64	Canonical proposal state	Qualify feedback and outcomes into governed datasets, prompts, agents, and model changes.
AI-C055	Operations	Deployment Parity Validator	Intelligence Layer	V4 Ch. 65, 66	Conformance evidence	Run common semantic, safety, policy, replay, and operations suites across profiles.
AI-C056	Operations	AI Cost and Capacity Optimizer	Intelligence Layer	V4 Ch. 23, 58	Planning projection	Forecast capacity and recommend approved routing without weakening task fitness or sovereignty.
AI-C057	Operations	Revocation and Retirement Service	Intelligence Layer	V4 Ch. 45, 63	Canonical lifecycle state	Stop new resolution and propagate withdrawal while preserving historical replay.

B Agent Architecture Catalog

The 44 governed agent types cover acquisition, understanding, knowledge, analysis, foresight, decision, domain intelligence, proactive intelligence, executive products, orchestration, governance, learning, and assurance. Agent names describe roles—not autonomous institutional authority.

ID	AGENT	FAMILY	LAYER	PRIMARY PRODUCT	AUTHORITY BOUNDARY	SUPERVISOR / GATE
AG-001	Observation Agent	Acquisition	World Observation Layer	Observation candidate	May detect and package observations; cannot admit canonical truth.	Collection Supervisor / source admission
AG-002	Crawler Agent	Acquisition	World Observation Layer	Acquisition batch	May traverse approved source scope under robots, rights, budget, and egress policy.	Collection Supervisor / source policy
AG-003	Search Agent	Acquisition	World Observation Layer	Search result set	May query approved internal and external indexes; cannot treat ranking as evidence authority.	Research Supervisor / policy-filtered retrieval
AG-004	Collection Agent	Acquisition	World Observation Layer	Evidence package	May collect exact permitted content with custody and coverage; cannot bypass source rights.	Collection Supervisor / evidence admission
AG-005	Cleaning Agent	Understanding	Intelligence Layer	Transformation proposal	May normalize and repair under reversible lineage; cannot erase source evidence.	Understanding Supervisor / transformation validation
AG-006	Classification Agent	Understanding	Intelligence Layer	Classification assessment	May assign governed candidate labels with confidence; cannot change access policy or ontology.	Understanding Supervisor / schema admission
AG-007	Summarization Agent	Understanding	Intelligence Layer	Evidence-bound summary	May compress authorized evidence; cannot introduce uncited material claims.	Understanding Supervisor / faithfulness gate
AG-008	NER Agent	Understanding	Intelligence Layer	Entity mention set	May detect typed mentions; cannot create or merge canonical entities.	Entity Supervisor / identity-resolution gate
AG-009	Identity Resolution Agent	Understanding	Knowledge Graph	Entity resolution proposal	May rank identity candidates and preserve ambiguity; cannot commit a merge or split.	Graph Supervisor / stewardship gate
AG-010	Relationship Agent	Understanding	Intelligence Layer	Relationship proposal	May extract typed directed candidate relations with evidence; cannot write graph truth.	Graph Supervisor / relationship admission
AG-011	Knowledge Graph Agent	Knowledge	Knowledge Graph	Graph change proposal	May assemble validated graph mutations; cannot bypass graph-owner validation and commit.	Graph Supervisor / graph commit gate
AG-012	Ontology Steward Agent	Knowledge	Knowledge Graph	Ontology change proposal	May analyze coverage and compatibility; cannot publish ontology authority without steward approval.	Governance Supervisor / ontology board
AG-013	Evidence Agent	Knowledge	Enterprise Memory	Evidence binding	May resolve claims to exact source versions and spans; cannot manufacture citations.	Reasoning Supervisor / provenance validation
AG-014	Provenance Agent	Knowledge	Enterprise Memory	Lineage graph update	May construct lineage and impact edges; cannot conceal incomplete production history.	Governance Supervisor / lineage completeness
AG-015	Memory Agent	Knowledge	Enterprise Memory	Memory commit proposal	May classify and package reusable records; cannot silently persist execution memory as enterprise truth.	Memory Supervisor / ownership and retention gate
AG-016	Digital Twin Reconciliation Agent	Knowledge	Digital Twins	Twin state proposal	May estimate and reconcile state; cannot cross observed, forecast, scenario, or simulated branches.	Twin Supervisor / reconciliation gate
AG-017	Reasoning Agent	Analysis	Intelligence Layer	Evidence-backed assessment	May form hypotheses and structured assessments; cannot elevate inference to observed fact.	Reasoning Supervisor / product admission

ID	AGENT	FAMILY	LAYER	PRIMARY PRODUCT	AUTHORITY BOUNDARY	SUPERVISOR / GATE
AG-018	Dissent Agent	Analysis	Intelligence Layer	Challenge assessment	May seek counter-evidence and alternative hypotheses; cannot suppress the primary result or decide.	Reasoning Supervisor / mandatory challenge profile
AG-019	Prediction Agent	Foresight	Prediction Engine	Forecast	May estimate calibrated distributions for approved horizons; cannot claim certainty or extend beyond fitness.	Foresight Supervisor / forecast fitness gate
AG-020	Scenario Agent	Foresight	Scenario Intelligence	Scenario branch	May construct coherent plausible branches; cannot label plausibility as probability or observed truth.	Foresight Supervisor / scenario coherence gate
AG-021	Simulation Agent	Foresight	Simulation Engine	Simulation run	May execute pinned models and interventions; cannot mutate observed twin state or select policy.	Simulation Supervisor / model validity gate
AG-022	Decision Agent	Decision	Decision Intelligence	Decision package	May compare options and recommend; final institutional choice remains human.	Decision Supervisor / named human gate
AG-023	Risk Agent	Domain Intelligence	Decision Intelligence	Risk assessment	May identify exposure, likelihood, consequence, and response options; cannot accept residual risk.	Decision Supervisor / risk-owner review
AG-024	Opportunity Agent	Domain Intelligence	Decision Intelligence	Opportunity assessment	May identify and test value opportunities; cannot commit resources or objectives.	Decision Supervisor / human decision gate
AG-025	Competitor Agent	Domain Intelligence	Digital Twins	Competitor assessment	May maintain evidence-bound competitor hypotheses and twin proposals; cannot assert private intent as fact.	Domain Supervisor / analytical review
AG-026	Supply Chain Agent	Domain Intelligence	Digital Twins	Supply-chain assessment	May map dependencies and disruption exposure; cannot alter operational plans or suppliers.	Domain Supervisor / operational owner gate
AG-027	Geopolitical Agent	Domain Intelligence	Scenario Intelligence	Geopolitical assessment	May analyze actors, mechanisms, events, and plausible paths; cannot make sovereign policy decisions.	Domain Supervisor / expert review
AG-028	Technology Agent	Domain Intelligence	Scenario Intelligence	Technology assessment	May assess trajectories, capabilities, adoption, and substitution; cannot present maturity estimates as observations.	Domain Supervisor / expert review
AG-029	Cyber Agent	Domain Intelligence	Decision Intelligence	Cyber-risk assessment	May analyze defensive evidence and simulate approved response options; cannot conduct offensive action.	Security Supervisor / security authority gate
AG-030	Financial Agent	Domain Intelligence	Prediction Engine	Financial assessment	May analyze markets and exposures within licensed data and task scope; cannot execute trades or make fiduciary decisions.	Domain Supervisor / authorized human gate
AG-031	Weak Signal Agent	Proactive Intelligence	Intelligence Layer	Weak-signal candidate	May detect unusual convergent change; cannot issue an executive warning without validation.	Warning Supervisor / materiality gate
AG-032	Early Warning Agent	Proactive Intelligence	Executive Operating System	Early warning	May prepare accountable warnings with window and owner; cannot declare emergency authority.	Warning Supervisor / executive review policy
AG-033	Executive Briefing Agent	Executive Products	Executive Operating System	Executive briefing	May render decision-ready views from canonical products; cannot omit material dissent or uncertainty.	Executive Supervisor / briefing admission
AG-034	Reporting Agent	Executive Products	Executive Operating System	Controlled report	May format and distribute policy-approved products; cannot create parallel analytical truth.	Executive Supervisor / publication gate
AG-035	Planner Agent	Orchestration	Intelligence Layer	Versioned execution plan	May decompose an admitted objective into bounded tasks; cannot grant itself authority or tools.	Supervisor Agent / plan admission

ID	AGENT	FAMILY	LAYER	PRIMARY PRODUCT	AUTHORITY BOUNDARY	SUPERVISOR / GATE
AG-036	Supervisor Agent	Orchestration	Intelligence Layer	Execution control record	May admit, constrain, suspend, recover, and terminate agents; cannot waive policy or human gates.	Workflow Agent / governance policy
AG-037	Workflow Agent	Orchestration	Intelligence Layer	Durable workflow transition	May advance committed workflow state under policy; cannot infer approval from timeout or model output.	Workflow engine / state-transition contract
AG-038	Governance Agent	Governance	Executive Operating System	Governance assessment	May test conformance and prepare cases; cannot approve its own release or risk exception.	Governance authority / separation of duties
AG-039	Evaluation Agent	Governance	Intelligence Layer	Evaluation result	May execute registered metrics and assist review; cannot be sole judge of its own artifact lineage.	Evaluation authority / independent review
AG-040	Explainability Agent	Governance	Decision Intelligence	Explanation product	May render preserved evidence and execution state; cannot invent reasons or hidden cognition.	Decision Supervisor / faithfulness gate
AG-041	Policy Guard Agent	Governance	Intelligence Layer	Policy advisory signal	May identify potential control conflicts; authoritative policy enforcement remains external.	Policy service / deny-by-default boundary
AG-042	Learning Agent	Learning	Enterprise Memory	Learning proposal	May assemble verified lessons and change candidates; cannot self-modify production behavior.	Learning Supervisor / release governance
AG-043	Red Team Agent	Assurance	Intelligence Layer	Adversarial finding	May probe approved sandboxes and artifacts; cannot target unapproved systems or retain exploit access.	Safety authority / exercise control
AG-044	Audit Agent	Assurance	Enterprise Memory	Audit reconciliation	May validate execution completeness and integrity; cannot alter historical evidence.	Audit authority / independent review

C Agent-to-Layer and Product Map

Every agent terminates in a typed product or proposal owned by a canonical layer. The map below makes product admission, supervision, and human authority explicit at the role boundary.

ID	AGENT	OWNING LAYER	PRIMARY PRODUCT	CONTROL BOUNDARY
AG-001	Observation Agent	World Observation Layer	Observation candidate	Collection Supervisor / source admission
AG-002	Crawler Agent	World Observation Layer	Acquisition batch	Collection Supervisor / source policy
AG-003	Search Agent	World Observation Layer	Search result set	Research Supervisor / policy-filtered retrieval
AG-004	Collection Agent	World Observation Layer	Evidence package	Collection Supervisor / evidence admission
AG-005	Cleaning Agent	Intelligence Layer	Transformation proposal	Understanding Supervisor / transformation validation
AG-006	Classification Agent	Intelligence Layer	Classification assessment	Understanding Supervisor / schema admission
AG-007	Summarization Agent	Intelligence Layer	Evidence-bound summary	Understanding Supervisor / faithfulness gate
AG-008	NER Agent	Intelligence Layer	Entity mention set	Entity Supervisor / identity-resolution gate
AG-009	Identity Resolution Agent	Knowledge Graph	Entity resolution proposal	Graph Supervisor / stewardship gate
AG-010	Relationship Agent	Intelligence Layer	Relationship proposal	Graph Supervisor / relationship admission
AG-011	Knowledge Graph Agent	Knowledge Graph	Graph change proposal	Graph Supervisor / graph commit gate
AG-012	Ontology Steward Agent	Knowledge Graph	Ontology change proposal	Governance Supervisor / ontology board
AG-013	Evidence Agent	Enterprise Memory	Evidence binding	Reasoning Supervisor / provenance validation
AG-014	Provenance Agent	Enterprise Memory	Lineage graph update	Governance Supervisor / lineage completeness
AG-015	Memory Agent	Enterprise Memory	Memory commit proposal	Memory Supervisor / ownership and retention gate
AG-016	Digital Twin Reconciliation Agent	Digital Twins	Twin state proposal	Twin Supervisor / reconciliation gate
AG-017	Reasoning Agent	Intelligence Layer	Evidence-backed assessment	Reasoning Supervisor / product admission
AG-018	Dissent Agent	Intelligence Layer	Challenge assessment	Reasoning Supervisor / mandatory challenge profile
AG-019	Prediction Agent	Prediction Engine	Forecast	Foresight Supervisor / forecast fitness gate
AG-020	Scenario Agent	Scenario Intelligence	Scenario branch	Foresight Supervisor / scenario coherence gate
AG-021	Simulation Agent	Simulation Engine	Simulation run	Simulation Supervisor / model validity gate
AG-022	Decision Agent	Decision Intelligence	Decision package	Decision Supervisor / named human gate
AG-023	Risk Agent	Decision Intelligence	Risk assessment	Decision Supervisor / risk-owner review
AG-024	Opportunity Agent	Decision Intelligence	Opportunity assessment	Decision Supervisor / human decision gate
AG-025	Competitor Agent	Digital Twins	Competitor assessment	Domain Supervisor / analytical review
AG-026	Supply Chain Agent	Digital Twins	Supply-chain assessment	Domain Supervisor / operational owner gate
AG-027	Geopolitical Agent	Scenario Intelligence	Geopolitical assessment	Domain Supervisor / expert review
AG-028	Technology Agent	Scenario Intelligence	Technology assessment	Domain Supervisor / expert review

ID	AGENT	OWNING LAYER	PRIMARY PRODUCT	CONTROL BOUNDARY
AG-029	Cyber Agent	Decision Intelligence	Cyber-risk assessment	Security Supervisor / security authority gate
AG-030	Financial Agent	Prediction Engine	Financial assessment	Domain Supervisor / authorized human gate
AG-031	Weak Signal Agent	Intelligence Layer	Weak-signal candidate	Warning Supervisor / materiality gate
AG-032	Early Warning Agent	Executive Operating System	Early warning	Warning Supervisor / executive review policy
AG-033	Executive Briefing Agent	Executive Operating System	Executive briefing	Executive Supervisor / briefing admission
AG-034	Reporting Agent	Executive Operating System	Controlled report	Executive Supervisor / publication gate
AG-035	Planner Agent	Intelligence Layer	Versioned execution plan	Supervisor Agent / plan admission
AG-036	Supervisor Agent	Intelligence Layer	Execution control record	Workflow Agent / governance policy
AG-037	Workflow Agent	Intelligence Layer	Durable workflow transition	Workflow engine / state-transition contract
AG-038	Governance Agent	Executive Operating System	Governance assessment	Governance authority / separation of duties
AG-039	Evaluation Agent	Intelligence Layer	Evaluation result	Evaluation authority / independent review
AG-040	Explainability Agent	Decision Intelligence	Explanation product	Decision Supervisor / faithfulness gate
AG-041	Policy Guard Agent	Intelligence Layer	Policy advisory signal	Policy service / deny-by-default boundary
AG-042	Learning Agent	Enterprise Memory	Learning proposal	Learning Supervisor / release governance
AG-043	Red Team Agent	Intelligence Layer	Adversarial finding	Safety authority / exercise control
AG-044	Audit Agent	Enterprise Memory	Audit reconciliation	Audit authority / independent review

AGENT RULE

An agent may prepare a proposal, assessment, forecast, scenario, simulation, explanation, warning, or decision package. Canonical admission remains with the owning layer; consequential choice and irreversible effect remain with an authorized human.

D Model Class and Task Catalog

Model classes are interchangeable method families behind stable task and product contracts. Approval is never inherited from a class: each exact artifact and composition requires scope-specific fitness, safety, rights, sovereignty, and operational evidence.

ID	MODEL CLASS	PRIMARY TASKS	REQUIRED EVALUATION	AUTHORITY BOUNDARY
MC-001	Large language model	Language understanding, generation, structured reasoning support	Task correctness, faithfulness, calibration, safety, latency	No direct decision or tool authority
MC-002	Vision-language model	Image, diagram, map, chart, and document-image interpretation	Grounding, spatial accuracy, OCR fidelity, safety	Policy-filtered images and region-level evidence
MC-003	Speech and audio model	Transcription, diarization, event and acoustic classification	Word and speaker error, timing, language, noise robustness	Consent, biometric, and retention controls
MC-004	OCR and document model	Layout, table, handwriting, form, and document structure extraction	Field accuracy, reading order, coverage, uncertainty	Source image remains authoritative evidence
MC-005	Embedding model	Semantic representation for retrieval, clustering, matching, and novelty	Recall, neighborhood quality, drift, privacy	Never substitutes for identity, policy, or provenance
MC-006	Reranking model	Task-aware ordering of retrieved candidates	NDCG, recall at cutoff, evidence diversity, latency	Runs after authorization; ranking is not truth
MC-007	Classifier	Governed category, relevance, sensitivity, topic, and routing labels	Precision, recall, calibration, abstention, fairness	Candidate label until owning schema admits
MC-008	NER and span extractor	Entity, event, claim, quantity, and temporal mention extraction	Span and type F1, novelty, multilingual robustness	Cannot create canonical identity
MC-009	Relationship extraction model	Typed directed relationship candidates with supporting spans	Endpoint, type, direction, time, evidence accuracy	Graph stewardship required for commit
MC-010	Graph model and algorithm	Traversal, link prediction, community, influence, and graph representation	Path faithfulness, leakage, temporal validity, robustness	Canonical graph remains source of identity and relations
MC-011	Time-series forecasting model	Point, interval, quantile, and distribution forecasts	Calibration, coverage, sharpness, horizon and drift	Exact cutoff and population required
MC-012	Probabilistic and Bayesian model	Uncertainty-aware inference, updating, and risk estimation	Posterior calibration, prior sensitivity, identifiability	Assumptions and priors explicit
MC-013	Causal model	Causal structure, effect estimation, counterfactual analysis	Identification, balance, sensitivity, transportability	Causal claim requires declared assumptions
MC-014	Anomaly and weak-signal model	Novelty, change-point, outlier, convergence, and precursor detection	Detection delay, false alarms, stability, lead time	Candidate signal until mechanism and materiality review
MC-015	Optimization model	Constrained resource, routing, scheduling, portfolio, and policy option optimization	Feasibility, optimality gap, sensitivity, robustness	Objectives and constraints human-approved
MC-016	Simulation model	Discrete event, agent-based, system dynamics, Monte Carlo, and domain simulation	Validity, reproducibility, sensitivity, uncertainty	Synthetic branch isolation mandatory
MC-017	Policy or control model	Bounded adaptive selection within an approved environment	Reward validity, safety, regret, stability, off-policy risk	No self-authorized goals or external effect
MC-018	Evaluator or judge model	Assist scoring of correctness, quality, safety, and explanation	Agreement, bias, calibration, robustness, contamination	Never sole authority for own lineage
MC-019	Safety and content model	Detect unsafe, restricted, sensitive, or policy-relevant content	Recall at severity, false denial, evasion robustness	External policy enforcement remains authoritative
MC-020	Translation and localization model	Cross-language translation and terminology-controlled rendering	Meaning preservation, named entities, numbers, caveats	Source-language evidence and uncertainty preserved

E Model Registry Field Dictionary

The registry resolves every production model artifact to exact identity, lineage, task scope, provider boundary, evaluation, safety, deployment, release, monitoring, and historical replay state.

FIELD	BINDING DEFINITION	PRESENCE RULE
model_id	Stable global model artifact identity	Required unless contractually inapplicable
model_version	Immutable semantic version	Required unless contractually inapplicable
artifact_digest	Content-addressed integrity digest	Required unless contractually inapplicable
model_class	Controlled model class identifier	Required unless contractually inapplicable
provider	Owning or supplying organization	Required unless contractually inapplicable
provider_model_id	Provider-specific reference isolated by adapter	Required unless contractually inapplicable
base_model_lineage	Parent and derivative model identities	Required unless contractually inapplicable
task_contracts	Approved task and product contract references	Required unless contractually inapplicable
consequence_scope	Maximum approved consequence classes	Required unless contractually inapplicable
domain_scope	Approved sectors, populations, languages, and geographies	Required unless contractually inapplicable
deployment_profiles	Eligible SaaS, private-cloud, and on-premise profiles	Required unless contractually inapplicable
sovereignty_profile	Residency, egress, key, and administrator constraints	Required unless contractually inapplicable
data_class_scope	Permitted input and output classifications	Required unless contractually inapplicable
training_data_statement	Governed description of training sources, rights, and cutoffs	Required unless contractually inapplicable
knowledge_cutoff	Declared accessible knowledge horizon	Required unless contractually inapplicable
license_and_rights	Use, distribution, derivative, and retention rights	Required unless contractually inapplicable
inference_contract	Canonical request, response, streaming, error, and deadline profile	Required unless contractually inapplicable
context_limits	Input, output, modality, and budget constraints	Required unless contractually inapplicable
adapter_id	Approved provider adapter artifact	Required unless contractually inapplicable
prompt_compatibility	Compatible prompt and instruction artifact ranges	Required unless contractually inapplicable
evaluation_profile	Controlling evaluation plan and evidence	Required unless contractually inapplicable
fitness_state	Approved, restricted, suspended, expired, failed, or indeterminate	Required unless contractually inapplicable
calibration_profile	Task, population, horizon, and validity calibration	Required unless contractually inapplicable
safety_case	Hazards, controls, evidence, and residual-risk reference	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
privacy_profile	Retention, training use, deletion, memorization, and logging controls	Required unless contractually inapplicable
security_profile	Threat model, injection, extraction, and supply-chain evidence	Required unless contractually inapplicable
performance_profile	Latency, throughput, resource, and capacity envelope	Required unless contractually inapplicable
cost_profile	Token, compute, energy, license, and provider economics	Required unless contractually inapplicable
monitoring_profile	Runtime indicators, thresholds, slices, owners, and response	Required unless contractually inapplicable
fallback_policy	Allowed substitute and explicit degradation behavior	Required unless contractually inapplicable
release_manifest	Dependency-closed production release reference	Required unless contractually inapplicable
rollback_target	Known compatible system configuration	Required unless contractually inapplicable
owner	Accountable technical and product owner	Required unless contractually inapplicable
approvers	Independent release, AI, security, data, and domain authorities	Required unless contractually inapplicable
effective_time	Start of permitted resolution	Required unless contractually inapplicable
expiry_time	Mandatory re-evaluation or withdrawal time	Required unless contractually inapplicable
revocation_state	Current withdrawal and propagation state	Required unless contractually inapplicable
historical_locator	Replay-safe immutable artifact location	Required unless contractually inapplicable

F Agent Manifest Field Dictionary

The manifest defines an agent as a controlled execution package. A role label without the fields below is not a deployable agent identity.

FIELD	BINDING DEFINITION	PRESENCE RULE
agent_id	Stable agent type identity	Required unless contractually inapplicable
agent_version	Immutable package version	Required unless contractually inapplicable
package_digest	Signed package integrity digest	Required unless contractually inapplicable
role	Controlled analytical or orchestration role	Required unless contractually inapplicable
owning_layer	Canonical layer that owns the produced product	Required unless contractually inapplicable
task_contracts	Permitted task and product schemas	Required unless contractually inapplicable
purpose_scope	Approved institutional purposes	Required unless contractually inapplicable
consequence_scope	Permitted consequence classes	Required unless contractually inapplicable
authority_boundary	Acts the agent may propose, prepare, or perform	Required unless contractually inapplicable
prohibited_actions	Explicit non-delegable and unsafe behaviors	Required unless contractually inapplicable
input_contracts	Accepted canonical objects and context	Required unless contractually inapplicable
output_contracts	Produced proposals and canonical product candidates	Required unless contractually inapplicable
model_resolution_policy	Eligible model classes and fitness constraints	Required unless contractually inapplicable
instruction_bundle	Signed instruction and prompt references	Required unless contractually inapplicable
retrieval_profile	Sources, methods, policy, temporal, and budget rules	Required unless contractually inapplicable
memory_profile	Run, enterprise, strategic, and experience-memory behavior	Required unless contractually inapplicable
tool_capabilities	Registered tools and operations that may be requested	Required unless contractually inapplicable
side_effect_class	Maximum reachable effect and required gate	Required unless contractually inapplicable
supervisor_policy	Admission, monitoring, escalation, stop, and recovery rules	Required unless contractually inapplicable
human_gates	Named gate types and authority requirements	Required unless contractually inapplicable
planning_policy	Allowed decomposition, fan-out, delegation, and replanning	Required unless contractually inapplicable
coordination_policy	Permitted agent peers, messages, and quorum patterns	Required unless contractually inapplicable
time_budget	Wall-clock, deadline, and wait bounds	Required unless contractually inapplicable
compute_budget	Token, inference, accelerator, and cost limits	Required unless contractually inapplicable
tool_budget	Call, retry, rate, and resource limits	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
sandbox_profile	File, network, process, secret, and execution constraints	Required unless contractually inapplicable
uncertainty_policy	Confidence, partial result, and abstention behavior	Required unless contractually inapplicable
explanation_profile	Required explanation modes and audience contracts	Required unless contractually inapplicable
evaluation_profile	Task, trajectory, safety, and system tests	Required unless contractually inapplicable
risk_profile	Hazards, misuse cases, affected populations, and controls	Required unless contractually inapplicable
monitoring_profile	Runtime invariants, telemetry, thresholds, and sampling	Required unless contractually inapplicable
checkpoint_policy	Durable state and resumability rules	Required unless contractually inapplicable
failure_policy	Retry, compensate, escalate, suspend, and terminal semantics	Required unless contractually inapplicable
owner	Accountable product and technical owner	Required unless contractually inapplicable
approval_state	Current scope-specific governance decision	Required unless contractually inapplicable
release_manifest	Dependency-closed production release	Required unless contractually inapplicable
effective_and_expiry	Permitted resolution window	Required unless contractually inapplicable
revocation_and_retirement	Withdrawal, migration, and historical replay state	Required unless contractually inapplicable

G Context Manifest Schema

The context manifest closes the information, temporal, provenance, policy, quality, and integrity boundary used for an inference or agent step.

FIELD	BINDING DEFINITION	PRESENCE RULE
context_id	Stable execution-context identity	Required unless contractually inapplicable
task_id	Owning admitted task	Required unless contractually inapplicable
tenant_and_domain	Isolation and policy scope	Required unless contractually inapplicable
requester_and_purpose	Authenticated actor and institutional purpose	Required unless contractually inapplicable
assembly_time	Trusted time of context closure	Required unless contractually inapplicable
as_of_system_time	Historical information boundary	Required unless contractually inapplicable
valid_time_scope	World-time applicability interval	Required unless contractually inapplicable
knowledge_cutoff	Model and source knowledge horizon	Required unless contractually inapplicable
policy_version	Resolved authorization and data-use policy	Required unless contractually inapplicable
classification_ceiling	Maximum permitted sensitivity	Required unless contractually inapplicable
residency_and_egress	Location and provider boundaries	Required unless contractually inapplicable
source_collections	Approved retrieval collections and revisions	Required unless contractually inapplicable
retrieval_queries	Versioned lexical, vector, graph, temporal, and structured queries	Required unless contractually inapplicable
retrieval_results	Authorized ranked candidates and scores	Required unless contractually inapplicable
graph_revision	Pinned graph and ontology versions	Required unless contractually inapplicable
twin_snapshots	Pinned twin definitions and state branches	Required unless contractually inapplicable
memory_references	Enterprise, strategic, experience, and run-memory pointers	Required unless contractually inapplicable
evidence_bindings	Claim, field, source, span, and lineage links	Required unless contractually inapplicable
truth_states	Observed, inferred, assessed, forecast, scenario, and simulated labels	Required unless contractually inapplicable
quality_state	Freshness, completeness, coverage, validity, and confidence	Required unless contractually inapplicable
conflicts_and_dissent	Contradictory evidence and alternative hypotheses	Required unless contractually inapplicable
missing_information	Known absent, withheld, stale, or unresolved context	Required unless contractually inapplicable
compression_steps	Summaries, truncation, prioritization, and preserved mapping	Required unless contractually inapplicable
token_and_modality_budget	Allocated and consumed context resources	Required unless contractually inapplicable
instruction_bundle	Trusted control artifacts supplied separately	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
untrusted_content_labels	Origins and trust states of content	Required unless contractually inapplicable
privacy_transforms	Redaction, tokenization, aggregation, and localization	Required unless contractually inapplicable
integrity_digest	Digest of ordered context and metadata	Required unless contractually inapplicable
expiry_and_invalidations	Refresh, withdrawal, correction, and dependency triggers	Required unless contractually inapplicable
audit_and_replay_locator	Protected complete manifest location	Required unless contractually inapplicable

H AI Execution and Message Envelope

The execution envelope preserves identity, authority, causation, artifact resolution, context, budgets, capabilities, human gates, effects, products, and terminal state across model, agent, workflow, and tool boundaries.

FIELD	BINDING DEFINITION	PRESENCE RULE
execution_id	Stable run identity	Required unless contractually inapplicable
parent_execution_id	Delegating or resuming run	Required unless contractually inapplicable
workflow_instance_id	Owning durable workflow	Required unless contractually inapplicable
task_contract_id	Exact task semantics	Required unless contractually inapplicable
agent_id_and_version	Resolved agent package	Required unless contractually inapplicable
release_manifest_id	Dependency-closed release	Required unless contractually inapplicable
tenant_domain_purpose	Isolation and intended use	Required unless contractually inapplicable
requester_identity	Authenticated initiating principal	Required unless contractually inapplicable
delegation_chain	Human and workload authority lineage	Required unless contractually inapplicable
consequence_class	Effective consequence control profile	Required unless contractually inapplicable
input_object_refs	Immutable canonical inputs	Required unless contractually inapplicable
context_manifest_id	Closed context boundary	Required unless contractually inapplicable
plan_id_and_revision	Approved execution plan	Required unless contractually inapplicable
step_id_and_attempt	Current durable step and try	Required unless contractually inapplicable
model_resolution	Selected models, adapters, prompts, and policies	Required unless contractually inapplicable
capability_grants	Short-lived tool and resource authority	Required unless contractually inapplicable
human_gate_state	Required, pending, approved, rejected, expired, or revoked	Required unless contractually inapplicable
deadline_and_expiry	Time limits and stale boundary	Required unless contractually inapplicable
budgets	Token, time, compute, tool, fan-out, retry, and cost limits	Required unless contractually inapplicable
checkpoint_ref	Last committed resumable state	Required unless contractually inapplicable
message_sequence	Ordered inter-agent causation	Required unless contractually inapplicable
tool_request_digest	Typed call identity and exact parameters	Required unless contractually inapplicable
tool_effect_receipt	Attempted and actual external effect	Required unless contractually inapplicable
output_product_refs	Proposed and admitted outputs	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
uncertainty_and_completion	Confidence, partial, abstained, degraded, or failed state	Required unless contractually inapplicable
explanation_ref	Faithful explanation product	Required unless contractually inapplicable
policy_decisions	Resolved allow, deny, and obligation evidence	Required unless contractually inapplicable
evaluation_results	Inline validation and safety checks	Required unless contractually inapplicable
trace_and_audit	Correlation, causation, telemetry, and ledger references	Required unless contractually inapplicable
terminal_state	Completed, failed, suspended, cancelled, compensated, or unresolved	Required unless contractually inapplicable
integrity_digest	Digest over material execution state	Required unless contractually inapplicable
retention_and_deletion	Lifecycle and legal-hold obligations	Required unless contractually inapplicable

I Tool Capability and Side-Effect Classes

Tool classes make reachable authority visible before execution. Grants become more restrictive as an operation moves from observation toward consequential and irreversible external effect.

CLASS	CAPABILITY	PERMITTED BEHAVIOR	PRIMARY RISK	MINIMUM GATE
TC-00	Observe only	Read public or approved non-sensitive data	No external mutation	Task admission
TC-01	Restricted read	Read protected internal object or source	Disclosure and inference risk	Purpose + object policy
TC-02	Search	Query approved indexes or external search	Query disclosure and ranking bias	Source, egress, budget policy
TC-03	Retrieve evidence	Fetch exact source content or binary	Rights, malware, and content risk	Custody + content isolation
TC-04	Graph query	Traverse governed knowledge graph	Protected relationship inference	Graph policy + budget
TC-05	Twin query	Read pinned twin state	Stale or branch-confused state	Snapshot + branch validation
TC-06	Compute	Run bounded deterministic analytics	Resource exhaustion or unsafe code	Sandbox + resource limits
TC-07	Model inference	Invoke approved model task	Data egress, unsafe or unfit output	Model resolver + output admission
TC-08	Create proposal	Write non-canonical draft or candidate	Misclassification as approved state	Proposal namespace + watermark
TC-09	Canonical state proposal	Submit object or graph change	Truth corruption	Owning-component validation
TC-10	Internal reversible write	Update run, queue, case, or workspace	Workflow corruption	Idempotency + rollback
TC-11	Internal consequential write	Change decision-active internal state	Material institutional reliance	Human or steward gate
TC-12	External communication draft	Prepare email, message, report, or notice	False attribution or disclosure	Publication review
TC-13	External communication send	Transmit approved content	Irreversible disclosure or commitment	Named human approval + receipt
TC-14	Financial or contractual action	Commit funds, trade, procure, or contract	Financial and legal consequence	Non-delegable human authority
TC-15	Operational control action	Change infrastructure, process, or physical operation	Safety, mission, or availability consequence	Authorized operator gate
TC-16	Security action	Change defensive control or isolate asset	Business interruption or access loss	Security authority + emergency policy
TC-17	Prohibited autonomous effect	Offensive, coercive, final high-impact, or unbounded act	Unacceptable authority transfer	Deny; no grant exists

J Evaluation Metric Catalog

Metrics establish a common vocabulary across model, agent, product, workflow, human-AI system, operational, and outcome evaluation. Each evaluation profile defines population, slice, estimator, uncertainty, threshold, evidence, and decision rule.

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
EM-001	Correctness	Exact task and field correctness against governed reference	Task + population + consequence + artifact + validity
EM-002	Evidence faithfulness	Material claims supported by resolved evidence	Task + population + consequence + artifact + validity
EM-003	Citation precision	Citations support the associated claim	Task + population + consequence + artifact + validity
EM-004	Citation coverage	Material claims with required evidence binding	Task + population + consequence + artifact + validity
EM-005	Retrieval recall	Relevant authorized evidence recovered before cutoff	Task + population + consequence + artifact + validity
EM-006	Retrieval diversity	Coverage across independent relevant evidence sources	Task + population + consequence + artifact + validity
EM-007	Grounding accuracy	Entities, relations, time, graph, and twin state correctly resolved	Task + population + consequence + artifact + validity
EM-008	Temporal integrity	No hindsight or cutoff contamination	Task + population + consequence + artifact + validity
EM-009	Calibration	Predicted confidence or probability matches observed frequency	Task + population + consequence + artifact + validity
EM-010	Interval coverage	Forecast intervals contain outcomes at declared rate	Task + population + consequence + artifact + validity
EM-011	Sharpness	Forecast concentration conditional on valid calibration	Task + population + consequence + artifact + validity
EM-012	Abstention quality	Selective risk and coverage at abstention thresholds	Task + population + consequence + artifact + validity
EM-013	Robustness	Performance under noise, ambiguity, missing data, and perturbation	Task + population + consequence + artifact + validity
EM-014	Out-of-distribution detection	Identification of unsupported inputs and operating conditions	Task + population + consequence + artifact + validity
EM-015	Consistency	Stable semantic behavior across equivalent inputs and deployments	Task + population + consequence + artifact + validity
EM-016	Agent task success	Completion of admitted objective within contract	Task + population + consequence + artifact + validity
EM-017	Trajectory validity	Plan, delegation, tools, gates, and state transitions remain conformant	Task + population + consequence + artifact + validity
EM-018	Tool-call accuracy	Correct operation, parameters, target, and preconditions	Task + population + consequence + artifact + validity
EM-019	Side-effect integrity	Executed effect exactly matches authorized request	Task + population + consequence + artifact + validity
EM-020	Safety violation rate	Unacceptable or controlled hazard occurrences	Task + population + consequence + artifact + validity
EM-021	Injection resistance	Resistance to direct and indirect control manipulation	Task + population + consequence + artifact + validity
EM-022	Privacy leakage	Disclosure, memorization, inference, or reconstruction risk	Task + population + consequence + artifact + validity
EM-023	Fairness by slice	Task-appropriate error and impact differences across affected groups	Task + population + consequence + artifact + validity
EM-024	Explanation faithfulness	Explanations match preserved execution and evidence	Task + population + consequence + artifact + validity
EM-025	Explanation comprehension	Representative users understand meaning and uncertainty	Task + population + consequence + artifact + validity

ID	METRIC	MEASURED MEANING	MANDATORY SCOPE
EM-026	Human override quality	Appropriate challenge, correction, and non-reliance behavior	Task + population + consequence + artifact + validity
EM-027	Decision utility	Improvement in governed human decision process and outcome proxies	Task + population + consequence + artifact + validity
EM-028	Latency	End-to-end and stage response distributions	Task + population + consequence + artifact + validity
EM-029	Reliability	Availability, completion, recovery, duplicate, and data-loss behavior	Task + population + consequence + artifact + validity
EM-030	Cost efficiency	Quality- and consequence-adjusted compute, provider, and human cost	Task + population + consequence + artifact + validity
EM-031	Drift sensitivity	Detection power and delay for material change	Task + population + consequence + artifact + validity
EM-032	Recovery fitness	Restoration of exact semantic and authority boundary	Task + population + consequence + artifact + validity

K AI Risk and Control Matrix

The baseline risk matrix connects recurrent AI failure and misuse modes to primary preventive, detective, containment, and recovery controls. Task safety cases refine severity and residual risk without removing these control families.

ID	RISK	REQUIRED CONTROL SET	EVIDENCE EXPECTATION
AR-001	Unsupported factual claim	Evidence binding, product admission, abstention, correction impact	Adversarial test + runtime monitor + owned response
AR-002	Fabricated or misaligned citation	Citation resolver, span validation, provenance gate	Adversarial test + runtime monitor + owned response
AR-003	Identity or relationship error	Candidate state, confidence, stewardship, merge/split history	Adversarial test + runtime monitor + owned response
AR-004	Temporal leakage or hindsight	As-of resolution, cutoff manifest, historical replay tests	Adversarial test + runtime monitor + owned response
AR-005	Uncalibrated certainty	Calibration profile, uncertainty schema, selective prediction	Adversarial test + runtime monitor + owned response
AR-006	Model or agent out of distribution	Operating envelope, OOD detection, abstention	Adversarial test + runtime monitor + owned response
AR-007	Prompt or indirect injection	Instruction separation, untrusted-content labels, tool enforcement	Adversarial test + runtime monitor + owned response
AR-008	Sensitive-data disclosure	Purpose binding, minimization, provider policy, output filter	Adversarial test + runtime monitor + owned response
AR-009	Model memorization or reconstruction	Privacy evaluation, restricted retention, model controls	Adversarial test + runtime monitor + owned response
AR-010	Cross-tenant or cross-domain leakage	Isolation, policy-filtered retrieval, cache and memory tests	Adversarial test + runtime monitor + owned response
AR-011	Poisoned source or feedback	Source authority, evidence custody, anomaly and dataset review	Adversarial test + runtime monitor + owned response
AR-012	Unsafe plan or goal drift	Plan admission, supervisor invariants, checkpoints, stop	Adversarial test + runtime monitor + owned response
AR-013	Agent coordination loop	Message graph limits, loop detection, budgets, escalation	Adversarial test + runtime monitor + owned response
AR-014	Capability escalation	Least-privilege grants, external policy enforcement, short expiry	Adversarial test + runtime monitor + owned response
AR-015	Tool parameter manipulation	Typed schema, allowlists, target and digest validation	Adversarial test + runtime monitor + owned response
AR-016	Approval bypass	Human gate, request binding, current authority, commit revalidation	Adversarial test + runtime monitor + owned response
AR-017	Unknown or duplicate external effect	Idempotency, receipts, reconciliation, compensation	Adversarial test + runtime monitor + owned response
AR-018	Observed/synthetic state contamination	Truth-state schema, branch isolation, owner admission	Adversarial test + runtime monitor + owned response
AR-019	Misleading executive compression	Explanation product, dissent preservation, faithfulness review	Adversarial test + runtime monitor + owned response
AR-020	Automation bias	Human-factor design, alternatives, uncertainty, contestability	Adversarial test + runtime monitor + owned response
AR-021	Evaluator bias or self-approval	Independent anchors, separation of duties, multiple methods	Adversarial test + runtime monitor + owned response
AR-022	Benchmark contamination	Dataset lineage, holdout control, temporal cutoff, reproduction	Adversarial test + runtime monitor + owned response
AR-023	Performance drift	Runtime monitoring, slice thresholds, restriction workflows	Adversarial test + runtime monitor + owned response
AR-024	Provider or dependency change	Immutable release, conformance, supply-chain evidence	Adversarial test + runtime monitor + owned response
AR-025	Silent semantic fallback	Task resolver, minimum quality, explicit degradation	Adversarial test + runtime monitor + owned response

ID	RISK	REQUIRED CONTROL SET	EVIDENCE EXPECTATION
AR-026	Deployment semantic divergence	Common golden suite, profile eligibility, parity gate	Adversarial test + runtime monitor + owned response
AR-027	Unrecoverable learned behavior	Versioned learning artifacts, checkpoints, rollback, retirement	Adversarial test + runtime monitor + owned response
AR-028	Audit or lineage gap	Append-only ledger, commit coupling, replay validation	Adversarial test + runtime monitor + owned response
AR-029	Revocation propagation failure	Resolver denylist, signed bundles, offline expiry, drills	Adversarial test + runtime monitor + owned response
AR-030	Incident impact uncertainty	Dependency graph, broad containment, explicit uncertainty	Adversarial test + runtime monitor + owned response

L Explanation Product Contract

The explanation product is a governed object generated from preserved evidence and execution state. Audience renderings may simplify presentation but cannot change meaning, uncertainty, limitations, dissent, or authority.

FIELD	BINDING DEFINITION	PRESENCE RULE
explanation_id	Stable explanation product identity	Required unless contractually inapplicable
product_and_version	Exact explained product	Required unless contractually inapplicable
decision_use	Permitted use and consequence boundary	Required unless contractually inapplicable
audience_profile	Role, access, language, and accessibility rendering	Required unless contractually inapplicable
conclusion	Recommendation, assessment, forecast, warning, or result	Required unless contractually inapplicable
claim_map	Material claims and exact product fields	Required unless contractually inapplicable
evidence_map	Supporting source, object, span, feature, and version	Required unless contractually inapplicable
counter_evidence	Material contradictory or weakening evidence	Required unless contractually inapplicable
method	Models, rules, graph methods, statistics, optimization, or simulation	Required unless contractually inapplicable
model_and_agent_roles	Contribution of each computational component	Required unless contractually inapplicable
assumptions	Explicit factual, structural, causal, and operational assumptions	Required unless contractually inapplicable
alternatives	Competing hypotheses, options, scenarios, or methods	Required unless contractually inapplicable
uncertainty	Probability, confidence, evidence quality, ambiguity, and disagreement	Required unless contractually inapplicable
calibration_basis	Applicable evaluation population, horizon, and validity	Required unless contractually inapplicable
sensitivity	Material input and assumption changes that alter the result	Required unless contractually inapplicable
counterfactuals	Conditions under which conclusion or choice would change	Required unless contractually inapplicable
limitations	Operating envelope, missing information, stale state, and known failures	Required unless contractually inapplicable
dissent	Model, agent, expert, or stakeholder disagreement	Required unless contractually inapplicable
policy_and_authority	Applied constraints, obligations, and required human gate	Required unless contractually inapplicable
expected_effects	Benefits, harms, reversibility, dependencies, and time window	Required unless contractually inapplicable
provenance_ref	Complete execution and production lineage	Required unless contractually inapplicable
information_boundary	As-of, valid time, graph, twin, and context revisions	Required unless contractually inapplicable
contest_path	How to challenge, correct, review, override, reopen, or appeal	Required unless contractually inapplicable
withheld_scope	Policy-protected evidence and reason for withholding	Required unless contractually inapplicable

FIELD	BINDING DEFINITION	PRESENCE RULE
faithfulness_state	Complete, partial, failed, contested, corrected, or superseded	Required unless contractually inapplicable
integrity_digest	Digest binding explanation to product and lineage	Required unless contractually inapplicable

M Deployment Semantic Parity Matrix

SaaS, Private Cloud, and On-Premise use one AI semantic and authority contract. The physical responsibility and provider path may move; the decision, product, evidence, safety, audit, correction, replay, and learning meaning remains invariant.

CAPABILITY	SAAS	PRIVATE CLOUD	ON-PREMISE	PARITY EVIDENCE
Task and product contracts	Identical	Identical	Identical	Golden semantic suite
Agent manifests	Same schema and authority	Same schema and authority	Same schema and authority	Manifest conformance
Human final authority	Mandatory	Mandatory	Mandatory	Gate and replay suite
Model providers	Cloud and approved hosted	Customer-approved hosted or local	Local or explicitly brokered	Task-fitness equivalence
Artifact registry	Managed control plane	Federated or dedicated	Local signed mirror	Digest and lifecycle reconciliation
Context and retrieval	Policy-filtered	Policy-filtered	Policy-filtered local	Authorization and evidence suite
Data residency	Contracted regions	Customer boundary	Customer site	Data-flow verification
Network egress	Controlled service paths	Customer-approved paths	Denied or allowlisted	Egress enforcement tests
Identity and delegation	Federated	Federated/dedicated	Local/federated offline	Authority-chain suite
Policy enforcement	Shared services	Dedicated services	Local signed policy	Decision parity suite
Tool capability grants	Central enforcement	Dedicated enforcement	Local enforcement	Grant and side-effect suite
Inference capacity	Elastic shared isolation	Reserved/dedicated	Installed accelerators	Queue and degradation suite
Evaluation	Central plus tenant slices	Dedicated plus common suite	Local common suite	Reproducible evidence manifest
Monitoring	Managed telemetry	Dedicated telemetry	Local telemetry/export	Indicator and blind-spot reconciliation
Audit	Tenant-isolated ledger	Dedicated ledger	Local append-only ledger	Integrity and export verification
Artifact update	Staged online	Controlled promotion	Signed offline import	Order, dependency, and rollback tests
Emergency revocation	Immediate resolver update	Federated/local update	Imported denylist plus expiry	Revocation drill
Disconnected operation	Not primary	Optional constrained	First-class profile	Offline authority and reconciliation
Recovery	Managed regional recovery	Customer topology recovery	Site recovery package	Application-level semantic validation
Export and exit	Portable governed artifacts	Portable governed artifacts	Customer-held artifacts	Round-trip portability and replay

N Requirement, Constitutional, and Engineering Traceability

Traceability resolves each stable AI requirement range upward to constitutional invariants and downward to Volume 4 engineering chapters and release evidence. Individual requirements inherit every anchor listed for their chapter.

CH .	AI SUBJECT	REQUIREMENT RANGE	COU NT	CONSTITUTIONAL INHERITANCE	ENGINEERING ANCHORS
1	AI Architecture Authority and Scope	AI-01-001–AI-01-005	5	C-001, C-003, C-004, C-008, C-028, C-030, C-052	V4 Ch. 1, V4 Ch. 2, V4 Ch. 4, V4 Ch. 6, V4 Ch. 14, V4 Ch. 43, V4 Ch. 45, V4 Ch. 71, V4 Ch. 72
2	AI Specification Hierarchy and Conformance	AI-02-001–AI-02-005	5	C-004, C-005, C-028, C-030, C-045, C-051, C-052	V4 Ch. 2, V4 Ch. 3, V4 Ch. 6, V4 Ch. 23, V4 Ch. 45, V4 Ch. 46, V4 Ch. 56, V4 Ch. 66, V4 Ch. 68, V4 Ch. 72
3	Human Decision Sovereignty	AI-03-001–AI-03-005	5	C-004, C-005, C-024, C-025, C-026, C-028, C-029	V4 Ch. 21, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 48, V4 Ch. 55
4	Explainability by Construction	AI-04-001–AI-04-005	5	C-005, C-010, C-012, C-021, C-024, C-035, C-047	V4 Ch. 20, V4 Ch. 21, V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 39, V4 Ch. 46, V4 Ch. 55, V4 Ch. 57
5	Model, Method, and Agent Pluralism	AI-05-001–AI-05-005	5	C-005, C-021, C-028, C-030, C-035, C-041, C-042	V4 Ch. 14, V4 Ch. 20, V4 Ch. 23, V4 Ch. 27, V4 Ch. 42, V4 Ch. 45, V4 Ch. 46, V4 Ch. 56, V4 Ch. 70
6	AI Traceability and Documentation Inheritance	AI-06-001–AI-06-005	5	C-005, C-011, C-012, C-025, C-045, C-051, C-052	V4 Ch. 3, V4 Ch. 6, V4 Ch. 20, V4 Ch. 23, V4 Ch. 28, V4 Ch. 30, V4 Ch. 46, V4 Ch. 48, V4 Ch. 68, V4 Ch. 72
7	AI System Context	AI-07-001–AI-07-005	5	C-001, C-003, C-004, C-008, C-012, C-035, C-039	V4 Ch. 4, V4 Ch. 7, V4 Ch. 8, V4 Ch. 10, V4 Ch. 11, V4 Ch. 13, V4 Ch. 15, V4 Ch. 43, V4 Ch. 45
8	AI Plane Decomposition	AI-08-001–AI-08-005	5	C-003, C-008, C-028, C-030, C-035, C-042, C-043	V4 Ch. 4, V4 Ch. 10, V4 Ch. 11, V4 Ch. 12, V4 Ch. 13, V4 Ch. 14, V4 Ch. 43, V4 Ch. 45, V4 Ch. 46
9	AI Product and Value Flow	AI-09-001–AI-09-005	5	C-002, C-005, C-010, C-011, C-012, C-025, C-044	V4 Ch. 15, V4 Ch. 20, V4 Ch. 21, V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 43, V4 Ch. 48
10	Model Control Plane	AI-10-001–AI-10-005	5	C-012, C-028, C-030, C-031, C-035, C-041, C-045	V4 Ch. 10, V4 Ch. 14, V4 Ch. 23, V4 Ch. 45, V4 Ch. 46, V4 Ch. 52, V4 Ch. 56, V4 Ch. 66, V4 Ch. 68
11	Agent Control Plane	AI-11-001–AI-11-005	5	C-004, C-028, C-029, C-031, C-035, C-045, C-046	V4 Ch. 10, V4 Ch. 23, V4 Ch. 41, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 46, V4 Ch. 55, V4 Ch. 56
12	Context Control Plane	AI-12-001–AI-12-005	5	C-005, C-009, C-010, C-011, C-012, C-015, C-040	V4 Ch. 20, V4 Ch. 24, V4 Ch. 25, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 29, V4 Ch. 33, V4 Ch. 34, V4 Ch. 45
13	Knowledge, Memory, and Twin Grounding Fabric	AI-13-001–AI-13-005	5	C-009, C-011, C-015, C-017, C-018, C-019, C-034	V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 33, V4 Ch. 34, V4 Ch. 35
14	Evaluation, Governance, and Learning Planes	AI-14-001–AI-14-005	5	C-004, C-028, C-031, C-035, C-044, C-045, C-051	V4 Ch. 2, V4 Ch. 13, V4 Ch. 46, V4 Ch. 48, V4 Ch. 55, V4 Ch. 56, V4 Ch. 63, V4 Ch. 68, V4 Ch. 71, V4 Ch. 72
15	Model Identity, Registry, and Lifecycle	AI-15-001–AI-15-005	5	C-012, C-030, C-035, C-038, C-041, C-045, C-048	V4 Ch. 23, V4 Ch. 28, V4 Ch. 45, V4 Ch. 46, V4 Ch. 52, V4 Ch. 54, V4 Ch. 56, V4 Ch. 64, V4 Ch. 68, V4 Ch. 70
16	Model Gateway, Routing, and Resolution	AI-16-001–AI-16-005	5	C-005, C-028, C-030, C-035, C-041, C-042, C-049	V4 Ch. 15, V4 Ch. 20, V4 Ch. 21, V4 Ch. 22, V4 Ch. 23, V4 Ch. 45, V4 Ch. 49, V4 Ch. 50, V4 Ch. 53, V4 Ch. 58, V4 Ch. 60
17	Model Adapters and Provider Abstraction	AI-17-001–AI-17-005	5	C-005, C-012, C-030, C-035, C-038, C-041, C-042	V4 Ch. 15, V4 Ch. 20, V4 Ch. 21, V4 Ch. 23, V4 Ch. 45, V4 Ch. 53, V4 Ch. 54, V4 Ch. 67

CH .	AI SUBJECT	REQUIREMENT RANGE	COU NT	CONSTITUTIONAL INHERITANCE	ENGINEERING ANCHORS
18	Prompt, Instruction, and Policy Artifacts	AI-18-001–AI-18-005	5	C-004, C-005, C-012, C-028, C-031, C-035, C-045	V4 Ch. 20, V4 Ch. 23, V4 Ch. 28, V4 Ch. 43, V4 Ch. 44, V4 Ch. 46, V4 Ch. 50, V4 Ch. 55, V4 Ch. 66, V4 Ch. 68
19	Context Assembly and Budgeting	AI-19-001–AI-19-005	5	C-005, C-010, C-011, C-012, C-014, C-024, C-040	V4 Ch. 21, V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 29, V4 Ch. 45, V4 Ch. 46, V4 Ch. 57, V4 Ch. 58
20	Structured Output and Product Admission	AI-20-001–AI-20-005	5	C-004, C-005, C-010, C-012, C-028, C-035, C-049	V4 Ch. 15, V4 Ch. 20, V4 Ch. 21, V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 32, V4 Ch. 43, V4 Ch. 46
21	Inference Security and Privacy Boundary	AI-21-001–AI-21-005	5	C-035, C-037, C-039, C-040, C-041, C-047, C-050	V4 Ch. 29, V4 Ch. 49, V4 Ch. 50, V4 Ch. 51, V4 Ch. 52, V4 Ch. 53, V4 Ch. 54, V4 Ch. 55, V4 Ch. 60
22	Inference Capacity, Latency, and Economics	AI-22-001–AI-22-005	5	C-014, C-021, C-032, C-042, C-043, C-049, C-050	V4 Ch. 45, V4 Ch. 46, V4 Ch. 57, V4 Ch. 58, V4 Ch. 59, V4 Ch. 60, V4 Ch. 62
23	Hybrid Retrieval Architecture	AI-23-001–AI-23-005	5	C-009, C-011, C-012, C-014, C-015, C-035, C-038	V4 Ch. 18, V4 Ch. 21, V4 Ch. 26, V4 Ch. 28, V4 Ch. 29, V4 Ch. 30, V4 Ch. 33, V4 Ch. 34, V4 Ch. 57
24	Graph Retrieval and Graph Reasoning	AI-24-001–AI-24-005	5	C-005, C-009, C-011, C-012, C-017, C-035, C-040	V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 34, V4 Ch. 46, V4 Ch. 50
25	Enterprise, Strategic, and Agent Memory Boundaries	AI-25-001–AI-25-005	5	C-011, C-015, C-016, C-025, C-031, C-040, C-044	V4 Ch. 12, V4 Ch. 24, V4 Ch. 26, V4 Ch. 28, V4 Ch. 29, V4 Ch. 30, V4 Ch. 33, V4 Ch. 43, V4 Ch. 48, V4 Ch. 64
26	Temporal and Version-Aware Context	AI-26-001–AI-26-005	5	C-005, C-011, C-020, C-021, C-022, C-023, C-025	V4 Ch. 18, V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 30, V4 Ch. 36, V4 Ch. 37, V4 Ch. 38, V4 Ch. 48
27	Entity and Relationship Grounding	AI-27-001–AI-27-005	5	C-005, C-009, C-010, C-011, C-012, C-035, C-045	V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 32, V4 Ch. 34, V4 Ch. 46
28	Digital Twin Grounding for AI	AI-28-001–AI-28-005	5	C-009, C-018, C-019, C-020, C-022, C-023, C-025	V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 35, V4 Ch. 36, V4 Ch. 37, V4 Ch. 38
29	Provenance, Citations, and Evidence Binding	AI-29-001–AI-29-005	5	C-005, C-010, C-011, C-012, C-025, C-035, C-036	V4 Ch. 20, V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 29, V4 Ch. 30, V4 Ch. 46, V4 Ch. 55
30	Agent Identity, Role, and Package	AI-30-001–AI-30-005	5	C-012, C-028, C-029, C-031, C-035, C-045, C-046	V4 Ch. 23, V4 Ch. 43, V4 Ch. 44, V4 Ch. 46, V4 Ch. 55, V4 Ch. 56, V4 Ch. 65, V4 Ch. 68, V4 Ch. 70
31	Agent Execution Lifecycle	AI-31-001–AI-31-005	5	C-004, C-011, C-012, C-025, C-028, C-029, C-043	V4 Ch. 21, V4 Ch. 22, V4 Ch. 28, V4 Ch. 41, V4 Ch. 43, V4 Ch. 44, V4 Ch. 55, V4 Ch. 59, V4 Ch. 60
32	Planner Agent Architecture	AI-32-001–AI-32-005	5	C-004, C-005, C-024, C-028, C-029, C-035, C-049	V4 Ch. 3, V4 Ch. 21, V4 Ch. 41, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 46, V4 Ch. 57
33	Supervisor Agent Architecture	AI-33-001–AI-33-005	5	C-004, C-028, C-029, C-035, C-036, C-043, C-049	V4 Ch. 10, V4 Ch. 41, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 55, V4 Ch. 59, V4 Ch. 60, V4 Ch. 62
34	Workflow Agent and Durable Orchestration	AI-34-001–AI-34-005	5	C-004, C-011, C-025, C-028, C-029, C-036, C-043	V4 Ch. 17, V4 Ch. 19, V4 Ch. 22, V4 Ch. 41, V4 Ch. 42, V4 Ch. 43, V4 Ch. 55, V4 Ch. 59, V4 Ch. 60
35	Agent Communication and Coordination Protocol	AI-35-001–AI-35-005	5	C-004, C-005, C-012, C-028, C-029, C-035, C-038	V4 Ch. 19, V4 Ch. 20, V4 Ch. 22, V4 Ch. 23, V4 Ch. 28, V4 Ch. 43, V4 Ch. 44, V4 Ch. 55
36	Tool, Capability, Sandbox, and Side-Effect Architecture	AI-36-001–AI-36-005	5	C-004, C-028, C-029, C-035, C-037, C-039, C-040	V4 Ch. 21, V4 Ch. 22, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 49, V4 Ch. 50, V4 Ch. 51, V4 Ch. 53, V4 Ch. 55
37	Human Gates and Mixed-Initiative Collaboration	AI-37-001–AI-37-005	5	C-004, C-005, C-024, C-025, C-027, C-047, C-048	V4 Ch. 21, V4 Ch. 39, V4 Ch. 40, V4 Ch. 42, V4 Ch. 43, V4 Ch. 48, V4 Ch. 55, V4 Ch. 71
38	Agent Context, Memory, Checkpointing, and Replay	AI-38-001–AI-38-005	5	C-011, C-012, C-025, C-028, C-029, C-035, C-043	V4 Ch. 22, V4 Ch. 24, V4 Ch. 28, V4 Ch. 41, V4 Ch. 43, V4 Ch. 55, V4 Ch. 60, V4 Ch. 61, V4 Ch. 64
39	Observation, Crawler, and Collection Agents	AI-39-001–AI-39-005	5	C-011, C-012, C-013, C-014, C-035, C-038, C-040	V4 Ch. 29, V4 Ch. 31, V4 Ch. 41, V4 Ch. 43, V4 Ch. 44, V4 Ch. 53, V4 Ch. 54, V4 Ch. 57

CH .	AI SUBJECT	REQUIREMENT RANGE	COU NT	CONSTITUTIONAL INHERITANCE	ENGINEERING ANCHORS
40	Search Agent	AI-40-001–AI-40-005	5	C-005, C-012, C-013, C-014, C-035, C-038, C-040	V4 Ch. 18, V4 Ch. 21, V4 Ch. 23, V4 Ch. 29, V4 Ch. 31, V4 Ch. 43, V4 Ch. 53, V4 Ch. 54, V4 Ch. 57
41	Cleaning and Classification Agents	AI-41-001–AI-41-005	5	C-010, C-011, C-012, C-014, C-035, C-040, C-045	V4 Ch. 21, V4 Ch. 27, V4 Ch. 28, V4 Ch. 29, V4 Ch. 31, V4 Ch. 32, V4 Ch. 43, V4 Ch. 46, V4 Ch. 57
42	Summarization Agent	AI-42-001–AI-42-005	5	C-005, C-010, C-011, C-012, C-015, C-035, C-047	V4 Ch. 20, V4 Ch. 21, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 32, V4 Ch. 43, V4 Ch. 46, V4 Ch. 57
43	Named Entity Recognition and Identity Resolution Agents	AI-43-001–AI-43-005	5	C-005, C-009, C-010, C-011, C-012, C-035, C-045	V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 32, V4 Ch. 34, V4 Ch. 43, V4 Ch. 46
44	Relationship Agent	AI-44-001–AI-44-005	5	C-005, C-009, C-010, C-011, C-012, C-017, C-035	V4 Ch. 24, V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 32, V4 Ch. 34, V4 Ch. 46
45	Knowledge Graph and Ontology Agents	AI-45-001–AI-45-005	5	C-009, C-011, C-012, C-017, C-035, C-038, C-051	V4 Ch. 23, V4 Ch. 26, V4 Ch. 28, V4 Ch. 30, V4 Ch. 34, V4 Ch. 46, V4 Ch. 48, V4 Ch. 67, V4 Ch. 72
46	Reasoning, Evidence, Provenance, and Dissent Agents	AI-46-001–AI-46-005	5	C-004, C-005, C-010, C-012, C-021, C-022, C-024	V4 Ch. 21, V4 Ch. 27, V4 Ch. 28, V4 Ch. 32, V4 Ch. 43, V4 Ch. 45, V4 Ch. 46, V4 Ch. 57
47	Governance, Evaluation, and Learning Agents	AI-47-001–AI-47-005	5	C-004, C-028, C-031, C-035, C-044, C-045, C-051	V4 Ch. 2, V4 Ch. 13, V4 Ch. 46, V4 Ch. 48, V4 Ch. 55, V4 Ch. 56, V4 Ch. 63, V4 Ch. 68, V4 Ch. 72
48	Prediction Agent	AI-48-001–AI-48-005	5	C-005, C-020, C-021, C-025, C-030, C-045, C-049	V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 30, V4 Ch. 35, V4 Ch. 36, V4 Ch. 45, V4 Ch. 46, V4 Ch. 48, V4 Ch. 57
49	Scenario Agent	AI-49-001–AI-49-005	5	C-005, C-019, C-020, C-021, C-022, C-024, C-026	V4 Ch. 26, V4 Ch. 27, V4 Ch. 28, V4 Ch. 35, V4 Ch. 36, V4 Ch. 37, V4 Ch. 45, V4 Ch. 46
50	Simulation Agent	AI-50-001–AI-50-005	5	C-005, C-018, C-019, C-022, C-023, C-024, C-025	V4 Ch. 22, V4 Ch. 26, V4 Ch. 28, V4 Ch. 35, V4 Ch. 37, V4 Ch. 38, V4 Ch. 45, V4 Ch. 46, V4 Ch. 58
51	Decision Agent	AI-51-001–AI-51-005	5	C-004, C-005, C-017, C-024, C-025, C-026, C-027	V4 Ch. 21, V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 39, V4 Ch. 40, V4 Ch. 42, V4 Ch. 43, V4 Ch. 48, V4 Ch. 55
52	Risk and Opportunity Agents	AI-52-001–AI-52-005	5	C-005, C-017, C-020, C-022, C-026, C-032, C-034	V4 Ch. 24, V4 Ch. 27, V4 Ch. 34, V4 Ch. 36, V4 Ch. 37, V4 Ch. 39, V4 Ch. 40, V4 Ch. 47, V4 Ch. 48, V4 Ch. 57
53	Competitor and Supply Chain Agents	AI-53-001–AI-53-005	5	C-005, C-009, C-012, C-018, C-022, C-032, C-040	V4 Ch. 29, V4 Ch. 34, V4 Ch. 35, V4 Ch. 36, V4 Ch. 37, V4 Ch. 39, V4 Ch. 47, V4 Ch. 50, V4 Ch. 54, V4 Ch. 57
54	Geopolitical and Technology Agents	AI-54-001–AI-54-005	5	C-005, C-009, C-012, C-013, C-022, C-032, C-035	V4 Ch. 23, V4 Ch. 27, V4 Ch. 28, V4 Ch. 34, V4 Ch. 35, V4 Ch. 36, V4 Ch. 37, V4 Ch. 47, V4 Ch. 50, V4 Ch. 57
55	Cyber and Financial Agents	AI-55-001–AI-55-005	5	C-004, C-005, C-011, C-028, C-035, C-039, C-050	V4 Ch. 21, V4 Ch. 26, V4 Ch. 27, V4 Ch. 36, V4 Ch. 39, V4 Ch. 42, V4 Ch. 44, V4 Ch. 47, V4 Ch. 53, V4 Ch. 55, V4 Ch. 63
56	Executive Briefing and Reporting Agents	AI-56-001–AI-56-005	5	C-005, C-012, C-025, C-027, C-034, C-047, C-048	V4 Ch. 24, V4 Ch. 27, V4 Ch. 28, V4 Ch. 34, V4 Ch. 40, V4 Ch. 47, V4 Ch. 48, V4 Ch. 55, V4 Ch. 57
57	Weak Signal and Early Warning Agents	AI-57-001–AI-57-005	5	C-002, C-005, C-020, C-022, C-032, C-033, C-034	V4 Ch. 34, V4 Ch. 36, V4 Ch. 37, V4 Ch. 39, V4 Ch. 40, V4 Ch. 47, V4 Ch. 48, V4 Ch. 57, V4 Ch. 62
58	AI Risk Taxonomy and Consequence Classes	AI-58-001–AI-58-005	5	C-004, C-005, C-024, C-028, C-029, C-043, C-052	V4 Ch. 2, V4 Ch. 6, V4 Ch. 14, V4 Ch. 21, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 45, V4 Ch. 56, V4 Ch. 68
59	Safe Action and Side-Effect Governance	AI-59-001–AI-59-005	5	C-004, C-024, C-025, C-026, C-027, C-028, C-029	V4 Ch. 14, V4 Ch. 21, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 48, V4 Ch. 55, V4 Ch. 63
60	Prompt, Instruction, and Tool Security	AI-60-001–AI-60-005	5	C-005, C-012, C-013, C-027, C-028, C-035, C-043	V4 Ch. 15, V4 Ch. 20, V4 Ch. 21, V4 Ch. 27, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 45, V4 Ch. 57

CH .	AI SUBJECT	REQUIREMENT RANGE	COUNT	CONSTITUTIONAL INHERITANCE	ENGINEERING ANCHORS
61	Data, Model, and Agent Privacy	AI-61-001–AI-61-005	5	C-005, C-013, C-014, C-015, C-016, C-027, C-035	V4 Ch. 9, V4 Ch. 15, V4 Ch. 21, V4 Ch. 24, V4 Ch. 27, V4 Ch. 29, V4 Ch. 33, V4 Ch. 45, V4 Ch. 54, V4 Ch. 64
62	Model and Agent Safety Architecture	AI-62-001–AI-62-005	5	C-004, C-024, C-027, C-028, C-029, C-043, C-049	V4 Ch. 14, V4 Ch. 21, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 44, V4 Ch. 45, V4 Ch. 48, V4 Ch. 56, V4 Ch. 63
63	Uncertainty, Calibration, Abstention, and Partial Results	AI-63-001–AI-63-005	5	C-005, C-019, C-021, C-022, C-023, C-024, C-047	V4 Ch. 20, V4 Ch. 23, V4 Ch. 27, V4 Ch. 36, V4 Ch. 37, V4 Ch. 38, V4 Ch. 39, V4 Ch. 46, V4 Ch. 56
64	Explanation, Contestability, and Human Factors	AI-64-001–AI-64-005	5	C-005, C-010, C-012, C-024, C-025, C-035, C-047	V4 Ch. 20, V4 Ch. 21, V4 Ch. 27, V4 Ch. 28, V4 Ch. 39, V4 Ch. 46, V4 Ch. 48, V4 Ch. 55, V4 Ch. 57
65	AI Audit, Non-Repudiation, and Accountability	AI-65-001–AI-65-005	5	C-004, C-010, C-025, C-026, C-028, C-036, C-048	V4 Ch. 11, V4 Ch. 12, V4 Ch. 18, V4 Ch. 21, V4 Ch. 39, V4 Ch. 43, V4 Ch. 44, V4 Ch. 48, V4 Ch. 51, V4 Ch. 55
66	AI Evaluation Architecture	AI-66-001–AI-66-005	5	C-005, C-021, C-022, C-028, C-030, C-047, C-052	V4 Ch. 20, V4 Ch. 23, V4 Ch. 27, V4 Ch. 39, V4 Ch. 45, V4 Ch. 46, V4 Ch. 47, V4 Ch. 56, V4 Ch. 66
67	Red Teaming and Safety Testing	AI-67-001–AI-67-005	5	C-013, C-024, C-027, C-028, C-029, C-043, C-049	V4 Ch. 15, V4 Ch. 21, V4 Ch. 39, V4 Ch. 42, V4 Ch. 43, V4 Ch. 45, V4 Ch. 54, V4 Ch. 57, V4 Ch. 63, V4 Ch. 68
68	Runtime Monitoring, Drift, and Fitness	AI-68-001–AI-68-005	5	C-021, C-022, C-023, C-028, C-030, C-043, C-049	V4 Ch. 23, V4 Ch. 38, V4 Ch. 45, V4 Ch. 47, V4 Ch. 56, V4 Ch. 58, V4 Ch. 59, V4 Ch. 60, V4 Ch. 61, V4 Ch. 63, V4 Ch. 68
69	Governed Learning and Adaptation	AI-69-001–AI-69-005	5	C-010, C-016, C-025, C-028, C-030, C-044, C-051	V4 Ch. 29, V4 Ch. 30, V4 Ch. 45, V4 Ch. 47, V4 Ch. 48, V4 Ch. 50, V4 Ch. 56, V4 Ch. 64, V4 Ch. 66, V4 Ch. 68
70	AI Release, Promotion, Rollback, and Retirement	AI-70-001–AI-70-005	5	C-004, C-025, C-028, C-030, C-041, C-042, C-049	V4 Ch. 3, V4 Ch. 14, V4 Ch. 23, V4 Ch. 42, V4 Ch. 45, V4 Ch. 46, V4 Ch. 56, V4 Ch. 63, V4 Ch. 66, V4 Ch. 68
71	AI Deployment Parity and Sovereignty	AI-71-001–AI-71-005	5	C-003, C-013, C-027, C-030, C-041, C-042, C-049	V4 Ch. 8, V4 Ch. 9, V4 Ch. 15, V4 Ch. 16, V4 Ch. 17, V4 Ch. 45, V4 Ch. 52, V4 Ch. 53, V4 Ch. 54, V4 Ch. 62, V4 Ch. 63, V4 Ch. 65
72	AI Incident, Continuity, and Governance Closure	AI-72-001–AI-72-005	5	C-004, C-025, C-028, C-029, C-049, C-051, C-052	V4 Ch. 6, V4 Ch. 11, V4 Ch. 14, V4 Ch. 23, V4 Ch. 43, V4 Ch. 45, V4 Ch. 48, V4 Ch. 51, V4 Ch. 55, V4 Ch. 63, V4 Ch. 68, V4 Ch. 71, V4 Ch. 72

Constitutional AI architecture matrix

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 5 ANCHORS
C-001	Category integrity	The Eye shall be conceived, designed, sold, and operated as a Strategic Intelligence Operating System. It shall not be reduced to a chatbot, dashboard, BI tool, copilot, or reporting product.	Ch. 1, Ch. 7, Ch. 19
C-002	Closed strategic loop	The product shall implement one continuous operating loop: Observe, Understand, Predict, Simulate, Decide, and Learn. No stage may be treated as a disconnected feature island.	Ch. 9, Ch. 57, Ch. 2, Ch. 20
C-003	Product boundary	Interfaces such as chat, dashboards, reports, and APIs may exist, but they are access modes to the operating system rather than the product definition.	Ch. 1, Ch. 7, Ch. 8, Ch. 71, Ch. 3, Ch. 21
C-004	Human decision sovereignty	A named human authority shall make every final strategic or high-impact decision. The system may advise, prepare, and execute approved workflows, but it shall not silently assume decision rights.	Ch. 1, Ch. 2, Ch. 3, Ch. 7, Ch. 11, Ch. 14, Ch. 70, Ch. 72, Ch. 58, Ch. 59, Ch. 62, Ch. 65, Ch. 46, Ch. 47, Ch. 51, Ch. 55, Ch. 31, Ch. 32, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 37, Ch. 18, Ch. 20, Ch. 4, Ch. 22

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 5 ANCHORS
C-005	Explain every recommendation	Every material forecast, scenario, simulation result, risk, opportunity, and recommendation shall expose evidence, assumptions, uncertainty, lineage, alternatives, and accountable system versions.	Ch. 2, Ch. 3, Ch. 4, Ch. 5, Ch. 6, Ch. 9, Ch. 12, Ch. 66, Ch. 58, Ch. 60, Ch. 61, Ch. 63, Ch. 64, Ch. 40, Ch. 42, Ch. 43, Ch. 44, Ch. 46, Ch. 48, Ch. 49, Ch. 50, Ch. 51, Ch. 52, Ch. 53, Ch. 54, Ch. 55, Ch. 56, Ch. 57, Ch. 32, Ch. 35, Ch. 37, Ch. 16, Ch. 17, Ch. 18, Ch. 19, Ch. 20, Ch. 24, Ch. 26, Ch. 27, Ch. 29, Ch. 23
C-006	Enterprise operating standard	The common platform shall satisfy enterprise requirements for governance, security, scale, auditability, reliability, and lifecycle management from its first production architecture.	Ch. 6, Ch. 24
C-007	Universal core, governed specialization	A single domain-neutral core shall serve all customer classes. Domain ontologies, models, workflows, agents, and policies may extend the core without forking its constitutional semantics.	Ch. 7, Ch. 25
C-008	Ten-layer architecture	The ten named architecture layers are the canonical product structure. Cross-cutting planes may support them but shall not replace, collapse, or rename them without constitutional amendment.	Ch. 1, Ch. 7, Ch. 8, Ch. 26
C-009	Knowledge Graph centrality	The Knowledge Graph shall be the connected semantic heart of The Eye and the shared context substrate for memory, twins, reasoning, prediction, scenarios, simulations, and decisions.	Ch. 12, Ch. 13, Ch. 43, Ch. 44, Ch. 45, Ch. 53, Ch. 54, Ch. 23, Ch. 24, Ch. 27, Ch. 28, Ch. 9
C-010	Truth-state separation	Observed evidence, extracted claims, inferred relationships, model outputs, human judgments, and synthetic scenario data shall remain explicitly distinguishable throughout the system.	Ch. 4, Ch. 9, Ch. 12, Ch. 69, Ch. 64, Ch. 65, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 46, Ch. 19, Ch. 20, Ch. 27, Ch. 29, Ch. 10, Ch. 28
C-011	Temporal truth	Material facts and relationships shall preserve event time, observation time, validity time, and system-record time so that past world states can be reconstructed without hindsight contamination.	Ch. 6, Ch. 9, Ch. 12, Ch. 13, Ch. 39, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 55, Ch. 31, Ch. 34, Ch. 38, Ch. 19, Ch. 23, Ch. 24, Ch. 25, Ch. 26, Ch. 27, Ch. 29, Ch. 11
C-012	End-to-end provenance	Every material intelligence object shall carry source, chain of custody, transformation history, ownership, rights, quality, and version lineage back to its evidence.	Ch. 4, Ch. 6, Ch. 7, Ch. 9, Ch. 10, Ch. 12, Ch. 60, Ch. 64, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 46, Ch. 53, Ch. 54, Ch. 56, Ch. 30, Ch. 31, Ch. 35, Ch. 38, Ch. 15, Ch. 17, Ch. 18, Ch. 19, Ch. 20, Ch. 23, Ch. 24, Ch. 27, Ch. 29
C-013	Observation breadth	The World Observation Layer shall support the full canonical source universe and allow governed custom sources without architectural rework.	Ch. 67, Ch. 71, Ch. 60, Ch. 61, Ch. 39, Ch. 40, Ch. 54, Ch. 13, Ch. 31
C-014	Freshness and quality contracts	Sources, derived data, twins, models, and intelligence products shall declare measurable freshness, completeness, quality, and expiry expectations.	Ch. 61, Ch. 39, Ch. 40, Ch. 41, Ch. 19, Ch. 22, Ch. 23, Ch. 14, Ch. 32
C-015	Durable institutional memory	The Eye shall preserve institutional context, decisions, assumptions, evidence, outcomes, and lessons as durable memory, subject to lawful retention, deletion, and customer policy.	Ch. 12, Ch. 13, Ch. 61, Ch. 42, Ch. 23, Ch. 25, Ch. 15, Ch. 33
C-016	Memory governance	Memory shall be permission-aware, tenant-isolated, versioned, retrievable, correctable, and protected from unverified model output becoming accepted organizational truth.	Ch. 69, Ch. 61, Ch. 25, Ch. 16, Ch. 34
C-017	Strategy Graph	Objectives, hypotheses, decisions, initiatives, capabilities, resources, indicators, risks, opportunities, and outcomes shall be connected in a governed Strategy Graph anchored to the Knowledge Graph.	Ch. 13, Ch. 44, Ch. 45, Ch. 51, Ch. 52, Ch. 24, Ch. 17, Ch. 35
C-018	Grounded Digital Twins	Every Digital Twin shall be grounded in evidence, graph entities, temporal state, declared assumptions, and calibrated models; it shall never present synthetic state as observed fact.	Ch. 13, Ch. 50, Ch. 53, Ch. 28, Ch. 18, Ch. 36
C-019	Twin versionability	Digital Twins shall support time travel, branching, comparison, replay, reconciliation, and versioned evolution across actual and simulated states.	Ch. 13, Ch. 63, Ch. 49, Ch. 50, Ch. 28, Ch. 19, Ch. 37
C-020	Canonical prediction horizons	The Prediction Engine shall support 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons with horizon-appropriate methods and uncertainty.	Ch. 48, Ch. 49, Ch. 52, Ch. 57, Ch. 26, Ch. 28, Ch. 20, Ch. 38
C-021	Probabilistic forecasting	Forecasts shall communicate distributions, confidence, calibration, drivers, sensitivity, and failure conditions rather than false precision or unsupported certainty.	Ch. 4, Ch. 5, Ch. 66, Ch. 68, Ch. 63, Ch. 46, Ch. 48, Ch. 49, Ch. 22, Ch. 26, Ch. 21, Ch. 39
C-022	Scenario plurality	Scenario Intelligence shall maintain multiple coherent futures, including baseline, upside, downside, disruption, and user-defined alternatives; no single forecast shall masquerade as destiny.	Ch. 66, Ch. 68, Ch. 63, Ch. 46, Ch. 49, Ch. 50, Ch. 52, Ch. 53, Ch. 54, Ch. 57, Ch. 26, Ch. 28, Ch. 22, Ch. 40

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 5 ANCHORS
C-023	Reproducible simulation	Every simulation shall declare initial state, assumptions, interventions, model versions, constraints, random seeds where applicable, outputs, and sensitivity so results can be reproduced and challenged.	Ch. 68, Ch. 63, Ch. 50, Ch. 26, Ch. 28, Ch. 23, Ch. 41
C-024	Decision completeness	Decision Intelligence shall compare viable options against explicit objectives, constraints, risks, opportunities, trade-offs, second-order effects, reversibility, and information value.	Ch. 3, Ch. 4, Ch. 67, Ch. 58, Ch. 59, Ch. 62, Ch. 63, Ch. 64, Ch. 46, Ch. 49, Ch. 50, Ch. 51, Ch. 32, Ch. 37, Ch. 19, Ch. 24, Ch. 42
C-025	Decision Replay	The Eye shall preserve the state of knowledge available at decision time and support replay against later evidence and outcomes without rewriting the historical record.	Ch. 3, Ch. 6, Ch. 9, Ch. 69, Ch. 70, Ch. 72, Ch. 59, Ch. 64, Ch. 65, Ch. 48, Ch. 50, Ch. 51, Ch. 56, Ch. 31, Ch. 34, Ch. 37, Ch. 38, Ch. 25, Ch. 26, Ch. 28, Ch. 29, Ch. 43
C-026	Objective alignment	Recommendations shall be evaluated against declared objectives, stakeholder obligations, risk appetite, policy, and strategic constraints rather than an opaque global score.	Ch. 3, Ch. 59, Ch. 65, Ch. 49, Ch. 51, Ch. 52, Ch. 26, Ch. 44
C-027	Executive Operating System	The executive surface shall organize strategic attention, decisions, scenarios, evidence, objectives, commitments, and outcomes. It shall not degrade into a passive dashboard collection.	Ch. 67, Ch. 71, Ch. 59, Ch. 60, Ch. 61, Ch. 62, Ch. 51, Ch. 56, Ch. 37, Ch. 27, Ch. 45
C-028	Bounded agent authority	Every agent shall have a declared purpose, tools, data scope, permissions, budgets, escalation rules, prohibited actions, evaluation suite, and accountable owner.	Ch. 1, Ch. 2, Ch. 3, Ch. 5, Ch. 8, Ch. 10, Ch. 11, Ch. 14, Ch. 66, Ch. 67, Ch. 68, Ch. 69, Ch. 70, Ch. 72, Ch. 58, Ch. 59, Ch. 60, Ch. 62, Ch. 65, Ch. 47, Ch. 55, Ch. 30, Ch. 31, Ch. 32, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 16, Ch. 18, Ch. 20, Ch. 28, Ch. 46
C-029	Supervised orchestration	Planner, Supervisor, and Workflow agents shall enforce policy, separation of duties, checkpoints, timeouts, retries, human gates, and complete execution traces across multi-agent work.	Ch. 3, Ch. 11, Ch. 67, Ch. 72, Ch. 58, Ch. 59, Ch. 62, Ch. 30, Ch. 31, Ch. 32, Ch. 33, Ch. 34, Ch. 35, Ch. 36, Ch. 38, Ch. 29, Ch. 47
C-030	Model pluralism	The platform shall support multiple models and reasoning approaches behind governed interfaces so customers retain control over cost, sovereignty, performance, and vendor dependence.	Ch. 1, Ch. 2, Ch. 5, Ch. 8, Ch. 10, Ch. 66, Ch. 68, Ch. 69, Ch. 70, Ch. 71, Ch. 48, Ch. 15, Ch. 16, Ch. 17, Ch. 30
C-031	No silent self-modification	Continuous learning shall never permit unreviewed production models, agents, prompts, policies, ontologies, or decision logic to modify themselves silently.	Ch. 10, Ch. 11, Ch. 14, Ch. 47, Ch. 30, Ch. 18, Ch. 25, Ch. 31, Ch. 49
C-032	Proactive intelligence	The system shall surface material changes, weak signals, risks, and opportunities before users ask, while respecting relevance, authority, attention, and notification policy.	Ch. 52, Ch. 53, Ch. 54, Ch. 57, Ch. 22, Ch. 32, Ch. 50
C-033	Early-warning discipline	Warnings shall be deduplicated, prioritized, evidence-backed, confidence-aware, time-bounded, and linked to affected objectives, assets, decisions, and response playbooks.	Ch. 57, Ch. 33, Ch. 51
C-034	Decomposable Strategic Health Score	Any Strategic Health Score shall expose component measures, weights, evidence, confidence, trend, and sensitivity. The score shall never replace the underlying analysis.	Ch. 13, Ch. 52, Ch. 56, Ch. 57, Ch. 34
C-035	Trust as a system property	Trust, provenance, identity, authorization, policy, safety, and audit shall operate across every layer and intelligence object, not as a downstream compliance add-on.	Ch. 4, Ch. 5, Ch. 7, Ch. 8, Ch. 10, Ch. 11, Ch. 14, Ch. 60, Ch. 61, Ch. 64, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 45, Ch. 47, Ch. 54, Ch. 55, Ch. 30, Ch. 32, Ch. 33, Ch. 35, Ch. 36, Ch. 38, Ch. 15, Ch. 16, Ch. 17, Ch. 18, Ch. 20, Ch. 21, Ch. 23, Ch. 24, Ch. 27, Ch. 29, Ch. 53
C-036	Non-repudiable accountability	Material access, transformations, agent actions, approvals, decisions, overrides, releases, and exports shall produce tamper-evident audit records attributable to identities and versions.	Ch. 65, Ch. 33, Ch. 34, Ch. 29, Ch. 36, Ch. 54
C-037	Customer data control	Customers retain governance over their data, models, twins, policies, retention, residency, encryption, exports, and deletion within the chosen deployment and contractual boundary.	Ch. 36, Ch. 21, Ch. 37, Ch. 55
C-038	Open interoperability	The Eye shall integrate through governed APIs, events, connectors, common data formats, and export paths; core customer knowledge shall not be trapped behind proprietary-only access.	Ch. 39, Ch. 40, Ch. 45, Ch. 35, Ch. 15, Ch. 17, Ch. 23, Ch. 38, Ch. 56
C-039	Zero-trust security	Identity, device, workload, agent, model, source, and data access shall be continuously authenticated, authorized, scoped, logged, and re-evaluated under least privilege.	Ch. 7, Ch. 55, Ch. 36, Ch. 21, Ch. 39, Ch. 57

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 5 ANCHORS
C-040	Privacy and minimization	Collection and use shall be purpose-bound, proportionate, policy-enforced, and minimized, with support for consent, legal basis, access, correction, retention, and deletion obligations.	Ch. 12, Ch. 39, Ch. 40, Ch. 41, Ch. 53, Ch. 36, Ch. 19, Ch. 21, Ch. 24, Ch. 25, Ch. 58
C-041	Sovereignty by design	Data residency, key ownership, network isolation, local inference, model choice, administrative control, and disconnected operation shall be supported where customer policy requires them.	Ch. 5, Ch. 10, Ch. 70, Ch. 71, Ch. 15, Ch. 16, Ch. 17, Ch. 21, Ch. 41, Ch. 59
C-042	Deployment semantic parity	SaaS, Private Cloud, and On-Premise deployments shall preserve the same core semantics, governance model, APIs, artifacts, and intelligence lifecycle, with documented capability differences only where unavoidable.	Ch. 5, Ch. 8, Ch. 70, Ch. 71, Ch. 16, Ch. 17, Ch. 22, Ch. 42, Ch. 60
C-043	Resilient operation	The platform shall fail safely, degrade intelligibly, recover predictably, preserve evidence and audit continuity, and expose when data, models, or dependencies are unavailable or stale.	Ch. 8, Ch. 67, Ch. 68, Ch. 58, Ch. 60, Ch. 62, Ch. 31, Ch. 33, Ch. 34, Ch. 38, Ch. 22, Ch. 43, Ch. 61
C-044	Governed learning loop	Learning from outcomes, feedback, overrides, and new evidence shall pass through evaluation, approval, release, monitoring, rollback, and tenant-isolation controls.	Ch. 9, Ch. 14, Ch. 69, Ch. 47, Ch. 25, Ch. 44, Ch. 62
C-045	Continuous evaluation	Models, agents, forecasts, extractors, ontologies, simulations, and recommendations shall be evaluated for accuracy, calibration, robustness, drift, bias, safety, cost, and operational fitness.	Ch. 2, Ch. 6, Ch. 10, Ch. 11, Ch. 14, Ch. 41, Ch. 43, Ch. 47, Ch. 48, Ch. 30, Ch. 15, Ch. 18, Ch. 27, Ch. 45, Ch. 63
C-046	Governed marketplaces	Scenario and Agent Marketplace packages shall be signed, versioned, inspectable, permission-scoped, licensed, evaluated, sandboxed, monitored, revocable, and traceable to publishers.	Ch. 11, Ch. 30, Ch. 46, Ch. 64
C-047	Experience integrity	The product experience shall expose context, uncertainty, provenance, decision state, and consequence at the point of use; critical meaning shall not be hidden behind ornamental visualization.	Ch. 4, Ch. 66, Ch. 63, Ch. 64, Ch. 42, Ch. 56, Ch. 37, Ch. 21, Ch. 47, Ch. 65
C-048	Role and accessibility clarity	Experiences shall be accessible and role-appropriate for executives, analysts, operators, domain experts, administrators, and governance teams without creating separate truths.	Ch. 65, Ch. 56, Ch. 37, Ch. 15, Ch. 48, Ch. 66
C-049	Operational quality contract	Reliability, performance, scale, freshness, accuracy, calibration, explainability, reproducibility, security, accessibility, and cost shall be measured as product behavior, not assumed as implementation detail.	Ch. 67, Ch. 68, Ch. 70, Ch. 71, Ch. 72, Ch. 62, Ch. 48, Ch. 32, Ch. 33, Ch. 16, Ch. 20, Ch. 22, Ch. 49
C-050	Responsible-use boundary	The Eye shall not autonomously execute decisions that materially determine life, liberty, legal status, access to essential services, employment, credit, or equivalent high-impact rights and interests.	Ch. 55, Ch. 21, Ch. 22, Ch. 50, Ch. 68
C-051	Constitutional change control	Changes to mission, product category, human authority, principles, canonical layers, core semantics, or constitutional invariants require explicit amendment, impact analysis, ratification, and versioned publication.	Ch. 2, Ch. 6, Ch. 14, Ch. 69, Ch. 72, Ch. 45, Ch. 47, Ch. 51
C-052	Documentation inheritance	Volumes 1 through 10 shall inherit this constitution, use its canonical terms, trace material requirements to invariant IDs, and identify rather than conceal any unresolved conflict.	Ch. 1, Ch. 2, Ch. 6, Ch. 66, Ch. 72, Ch. 58, Ch. 52, Ch. 70

O AI Architecture Decision Baseline

These decisions freeze the governing AI architecture beneath Volumes 0, 3, and 4. Implementations may change technologies and providers behind them but cannot treat a local exception as authority to change their semantics.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
AI-ADR-001	AI remains subordinate to the ten canonical layers	No AI-owned parallel product or truth plane	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-002	Human final authority is non-delegable	Consequential decisions and irreversible effects require named human action	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-003	Task contracts are provider-independent	Models and tools are replaceable dependencies behind canonical semantics	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-004	Knowledge Graph is the shared semantic heart	Identity and relationships resolve through governed graph contracts	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-005	Context is a signed execution artifact	Every result binds a reproducible temporal, policy, and evidence boundary	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-006	Enterprise, strategic, run, and experience memory remain distinct	Authority, retention, correction, and reuse cannot blur	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-007	Agents are versioned packages, not personas	Identity, role, authority, tools, controls, and evidence are explicit	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-008	Supervision and policy enforcement remain outside governed agents	No agent self-authorizes or serves as its sole safety boundary	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-009	Workflow state is durable	Long-running execution, waits, gates, retries, and recovery are replayable	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-010	Tool use is capability-based	Least-privilege grants and typed adapters govern every material operation	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-011	Preparation and effect are separate	External action revalidates authority and produces a receipt	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-012	Inference never commits canonical truth directly	Owning layers validate and admit model and agent proposals	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-013	Uncertainty and abstention are product state	Incomplete fitness cannot be hidden by fluent output	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-014	Explanation is built from execution evidence	Post-hoc persuasive narratives cannot substitute for faithful lineage	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-015	Evaluation addresses the complete human-AI system	Component scores alone cannot authorize consequential use	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.

ADR	DECISION	BINDING CONSEQUENCE	IMPLEMENTATION OBLIGATION
AI-ADR-016	Evaluator models are governed dependencies	Independent human anchors and separation of duties remain mandatory	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-017	Learning produces released artifacts	Runtime feedback never silently changes production behavior	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-018	One release manifest closes behavioral dependencies	Model-only promotion and rollback are prohibited	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-019	Runtime fitness can withdraw authorization	Monitoring breaches create durable restriction, not alerts alone	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-020	Deployment models preserve semantic parity	Sovereignty changes placement and operation, not product meaning	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-021	Disconnected operation is controlled and reconcilable	Signed bundles, local enforcement, expiry, and later reconciliation are required	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-022	Audit and Decision Replay share immutable execution identity	Original information boundaries survive later correction and learning	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-023	AI incidents traverse product and decision impact	Containment includes outputs, commitments, exports, and learned artifacts	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.
AI-ADR-024	No provider, model, or agent is constitutionally privileged	Selection follows current task fitness, policy, and sovereignty	Implement through owned components, versioned contracts, external controls, conformance evidence, release state, monitoring, rollback, and audit.

P AI Glossary

These terms are canonical for models, agents, context, evaluation, safety, governance, delivery, and AI operations. Domain specifications may add narrower terms but may not redefine the concepts below.

TERM	AI ARCHITECTURE DEFINITION
Abstention	Explicit refusal to produce or admit a result because fitness, evidence, policy, calibration, or integrity is insufficient.
Agent	Versioned execution package that performs a bounded role under task, context, models, tools, policy, supervision, budgets, and stop conditions.
Agent package	Immutable agent manifest, instructions, code, schemas, dependencies, evaluation, controls, and release metadata.
Agent trajectory	Ordered plan, messages, context, model calls, tools, checkpoints, reviews, outputs, and state transitions of one execution.
AI artifact	Versioned model, adapter, prompt, agent, evaluator, dataset, calibrator, rule, policy proposal, or learned configuration.
AI product	Canonical intelligence object produced or assisted by AI and admitted through its owning layer.
AI release	Dependency-closed approved configuration of all behavior-affecting AI artifacts and controls.
Calibration	Measured relationship between predicted confidence or probability and observed frequency for a declared population.
Capability grant	Short-lived least-privilege authorization for an exact tool, operation, target, purpose, limit, identity, and time.
Consequence class	Control class derived from potential harm, scale, irreversibility, vulnerability, and reachable effects of a precise use.
Context	Authorized, temporal, provenance-bearing information and control boundary supplied to an AI task.
Context manifest	Integrity-bound record of sources, retrieval, time, graph, twins, memory, evidence, instructions, policy, quality, and omissions.
Contest	Durable challenge to a material AI claim, source, method, result, recommendation, or policy application.
Decision sovereignty	Principle that a named authorized human retains final authority for consequential institutional decisions.
Dissent	Preserved material disagreement among evidence, models, agents, experts, or stakeholders.
Drift	Material change in data, concepts, populations, behavior, calibration, tools, policy, providers, workload, or outcomes.
Evaluator	Governed method or reviewer that measures an exact AI configuration against a declared evaluation claim.
Evidence binding	Resolvable connection from a claim or field to exact supporting source or object state and transformation lineage.
Execution envelope	Typed identity, authority, task, context, artifact, budget, tool, policy, evidence, and lifecycle contract for a run.
Experience memory	Approved reusable task patterns and lessons released through governance, distinct from raw execution history.
Explainability	Ability to account faithfully for evidence, methods, assumptions, uncertainty, alternatives, controls, and limitations.
Explanation product	Versioned structured object that binds an AI result to its decision-relevant evidence and execution state.
Fitness	Scope-specific determination that an exact configuration is suitable for a task, population, consequence, deployment, and time.
Grounding	Binding model or agent work to governed entities, relationships, memory, graph, twins, time, evidence, and truth states.
Human gate	Durable boundary at which a named authorized human approves, rejects, changes, abstains, or escalates an exact act.

TERM	AI ARCHITECTURE DEFINITION
Hybrid retrieval	Governed combination of lexical, semantic, graph, temporal, geospatial, structured, and memory retrieval.
Inference	Computation that produces a probabilistic, assessed, or generated result without independently establishing canonical truth.
Instruction authority	Ordered trust hierarchy separating controlling artifacts and approved directions from untrusted content.
Model adapter	Provider-specific binding that translates a canonical task contract while preserving semantics and controls.
Model gateway	Governed interface that resolves, invokes, observes, and normalizes approved model artifacts.
Model pluralism	Use of replaceable model and method families behind stable task and product contracts.
Operating envelope	Evaluated bounds of task, data, population, quality, uncertainty, safety, deployment, and operational conditions.
Partial result	Output that explicitly identifies completed work, missing dependencies, affected claims, and permitted downstream use.
Policy guard	Advisory agent that detects possible control conflicts while authoritative enforcement remains external.
Product admission	Owning-layer validation and commit that converts a proposal into governed canonical state.
Prompt artifact	Signed versioned instruction or template resolved for a precise task, model, policy, and release.
Provider abstraction	Isolation of provider protocols and semantics behind canonical model and tool interfaces.
Red teaming	Authorized adversarial testing of the complete AI system, controls, people, and operational response.
Replay	Reconstruction of an execution or decision using its exact historical information, artifacts, controls, and authority.
Residual risk	Risk remaining after verified controls, accepted by a named authority for a bounded scope and time.
Risk acceptance	Explicit time-bounded accountable decision to operate with stated residual risk and monitoring.
Run memory	Tenant- and execution-scoped plans, messages, tools, checkpoints, intermediate products, and reviews.
Safety case	Structured argument connecting hazards, controls, assumptions, evidence, monitoring, residual risk, and release approval.
Side effect	Material change outside analytical execution, including communication, commitment, financial, cyber, physical, or operational action.
Strategic Memory	Connected governed record of strategic evidence, assumptions, objectives, decisions, commitments, outcomes, and lessons.
Structured output	Typed model or agent result validated against canonical schema and semantic admission rules.
Supervisor	External runtime authority that admits, constrains, monitors, suspends, recovers, and terminates agent execution.
Synthetic branch	Scenario or simulated state isolated from observed and estimated canonical state.
Task contract	Provider-independent definition of input, output, method constraints, consequence, controls, quality, and failure behavior.
Tool	Registered typed capability adapter through which an agent reads, computes, proposes, or requests an effect.
Tool authority	Exact permitted operation, target, purpose, parameter, limit, identity, time, and human-gate boundary.
Truth state	Observed, claimed, inferred, assessed, forecast, scenario, simulated, disputed, corrected, or withdrawn status.
Uncertainty	Explicit representation of randomness, lack of knowledge, ambiguity, disagreement, model limits, and incompleteness.
Weak signal	Low-strength but potentially meaningful evidence of emerging change whose mechanism and materiality remain under assessment.
Workflow	Durable versioned coordination of tasks, waits, gates, retries, compensation, evidence, and terminal state.

Baseline closure

Volume 5 closes when every AI behavior can identify its controlling constitutional invariant, canonical layer, engineering owner, task and product contract, model and agent artifacts, context and evidence boundary, consequence class, policy and capability controls, human gate, uncertainty, explanation, evaluation, safety case, release, monitoring, deployment-parity evidence, incident path, learning path, and retirement state.

END STATE

The Eye operates here as one governed Strategic Intelligence Operating System: continuously observing the world, constructing evidence-bound understanding, remembering institutional truth, connecting knowledge, reconciling living twins, forecasting and simulating possible futures, preparing explainable decision packages for named humans, governing effects and commitments, measuring outcomes, and learning only through accountable release.

CONSTITUTIONAL INHERITANCE / C-001 · C-004 · C-005 · C-024 · C-028 · C-035 · C-042 · C-044 · C-051 · C-052